

Chapter 1

Introduction to Business Research

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1.1 Research: Definitions

When listening to the radio, watching the television or reading a daily newspaper it is difficult to avoid the term ‘research’. The results of ‘research’ are all around us. A debate about the findings of a recent poll of people’s opinions inevitably includes a discussion of ‘research’, normally referring to the way in which the data were collected. Politicians often justify their policy decisions on the basis of ‘research’. Newspapers report the findings of research companies’ surveys. Documentary programmes tell us about ‘research findings’, and advertisers may highlight the ‘results of research’ to encourage you to buy a particular product or brand. However, what these examples really emphasise is the wide range of meanings given to the term ‘research’ in everyday speech.

Now, let’s have few definitions provided by authors. You would, at the end, realize that the definitions (the authors) see research from various perspectives and they are complementing each other rather than conflicting. And there would be as many definitions of (business) research as there are research books (authors). Attempt to appreciate the main describing characteristics of research in each definition. You would finally develop the distinguishing features of research and of course definition of your own.

Definition 1: *Redman and Mory*

“Research is a systematized effort to gain new knowledge.”

Definition 2

Research is also defined as a movement- movement from known to unknown.

Definition 3

Research is the application of human intelligence in a systematic manner to a problem whose solution is not immediately available.

Definition 4: *Clifford Woody*

“Research comprises of defining and redefining problems, formulating hypothesis or suggested solutions; making deductions and reaching conclusions; and at last carefully testing the conclusions to determine whether they fit the formulating hypothesis”.

Definition 5 (Saunders, Lewis and Thornhill)

Saunders et.al (2009) define **research** as something that people undertake in order to find out things in a systematic way, thereby increasing their knowledge. Two phrases are important in this definition: ‘systematic way’ and ‘to find out things’.

- a) ‘Systematic’ suggests that research is based on logical relationships and not just beliefs. As part of this, your research will involve an explanation of the methods used

to collect the data, will argue why the results obtained are meaningful, and will explain any limitations that are associated with them.

- b) ‘To find out things’ suggests there is a multiplicity of possible purposes for your research. These may include describing, explaining, understanding, criticizing and analyzing. However, it also suggests that you have a clear purpose or set of ‘things’ that you want to find out, such as the answer to a question or number of questions.

Definition 6

Walliman (2005) argues that many of everyday uses of the term ‘research’ are not research in the true meaning of the word. As part of this, he highlights ways in which the term is used wrongly:

- ✓ just collecting facts or information with no clear purpose;
- ✓ reassembling and reordering facts or information without interpretation;
- ✓ As a term to get your product or idea noticed and respected.

The first of these highlights the fact that, although research often involves the collection of information, it is more than just reading a few books or articles, talking to a few people or asking people questions. While collecting data may be part of the research process, if it is not undertaken in a systematic way, on its own and, in particular, with a clear purpose, it will not be seen as research. The second of these is commonplace in many reports. Data are collected, perhaps from a variety of different sources, and then assembled in a single document with the sources of these data listed. However, there is no interpretation of the data collected. Again, while the assembly of data from a variety of sources may be part of the process of research, without interpretation it is not research. Finally, the term ‘research’ can be used to get an idea or product noticed by people and to suggest that people should have confidence in it. In such instances, when you ask for details of the research process, these are either unclear or not forthcoming.

Based upon this brief discussion we can already see that research has a number of characteristics:

- Data are collected systematically.
- Data are interpreted systematically
- There is a clear purpose: to find things out.

Definition 7: Rusk

“Research is a point of view, an attitude of inquiry or a frame of mind. It asks questions which have hitherto not been asked, and it seeks to answer them by following a fairly definite procedure. It is not a mere theorising, but rather an attempt to elicit facts and to face them

once they have been assembled. Research is likewise not an attempt to bolster up pre-conceived opinions, and it implies a readiness to accept the conclusions to which an inquiry leads, no matter how unwelcome they may prove. When successful, research adds to the scientific knowledge of the subject.”

Definition 8: George J. Mouly

He defines research as, “The systematic and scholarly application of the scientific method interpreted in its broader sense, to the solution of social problems; conversely, any systematic study designed to promote the development of social studies as a science can be considered research.”

Definition 9: Francis G. Cornell

“To be sure the best research is that which is reliable, verifiable and exhaustive, so that it provides information in which we have confidence. The main point here is that research is, literally speaking, a kind of human behavior, an activity in which people engage. By this definition, all intelligent human behavior involves some research.”

Definition 10: James Harvey Robinson

“Research is but diligent search which enjoys the high flavor or primitive hunting.”

Definition 11: Encyclopaedia of Social Science

“Research is the manipulation of things concepts or symbols for the purpose of generalizing to extend, correct or verify knowledge, whether that knowledge aids in the practice of an art.”

Definition 12:P.M. Cook

“Research is an honest, exhaustive, intelligent searching for facts and their meanings or implications with reference to a given problem. The product or findings of a given piece of research should be an authentic, verifiable and contribution to knowledge in the field studied.” He has emphasized the following characteristics of research in his definition:

- a) It is an honest and exhaustive process.
- b) The facts are studied with understanding.
- c) The facts are discovered in the light of problem. Research is problem-centered.
- d) The findings are valid and verifiable.
- e) Research work should contribute new knowledge in that field.

1.2 Business Research: Definitions, Scope and When needed

It is defined as the systematic and objective process of gathering, recording, and analyzing data for aid in making business decisions (Zikmund, 2003).

Business research can be described as a systematic and organized effort to investigate a problem encountered in the work setting that needs a solution (Sekaran, 2006: pp 5)

Business research is the application of the scientific method in searching for the truth about business phenomena (Zikmund et al., 2008). These activities include defining business opportunities and problems, generating and evaluating alternative courses of action, and monitoring employee and organizational performance.

Points from the definitions

1. Systematic

- Research information is neither intuitive nor haphazardly gathered
- It connects patient study and scientific investigation where in the researcher taken another, more careful look at discover all that can be known about the subject of the study

2. Objective

- The role of the researcher is to be detached and impersonal rather than engaging in a biased attempt to prove preconceived ideas.
- If bias enters the research process, the value of the data collected and analyzed or the information generated is considerably reduced.
- Two main dangerous involved if the research endeavor is not objective
 - ✎ Possible loss of credibility
 - ✎ Misleading and wrong decision

3. Objective of business research

- The definitions point out that its objective is to facilitate the managerial decision making process for all aspects of a business- finance, marketing, personnel and so on
- As essential tool for management in its problem solving and decision making activities. Business research generates and provides the necessary qualitative or quantitative information up on which to base decision
- By reducing the uncertainty of decisions, research reduces the risk of making wrong decision

Note

Research should be an aid to managerial judgment, not a substitute for it

4. Relevancy

It furnishes three important tasks:

- It avoids collection of irrelevant information and saves time and money
- It compares the information to be collected with researcher's criteria for action
- It enables to see whether the research is proceeding in the right direction

5. Reproducible

A reproducible research procedure is one, which an equally competent researcher could duplicate, and from it deduces approximately the same results. Precise information regarding samples-methods, collection etc., should be specified.

6. Control:

Research is not only affected by the factors, which one is investigating but some other extraneous factors also. It is impossible to control all the factors. All the factors that we think may affect the study have to be controlled and accounted for.

Scope of Business Research

- Research regarding production, finance, marketing, and management in for-profit corporations is business research.

Business Research= marketing research+ Financial research+ Operations/production Research+ Human resources research + Materials mgt research+ Risk Mgt research+ other business functions research

Business research encompasses all of these functional specialties. While researchers in different functional areas may investigate different phenomena, they are similar to one another because they share similar research methods.

- Business research covers a wide range of phenomena. For managers, the purpose of research is to provide knowledge regarding the organization, the market, the economy, or another area of uncertainty. A financial manager may ask, “Will the environment for long-term financing be better two years from now?” A personnel manager may ask, “What kind of training is necessary for production employees?” or “What is the reason for the company’s high employee turnover?” A marketing manager may ask, “How can I monitor my retail sales and retail trade activities?” Each of these questions requires information about how the environment, employees, customers, or the economy will respond to executives’ decisions. Research is one of the principal tools for answering these practical questions.
- Business research also includes efforts that assist non-profit organizations and governmental agencies, which perform many functions that are similar, if not identical, to those of for-profit business organizations.

When Is Business Research Needed?

The need to make intelligent, informed decisions ultimately motivates an organization to engage in business research. Not every decision requires research. Thus, when confronting a key decision, a manager must initially decide whether or not to conduct business research.

The determination of the need for research centers on (1) time constraints, (2) the availability of data, (3) the nature of the decision to be made, and (4) the value of the research information in relation to costs.

Time Constraints

Systematic research takes time. In many instances, management believes that a decision must be made immediately, allowing no time for research. Decisions sometimes are made without adequate information or thorough understanding of the business situation. Although making decisions without researching a situation is not ideal, sometimes the urgency of a situation precludes the use of research. The urgency with which managers usually want to make decisions conflicts with researchers' desire for rigor in following the scientific method.

Availability of Data

Often managers already possess enough data, or information, to make sound decisions without additional research. When they lack adequate information, however, research must be considered. This means that data need to be collected from an appropriate source. If a potential source of data exists, managers will want to know how much it will cost to get the data.

Nature of the Decision

The value of business research will depend on the nature of the managerial decision to be made. A routine tactical decision that does not require a substantial investment may not seem to warrant a substantial expenditure for research. For example, a computer company must update its operator's instruction manual when it makes minor product modifications. The research cost of determining the proper wording to use in the updated manual is likely to be too high for such a minor decision. The nature of the decision is not totally independent of the next issue to be considered: the benefits versus the costs of the research. In general, however, the more strategically or tactically important the decision, the more likely it is that research will be conducted.

Benefits versus Costs

Earlier we discussed some of the managerial benefits of business research. Of course, conducting research to obtain these benefits requires an expenditure of money. In any decision-making situation, managers must identify alternative courses of action and then weigh the value of each alternative against its cost. Business research can be thought of as an investment alternative. When deciding whether to make a decision without research or to postpone the decision in order to conduct research, managers should ask three questions:

1. Will the payoff or rate of return be worth the investment?

2. Will the information gained by business research improve the quality of the managerial decision enough to warrant the expenditure?
3. Is the proposed research expenditure the best use of the available funds?

1.3 Objectives of research

The research has the following three objectives:

- a) Theoretical objective
- b) Factual objective and
- c) Application objective.

A. Theoretical Objective

Those researches whose objectives are theoretical formulate the new theories, principles or laws. Such type of research is explanatory because it explains the relationships of certain variables. These researches contribute some basic knowledge to the human knowledge. The researches in different disciplines i.e., Physics, Chemistry, Mathematics etc. have the theoretical objective.

B. Factual Objective

Those researches whose objective is factual find out new facts. This objective is by nature descriptive. These researches describe facts or events which happened previously. Such type of research is done in history.

C. Application Objective

The research having application objective does not contribute a new knowledge in the fund of human knowledge but suggests new applications. By application we mean improvement and modification in practice. For example if anyone gives a new application of electricity then such type of research has application objective.

1.4. Types of Business Research

1. On the basis of the outcome of research

Basic/pure / fundamental research:-

- Basic research attempts to expand the limits of knowledge.
- It does not directly involve the solution to a particular pragmatic problem.
- Basic research findings generally cannot be immediately implemented.

Applied research:-

- **Applied research** encompasses those studies undertaken to answer questions about specific problem or to make decisions about a particular course of action or policy.

2. On the basis of purpose

a. Exploratory research/preliminary research (pilot survey)

- Conducted to clarify ambiguous problems need to gain better understanding of the dimension of problem(s)
- Uncovering conclusive evidence to determine a particular course of action is not the purpose of exploratory research
- Usually, exploratory research is conducted with the expectation that subsequent research will be required to provide conclusive evidence. The area of investigation may be so new or so vague that a researcher needs to do an exploration just to learn something about the area.

b. **Descriptive research:** - the major purpose of descriptive research is to describe characteristic of a population or phenomenon. Descriptive researches seek to determine the answers to **who, what, when, where and how questions**. Unlike exploratory research, descriptive studies are based on some previous understanding of the nature of the research problem.

c. **Causal research:** - the main goal of causal research is to identify **cause-and-effect** relationships among variables.

e.g.

- ✎ Influence of price, packaging, advertising on sales,
- ✎ Training Vs productivity.

3. On the basis of data analysis

- a. **Qualitative research:-** applicable for phenomenon that cannot be expressed in terms of quantity- things related to kind and quality
- b. **Quantitative research:** - based on the measurement of quantity or amount. It is applicable for phenomenon that can be expressed in terms of quantity.

4. On the basis of environment in which the research is carried out

- a. **Field research-** carried out in fields. They are also common in social science, Agricultural science, History and Archeology.
- b. **Laboratory research:-** carried out in laboratory; they are commonly experimental researches
- c. **Simulation research**

5. On the basis of time required to complete the research

- a. **One time research:-** research limited to a single time period

- b. **Longitudinal research:** - research carried out over several time periods. Such research is also called ongoing research

1.5. Research process: An overview

Business research follows a general pattern what is described by its process. The stages are

1. Defining the research problem
2. Planning a research design
3. Planning sample
4. Collecting data
5. Analyzing the data
6. Formulating conclusions and preparing the report

Note:

- **Forward linkage Vs Backward linkage**

Forward linkage implies that the earlier stages of research will influence the design of the latter stages. Eg. Objective of the research influence sample selection and data collection tool; sampling units will affect wording of questions. **Backward linkage** implies that the latter steps have an influence on the earlier stages in the research process.

- The stages of the research process overlap continuously, and it is clearly an oversimplification to state that every research project has exactly the same ordered sequence of activities.

1. **Formulating (defining) the research problem.**

Two steps are involved in formulating the research problem

- a. Understanding the problem thoroughly
- b. Rephrasing the same into meaningful terms from an analytical point of view

A problem well defined

- ✎ “A well defined problem is a problem half solved”.
- ✎ “The formulation of a problem is often more essential than its solution” (Albert Einstein)
- ✎ In business research if the data are collected before the nature of the business problem is carefully thought out, the data probably will not help to solve the problem.

2. **Planning the research design**

Research design is a master plan specifying the method and procedures for collecting and analyzing the need information. Research design provides a framework or plan of action for the research. Objectives of the study determined during the early stages of research are included in the design to ensure that the information collected is appropriate for solving the problem. It is an outline of

- Objective of the study
- Source of information
- Sampling methodology
- The research method or technique
- Schedule of the research
- Cost of the research

3. Sampling

Sampling involves any procedure that uses a small number of items or a portion of a population to make a conclusion regarding the whole population.

The first sampling question to ask is “**Who is to be sampled?**” The answer to this primary question requires the identification of a *target population*. Who do we want the sample to reflect? Defining this population and determining the sampling units may not be so easy. If, for example, a savings and loan association surveys people who already have accounts for answers to image questions, the selected sampling units may represent *current* customers but will not represent *potential* customers. Specifying the target population is a crucial aspect of the sampling plan.

The next sampling issue concerns sample size. **How big should the sample be?** Although management may wish to examine every potential buyer of a product or service, doing so may be unnecessary as well as unrealistic. Other things equal, larger samples are more precise than smaller ones. However, proper probability sampling can allow a small proportion of the total population to give a reliable measure of the whole. A later discussion will explain how large a sample must be in order to be truly representative of the universe or population. Essentially, this is a question of how much variance exists in the population.

The final sampling decision is **how to select the sampling units**. Simple random sampling may be the best known type, in which every unit in the population has an equal and known chance of being selected. However, this is only one type of sampling. For example, if members of the population are found in close geographical clusters, a cluster sampling procedure (one that selects area clusters rather than individual units in the population) will reduce costs. Rather than selecting 1,000 individuals throughout the United States, it may be more economical to first select 25 counties and then sample within those counties. This will substantially reduce travel, hiring, and training costs. In determining the appropriate sampling plan, the researcher will have to select the most appropriate sampling procedure for meeting the established study objectives.

4. Collecting Data

The data gathering stage begins once the sampling plan has been formalized. Data gathering is the process of gathering or collecting information. Data may be gathered by human observers or interviewers, or they may be recorded by machines as in the case of scanner data and Web-based surveys.

Obviously, the many research techniques involve many methods of gathering data. Surveys require direct participation by research respondents. This may involve filling out a questionnaire or interacting with an interviewer. In this sense, they are obtrusive. **Unobtrusive methods** of data gathering are those in which the subjects do not have to be disturbed for data to be collected. They may even be unaware that research is going on at all. For instance, a simple count of motorists driving past a proposed franchising location is one kind of data gathering method. However the data are collected, it is important to minimize errors in the process. For example, the data gathering should be consistent in all geographical areas. If an interviewer phrases questions incorrectly or records a respondent's statements inaccurately (not verbatim), major data collection errors will result.

5. Processing and analyzing data

EDITING AND CODING

After the fieldwork has been completed, the data must be converted into a format that will answer the manager's questions. This is part of the data processing and analysis stage. Here, the information content will be mined from the raw data. Data processing generally begins with editing and coding the data. Editing involves checking the data collection forms for omissions, legibility, and consistency in classification. The editing process corrects problems such as interviewer errors (an answer recorded on the wrong portion of a questionnaire, for example) before the data are transferred to the computer.

Before data can be tabulated, meaningful categories and character symbols must be established for groups of responses. The rules for interpreting, categorizing, recording, and transferring the data to the data storage media are called codes. This coding process facilitates computer or hand tabulation. If computer analysis is to be used, the data are entered into the computer and verified. Computer-assisted (online) interviewing is an example of the impact of technological change on the research process.

DATA ANALYSIS

Data analysis is the application of reasoning to understand the data that have been gathered. In its simplest form, analysis may involve determining consistent patterns and summarizing

the relevant details revealed in the investigation. The appropriate analytical technique for data analysis will be determined by management's information requirements, the characteristics of the research design, and the nature of the data gathered. Statistical analysis may range from portraying a simple frequency distribution to more complex multivariate analyses approaches, such as multiple regressions.

6. Drawing Conclusions and Preparing a Report

One of the most important jobs that a researcher performs is communicating the research results. This is the final stage of the research project, but it is far from the least important. The conclusions and report preparation stage consists of interpreting the research results, describing the implications, and drawing the appropriate conclusions for managerial decisions. These conclusions should fulfil the deliverables promised in the research proposal. In addition, it's important that the researcher consider the varying abilities of people to understand the research results. The report shouldn't be written the same way to a group of Ph.D.'s as it would be to a group of line managers.

All too many applied business research reports are overly complicated statements of technical aspects and sophisticated research methods. Frequently, management is not interested in detailed reporting of the research design and statistical findings, but wishes only a summary of the findings. If the findings of the research remain unread on the manager's desk, the study will have been useless. The importance of effective communication cannot be overemphasized. Research is only as good as its applications.

1.6. Overview of science and the scientific method

Science

In simple terms, *science* can be defined as a methodological and systematic approach to the acquisition of new knowledge. This definition of science highlights some of the key differences between how scientists and non-scientists go about acquiring new knowledge. Specifically, rather than relying on mere casual observations and an informal approach to learn about the world, scientists attempt to gain new knowledge by making careful observations and using systematic, controlled, and methodical approaches

Scientific Method

- The *scientific method* is best thought of as an approach to the acquisition of new knowledge, and this approach effectively distinguishes science from non-science.
- The scientific method is not actually a single method, as the name would erroneously lead one to believe, but rather an overarching perspective on how scientific

investigations should proceed. It is a set of research principles and methods that help researchers obtain valid results from their research studies.

- Because the scientific method deals with the general *approach* to research rather than the *content* of specific research studies, it is used by researchers in all different scientific disciplines. As will be seen in the following sections, the biggest benefit of the scientific method is that it provides a set of clear and agreed upon guidelines for gathering, evaluating, and reporting information in the context of a research study (Cozby, 1993).

Elements of the Scientific Method

There has been some disagreement among researchers over the years regarding the elements that compose the scientific method. In fact, some researchers have even argued that it is impossible to define a universal approach to scientific investigation. Nevertheless, for over 100 years, the scientific method has been the defining feature of scientific research.

Researchers generally agree that the scientific method is composed of the following key elements: *an empirical approach, observations, questions, hypotheses, experiments, analyses, conclusions, and replication.*

1. Empirical Approach

The scientific method is firmly based on the empirical approach. The *empirical approach* is an evidence-based approach that relies on direct observation and experimentation in the acquisition of new knowledge (Kazdin, 2003a). In the empirical approach, scientific decisions are made based on the data derived from direct observation and experimentation.

2. Observations

An important component in any scientific investigation is observation. In this sense, *observation* refers to two distinct concepts—being aware of the world around us and making careful measurements.

- a) Observations of the world around us often give rise to the questions that are addressed through scientific research. For example, the Newtonian observation that apples fall from trees stimulated much research into the effects of gravity. Therefore, a keen eye to your surroundings can often provide you with many ideas for research studies.
- b) *Observation* also refers to the process of making careful and accurate measurements, which is a distinguishing feature of well-conducted scientific investigations. When making measurements in the context of research, scientists typically take great precautions to avoid making biased observations.

3. Questions

After getting a research idea, perhaps from making observations of the world around us, the next step in the research process involves translating that research idea into an answerable question. The term “answerable” is particularly important in this respect, and it should not be overlooked. It would obviously be a frustrating and ultimately unrewarding endeavour to attempt to answer an unanswerable research question through scientific investigation.

4. Hypotheses

The next step in the scientific method is coming up with a *hypothesis*, which is simply an educated—and testable—guess about the answer to your research question. A hypothesis is often described as an attempt by the researcher to explain the phenomenon of interest.

Hypotheses can take various forms, depending on the question being asked and the type of study being conducted. A key feature of all hypotheses is that each must make a *prediction*.

5. Experiments

After articulating the hypothesis, the next step involves actually conducting the experiment (or research study). For example, if the study involves investigating the effects of exercise on levels of cholesterol, the researcher would design and conduct a study that would attempt to address that question. As previously mentioned, a key aspect of conducting a research study is measuring the phenomenon of interest in an *accurate* and *reliable* manner

6. Analyses

After conducting the study and gathering the data, the next step involves analyzing the data, which generally calls for the use of statistical techniques. The type of statistical techniques used by a researcher depends on the design of the study, the type of data being gathered, and the questions being asked. Although a detailed discussion of statistics is not the intent of this section, it is important to be aware of the role of statistics in conducting a research study. In short, statistics help researchers minimize the likelihood of reaching an erroneous conclusion about the relationship between the variables being studied.

7. Conclusions

After analyzing the data and determining whether to reject the hypothesis, the researcher is now in a position to draw some conclusions about the results of the study. For example, if the researcher rejected the null hypothesis, the researcher can conclude that the phenomenon being studied had an effect—a *statistically significant* effect, to be more precise.

8. Replication

Replication essentially means conducting the same research study a second time with another group of participants to see whether the same results are obtained (Kazdin, 1992;

Shaughnessy & Zechmeister, 1997). The same researcher may attempt to replicate previously obtained results, or perhaps other researchers may undertake that task. Replication illustrates an important point about scientific research—namely, that researchers should avoid drawing broad conclusions based on the results of a single research study because it is always possible that the results of that particular study were an aberration. In other words, it is possible that the results of the research study were obtained by chance or error and, therefore, that the results may not accurately represent the actual state of things. However, if the results of a research study are obtained a second time (i.e., replicated), the likelihood that the original study's findings were obtained by chance or error is greatly reduced.

1.7. Motivations in research

The possible motives for doing research may be either one or more of the following

- Desire to get a research degree along with its consequential benefits
- Desire to face the challenge in solving the unsolved problems
- Desire to get intellectual joy of doing some creative work
- Desire to be of service to society
- Desire to get respectability
- Directives of the government, employment conditions, curiosity about new things
- desire to understand causal relationships, social thinking and awakening

Chapter Two

Problem Definition/Formulation

Chapter Outline

2.1 Research Problem and problem

Definition: Meaning and Importance

2.2 The Problem-Definition Process

- Understand the business situation
- Identify key problem(s) from symptoms
- Identify the relevant variables
- Write research questions and/or research hypotheses
- Determine the unit of analysis

Chapter Objectives

Chapter Objectives

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2.1. Research Problem and problem Definition: Meaning and Importance

In the research process, the first and foremost step happens to be that of selecting and properly defining a research problem. It is crucial that the focus of the focus of further research, in other words, the problem, be unambiguously identified and defined. No amount of good research can find solutions to the situation, if the critical issue or the problem is not clearly pinpointed.

What is a research problem?

A problem does not necessarily mean that something is seriously wrong with a current situation that needs to be rectified immediately. A problem could simply indicate an interest in an issue where finding the right answers might help to improve an existing situation. Thus, a problem can be best described as any situation where a gap exists between the actual and the desired ideal states.

A **problem** occurs when there is a difference between the current conditions and a more preferable set of conditions. In other words, a gap exists between the way things are now and a way that things could be better. The gap can come about in a number of ways:

1. Business performance is worse than expected business performance. For instance, sales, profits, and margins could be below targets set by management. This is a very typical type of problem analysis. Think of all the new products that fail to meet their targeted

goals. Trend analysis would also be included in this type of problem. Management is constantly monitoring key performance variables. Previous performance usually provides a benchmark forming expectations. Sales, for example, are generally expected to increase a certain percentage each year. When sales fall below this expectation, or particularly when they fall below the previous year's sales, management usually recognizes that they have a potential problem on their hands.

2. Actual business performance is less than possible business performance. Realization of this gap first requires that management have some idea of what is possible. This may form a research problem in and of itself. Opportunity-seeking often falls into this type of problem-definition process. Many American and European Union companies have redefined what possible sales levels are based upon the expansion of free markets around the world. China's Civil Aviation Administration has relaxed requirements opening the Chinese air travel market to private airlines. Suddenly, the possible market size for air travel has increased significantly, creating opportunities for growth.
3. Expected business performance is greater than possible business performance. Sometimes, management has unrealistic views of possible performance levels—either too high or too low. One key problem with new product introductions involves identifying realistic possibilities for sales. While you may have heard the old adage that 90 percent of all new products fail, how many of the failures had a realistic sales ceiling? In other words, did the company know the possible size of the market? In this case, the problem is not with the product but with the plan. Some product “failures” may actually have been successful if management had a more accurate idea of the total market potential. Management can close this gap through decision making. Researchers help managers make decisions by providing relevant input.

Problem-Definition (Problem Statement)

A **problem definition** indicates a specific managerial decision area to be clarified or problem to be solved. It specifies the research questions to be answered and the objective of the research. “To define a problem means to put a fence round it, to separate it by careful distinction from like questions found in related situations of need.” (Whitney cited in Koul, 1984). According to Manroe and Engelhart(cited in Koul,1984) “to define a problem means to specify it in detail and with precision each question and subordinate questions to be answered.”

A problem statement is the description of an issue currently existing which needs to be addressed. It provides the context for the research study and generates the questions which

the research aims to answer. The statement of the problem is the focal point of any research. A good problem statement is just one sentence (with several paragraphs of elaboration).

The problem statement is therefore the axis, which the whole research revolves around, because it explains in short the aim of the research. The statement of the problem involves the demarcation and formulation of the problem, i.e., the who/what, where, when, why.

What are key components of the statement problem?

Problem statements often have three elements:

1. The problem itself, stated clearly and with enough detail to establish why it is important
2. The method of solving the problem, often stated as a claim or a working thesis
3. The purpose, statement of objective and scope of the project being proposed.

These elements should be brief so that the reader can get the idea easily. One page is enough for a statement problem.

Criteria for Research Problem Statements:

- The statement of the problem should clearly indicate what is to be investigated.
- The actual statement may be in a declarative or in a question form.
- The statement should indicate the variables of interest and the specific relationship between the variables that are to be studied

Examples of Well- defined problems:

- "The frequency of job layoffs is creating fear, anxiety, and a loss of productivity in middle management workers."
- What factors were responsible for the lower labour productivity of Ethiopia's manufacturing industries during the time 20-to 20-relative to other African countries?
- To what extent has the new advertising campaign been successful in creating the high quality, customer-cantered corporate image that it was intended to produce?
- What are the effects of downsizing on the long-range patterns of companies?
- Does expansion of internal operations result in an enhancement of the firm's image and value?

2.2. The Problem-Definition Process

The problem-definition process involves several interrelated steps. Sometimes, the boundaries between each step aren't exactly clear. But generally, completing one step leads to the other and by the time the problem is defined, each of these steps has been addressed in some way. The steps are

1. Understand the business situation—identify key symptoms

2. Identify key problem(s) from symptoms
3. Determine the relevant variables
4. Write research questions and/or research hypotheses, and research objectives
5. Determine the unit of analysis

A separate section deals with each stage below.

2.2.1 Understand the background of the problem

Situation analysis

- Preliminary investigation/informative gathering of background information to familiarize researchers managers with the decision area
- Gaining an awareness of organizational/environmental conditions
- Discussing with those who first raised in order to find out how the problem originally came about and with what objectives in view
- Discussing with those who have a good knowledge of the problem concerned

2.2.2 Isolate and identify the problems, NOT the symptoms

A researcher should know what exactly the issue is, for which one seeks answers. It is important that symptoms of problems are not defined as the real problem. Finding the “right answer” to the “wrong” problem definitions will not help.

Example 1:

When a firm has a problem with advertising effectiveness, the possible causes of this problem may be low brand awareness, the wrong brand image, use of the wrong media, or perhaps too small a budget. Certain occurrences that appear to be the problem may be only symptoms of a deeper problem.

Example 2

A manager might have tried to increase productivity by increasing the piece rate, but with little success. Here the real problem may be the low morale and motivation of employees who feel they are not being recognized as valuable contributors to the system and get no praise for the good work that they do. The low productivity may merely be a symptom of the deep-rooted morale and motivation problem.

Example 3

- Women employees complain that salaries are too low
 - Problem definition: salaries need to be compared to industry average
 - True problem: benefit program is not suited to women’s needs

Example 4

- Brewery company- consumer prefer taste of competitor's product
 - Problem definition: taste of brewery product needs to be reformulated
 - True problem – old fashioned package is influencing taste perception.

2.2.3 Determine the relevant variables

What things should be studied to address a decision statement? Researchers answer this question by identifying key variables. The term variable is an important one in research.

A variable is defined as anything that varies or changes in value. Variables can exhibit differences in value, usually in magnitude or strength, or in direction. In research, a variable is either observed or manipulated, in which case it is an experimental variable.

The converse of a variable is a **constant**. A constant is something that does not change. Constants are not useful in addressing research questions. Since constants don't change, management isn't very interested in hearing the key to the problem is something that won't or can't be changed.

Types of Variables

1. Quantitative vs. Qualitative variables

Quantitative variable (Discrete and Continuous variables)

A discrete random variable may assume either a finite number of values or an infinite sequence of values. A **continuous variable** is one that can take on a range of values that correspond to some quantitative amount. Sales volume, profits, and margin are common business metrics that represent continuous variables.

Qualitative / Categorical variable

Categorical variable is one that indicates membership in some group. The term **classificatory Variable** is sometimes also used and is generally interchangeable with *categorical variable*.

2. Four other types of variables

- a) dependent variable (also known as the criterion variable)
- b) independent variable (also known as the predictor variable)
- c) Moderating variable
- d) Intervening variable

This distinction becomes very important in understanding how business processes can be modeled by a researcher. The distinction must be clear before one can correctly apply certain

statistical procedures like multiple regression analysis. In some cases, however, such as when only one variable is involved in a hypothesis, the researcher need not make this distinction.

A) Dependant Variable

The dependant variable is the variable of primary interest to the researcher. The researcher's goal is to understand and describe the dependant variable, or to explain its variability or to predict it. In other words, it is the main variable that lends itself for investigation as a viable factor. Through the analysis of the dependant variable (i.e. finding what variables influence it) it is possible to find answers or solutions to the problem. For this purpose, the researcher will be interested in quantifying and measuring the dependant variable as well as the other variables that influence this variable.

Example 1

A manager is concerned that the sales of a new product introduced after test marketing it do not meet with his expectations. The dependant variable here is sales. Since the sales of the product can vary-can be low, medium, or high-it is a variable; since sales is the main focus of interest to the manager, it is the dependant variable.

Example 2

A vice president is concerned that employees are not loyal to the organization, and in fact, seem to switch their loyalty to other institutions. The dependant variable in this case is would be organizational loyalty.

B) Independent Variable

An independent variable is one that influences the dependant variable in either a positive or negative way. That is, when the independent variable is present, the dependent variable is also present, and with each unit of change in the independent variable, there is a change in the dependent variable also.

Example 1

Research indicates that successful new product development has an influence on the stock market price the company. That is, the more successful the new product turns out to be, the higher will be the stock market price of that firm. Therefore, the success of the new product is the independent variable, and the stock market price the dependent variable.

Example 2

Cross-cultural research indicates that managerial values govern the power distance between superiors and subordinates. Here, power distance is the subject of interest and hence the

dependant variable. Managerial values that explain the variance in power distance is the independent variable.

C) Moderating Variable

The moderating variable is the one that has a strong contingent effect on the independent-dependent relationship. That is, the presence of a third variable (the moderating variable) modifies the original relationship between the independent variable and the dependent variable.

Example

A prevalent theory is that the diversity of the workforce (comprising people of different ethnic origins, races, and nationalities) contributes more to organizational effectiveness because each group brings its own special expertise and skills to the workforce. In this case, organizational effectiveness is the dependant variable which is positively influenced by workforce diversity-the independent variable. However, the effective utilization of different talents, perspectives, and electric-problem solving capabilities for enhanced organizational effectiveness is contingent on the skill of managers in acting as a catalyst. This managerial expertise then becomes the moderating variable.

D) Intervening variable

An intervening variable is one that surfaces between the time the independent variable start operating to influence the dependant variable and the time their impact is felt on it. There is thus a temporal quality or time dimension to the intervening variable.

Example

Where the independent variable workforce diversity influence influences the dependent variable organizational effectiveness, the intervening variable that surfaces as a function of the diversity in the workforce is creative synergy. This helps us to understand how organizational effectiveness can result from having diversity in the workforce.

2.2.4 State the research questions and/or research Hypotheses, and research objectives

Both manager and researchers expect problem definition efforts to result in statements of research questions and research objectives

Research questions

A research question is the researcher's translation of the business problem into a specific need for inquiry. These research questions assume two forms: a central question and associated sub questions. The central question is a statement of the question being examined in the study in its most general form. A researcher asks one or two central questions followed by no more than five to seven sub questions. Several sub questions follow each general central question and the sub question narrow the focus of the study but leave open the questioning.

Guidelines (mainly for qualitative Research)

- Begin the research questions with the words “**what**” or “**how**” to convey an open and emerging design. “Why” suggests cause and effect, an approach consistent with quantitative research.
- Focus on a single phenomenon or concept.
- Use exploratory verbs that convey the language of emerging design of research. These verbs tell the reader that the study will
 - Discover (e.g., grounded theory)
 - Seek to understand (e.g., ethnography)
 - Explore a process (e.g., case study)
 - Describe the experiences (e.g., phenomenology)
 - Report the stories (e.g., narrative research)
- Use no directional language. Delete words that suggest or infer a quantitative study, words with a directional orientation such as “affect,” “influence,” “impact,” “determine,” “cause,” and “relate.”
- Expect the research questions to evolve and to change during the study in a manner consistent with the assumptions of an emerging design
- Use open-ended questions without reference to the literature or theory unless otherwise indicated by a qualitative strategy of inquiry.

Example 1

“The problem is to determine the best ways our company can train existing and potential users of networked personal computers.” (Problem Statement)

This problem statement led to the following research questions:

RQ1: How familiar are employees with the various software applications?

RQ2: what attitudes do employees have toward these software packages?

RQ3: How important are the various factors for evaluating the use of a PC?

RQ 4: How effective are training efforts in increasing knowledge and use of the new applications?

Example 2

Should the organization offer outplacement services? (Research problem/central question)

RQ 1: Are managers aware of out placement questions?

RQ 2: How concerned are managers about out placement services?

RQ 3: Do employees prefer out placement services?

Research Hypothesis

Hypothesis is an unproven proposition or possible solution to a problem which can or may not be real solutions. They are declarative statements that can be empirically tested.

Statements of Hypotheses: Formats

1. *If---then statements*

Example

- If employees are healthier, then they will take sick leave less frequently.
- If worker's attitudes towards an organization climate are changed in positive direction, there will be an increase in organizational effectiveness

2. *Directional and non-directional Hypotheses*

If, in stating the r/ship b/n variables or comparing two groups, terms such as positive, negative, more than, less than and the like are used, then these hypotheses are directional because the direction of the r/ship b/n the variables is indicated.

Example

- Sales person who show the highest job satisfaction will be the most productive
- Opinion leaders are more effected by mass media communication sources than are non leaders

On the other hand, non-directional hypotheses are those that do not postulate a r/ship or difference, but offer no indication of the direction of these r/ships or differences.

Example

- There is a r/ship b/n age and job satisfaction
- There is a d/nce b/n the work ethnic values of American and Asian employees.

3. *Null and alternate Hypotheses*

Null hypothesis

It states that the population correlation b/n two variables is equal to zero or that the difference in the mean of two groups in the population is equal to zero (or some definite number). In general, the null hypothesis is expressed as no (significant) r/ship b/n two variables or no (significant) difference b/n two groups.

Alternate Hypothesis

It is, the opposite of null, a statement expressing a r/ship b/n two variables or indicating d/nces b/n two groups.

Example (Standard use of languages in stating null hypotheses)

1. There is no relationship between utilization of ancillary support services and academic persistence for non-traditional women college student.
2. There is no relationship between family support systems and academic persistence for non-traditional aged college women.
3. There is no relationship between ancillary support services and family support systems for non-traditional college women.

Research hypothesis differs from research questions in that

- Hypothesis predict the nature of the answer
- Hypothesis proposes a relationship between two or more variables
- Questions are interrogative, whereas research hypotheses are declarative.
- At times in research, particularly in exploratory research, a proposal can only offer research questions. Research hypotheses are much more specific and therefore require considerably more theoretical support

Characteristics of hypothesis: Hypothesis must possess the following characteristics:

- Hypothesis should be ***clear and precise***. If the hypothesis is not clear and precise, the inferences drawn on its basis cannot be taken as reliable.
- Hypothesis should be ***capable of being tested***
- Hypothesis should state ***relationship between variables***, if it happens to be a relational hypothesis.
- Hypothesis should ***be limited in scope and must be specific***. A researcher must remember that narrower hypotheses are generally more testable and he should develop such hypotheses.
- Hypothesis should be stated as far as possible in ***most simple terms*** so that the same is easily understandable by all concerned. But one must remember that simplicity of hypothesis has nothing to do with its significance.
- Hypothesis should be consistent ***with most known facts***, i.e., it must be consistent with a substantial body of established facts. In other words, it should be one which judges accept as being the most likely.
- Hypothesis should be amenable to ***testing within a reasonable time***. One should not use even an excellent hypothesis, if the same cannot be tested in reasonable time for one cannot spend a life-time collecting data to test it.
- Hypothesis must explain the facts that gave rise to the ***need for explanation***. This means that by using the hypothesis plus other known and accepted generalizations, one should

be able to deduce the original problem condition. Thus hypothesis must actually explain what it claims to explain; it should have empirical reference.

Research Objectives

- The research objective is the researcher's version of the business problem. Once the research question and /or hypotheses have been stated, the research objectives are derived. These objectives explain the purpose of research in measurable terms and define standard of what the research should accomplish.
- In some instances the business problem and the research objectives are the same. Objectives specify the information needed to make a decision.
- The number of research objectives should be limited to a manageable quantity. The fewer the study objectives, the easier to ensure that each will be addressed adequately.
- Statement of the business problem influences the research objectives. The specific objectives, in turn, are the basis for the research design.

Example 1

“The problem is to determine the best ways our company can train existing and potential users of networked personal computers.” (Problem Statement stated as a broad/general objective)

RO 1: To identify how far employees are familiar with the various software applications?

RO 2: To describe employees' attitudes towards the available software applications?

RO 3: To determine the importance of factors that affect -----?

RO 4: To measure the effectiveness of the various training methods/efforts in increasing the knowledge and use of new applications?

Example 2

Should the organization offer outplacement services? (Research problem/central question)

RO 1: To determine managers' awareness using aided call

RO 2: To measure managers' satisfaction with existing personnel policies

RO 3: To identify employees' preferences in outplacement concern.

2.2.5 Determine the unit of analysis

Unit of analysis (UOA) indicates what or who should provide the data and at what level of aggregation. Researchers specify whether an investigation will collect data about individuals (such as customers, employees, and owners), households (families, extended families, and so forth), organizations (businesses and business units), departments (sales, finance, and so forth), geographical areas, or objects (products, advertisements, and so forth). Our research questions determine the unit of analysis.

Examples

- The chief financial officer of a manufacturing firm wants to know how many of the staff would be interested in attending a 3-day seminar on making appropriate investment decisions. For this purpose, data will be collected from each individual staff member and the unit of analysis is the individual
- In studies of home buying the husband/wife dyad typically is the unit of analysis rather than the individual because many purchase decisions are made jointly by husband and wife.
- In studies of organizational behaviour (group level), cross-functional teams rather than individual employees may be selected as the unit of analysis.
- Proctor and Gamble wants to see which of its various divisions (soap, paper, oil, etc) have made profits of over 12% during the current year. Here, the profits each of the divisions will be examined and the information aggregated across the various geographical units of the division. Hence the unit of analysis will be the divisions.
- An employment analyst wants to see the proportion of the workforce employed by the health care, utilities, transportation and manufacturing industries. Thus the unit of analysis, in this scenario, are the industries.
- The CEO of a multinational corporation wants to know the profits made during the past 5 years by each of subsidiaries in England, Germany, France and Ethiopia. Unit of analysis- countries.

Chapter Three

Research Design

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3.2. Purposes of studies	
3.3. Research Strategies	

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3.1. Research Design: Meaning, Importance and Elements

After the researcher has formulated the research problem, he or she must develop the research design as part of the research design stage.

Research design constitutes the blue print for the collection measurement and analysis of data. Research design is the plan and structure of the investigation so conceived as to obtain answers to research questions. The plan is the overall scheme or program of the research. It includes an outline of what the investigator will do from writing hypotheses and their operational implications to the final analysis of data.

Research design expresses both the structure of the problem and the plan of investigation used to obtain empirical evidence on those relationship. These definitions differ in detail, but together they give the essentials of research design.

- o An activity and time based plan
- o A plan always based on the research questions
- o A guide for selecting sources, and types of information
- o A for specifying the relationships among the study's variables
- o A procedural outline for every research activity

In brief, research design must, at least, contain

- o A clear statement of the research problem
- o Procedures and techniques to be used for gathering information
- o The population to be studied
- o Sample design
- o Methods to be used in processing and analyzing data
- o Time and cost budgets

Need for research design

- o It facilitates the smooth operation of various research activities
- o It makes research as efficient as possible yielding maximum information with minimum expenditure
- o It helps the researcher to organize ideas in a form whereby it will be possible for him/her to look for flaws and inadequacies

- Design can be given to others for their comments and critical evaluation. In the absence of such a course of action, it will be difficult for the critics to provide a comprehensive review of the proposed study

Elements of Research Design

Research design involves a series of rational decision-making choices and issues. The various decision areas (elements) involved in research design can be grouped into the following (sekeran, 2006; saunders et al, 2009; cooper and S 2006

A	E
B	F
C	G
	H
D	I

The first five of the research design issues are discussed in great detail in the following sections. A chapter is devoted for each of the last four design issues- only a preview is made in the last sub section.

3.2. Purpose of the Research

The way in which you asked your research question would result in descriptive, descriptive and explanatory, or explanatory answers. In thinking about your research question, you inevitably have begun to think about the purpose of your research. The classification of research purpose most often used in the research methods' literature is the threefold one of *exploratory, descriptive and explanatory*. However, in the same way as your research question can be both descriptive and explanatory, so your research project may have more than one purpose.

1. Exploratory Study

- Initial research conducted to clarify and define the nature of a problem
- Does not provide conclusive evidence
- Subsequent research expected
- Much exploratory research provides qualitative data
- Exploratory research does not involve rigorous mathematical analysis.
- The focus is on words, and observations: stories, visual portrayals, characterizations.

Categories of Exploratory Research

*A researcher may choose from four general categories of exploratory research methods-**experience survey, secondary data analysis, case studies and pilot studies.***

Experience Surveys

- Ask knowledgeable/experienced individuals who have had personal experience in the field.
 - most are quite willing
 - Quite informal; may not much more than conversation
 - May be conducted by the line manager
 - Not expected to be conclusive
- Not necessary to establish a representative probable sample.

Secondary Data Analysis

Data collected for a purpose other than the project at hand

- Economical
- Quick source for background information
- Extensive review of literature is conducted mainly in basic research.
- May be equally important in applied research.
- It is also a technique for descriptive and conclusive research.

Case Studies

- To Intensely investigate one or a few situations similar to the researcher's problem situation.
- Investigate in depth and attention to detail.
- Careful study
- May require cooperation
- The results should be seen as tentative; making generalizations from a few cases can be misleading/dangerous.

Pilot Studies

- A collective/generic term
- Any small scale exploratory study that uses sampling
- They do not apply rigorous standards of sampling and analysis.
- Generates primary data, usually for qualitative analysis
- The primary data usually are collected from employees, consumers, voters, or other subjects of ultimate concern

Example

The following is an example where exploratory research would be appropriate.

The manager of a multinational corporation is curious to know if the work ethical values of employees working in its subsidiary in Pennathur city would be different from those of Americans. There is very little information about Pennathur (except that it is a small city in Southern India), and since there is a considerable controversy about what work ethic values mean to people in other cultures, the manager's curiosity can be satisfied only by an exploratory study, interviewing the employees in organizations.

2. Descriptive research

The goal of descriptive research is to offer to the researcher a profile or to describe relevant aspects of the phenomenon of interest from an individual, organizational, industry-oriented perspective. The object of descriptive research is 'to portray an accurate profile of persons, events or situations' (Robson 2002:59).

Below are examples of situations warranting a descriptive research

Example 1: A bank manager wants to have a profile of the individual who have loan payments outstanding for 6 months and more. It would include details of their average age, earnings, nature of occupation, and the like. This might help him to elicit further information or decide right away on the types of individuals who should be made ineligible for loans in the future.

Example 2: A CEO may be interested in having a description of organizations in her industry that follow the LIFO system. In this case, the report might include the age of the organization, their locations, their production levels, assets, sales, inventory levels, suppliers, and profits. Such information might allow comparison of the performance levels of specific types of companies.

Example 3: A marketing manager might want to develop a pricing, sales, distribution, and advertising strategy for her product. With this in mind, she might ask for the information regarding the competitors, with respect to the following:

- The percentage of sales groups organized by product line, by accountants, and by region
- The types of distribution channels used
- Percentage of competitors spending more dollars on promotion
- Percentage of those using the web to sell this product.

3. Explanatory Research

Studies that establish causal relationships between variables may be termed **explanatory research**. The emphasis here is on studying a situation or a problem in order to explain the relationships between variables.

Examples of such studies are given below

- Analysis of quantitative data on manufacturing scrap rates shows a relationship between scrap rates and the age of the machine being operated. You could go ahead and subject the data to statistical tests such as correlation in order to get a clearer view of the relationship.
- You might collect qualitative data to explain the reasons why customers of your company rarely pay their bills according to the prescribed payment terms.
- A marketing manager wants to know if the sales of the company will increase if he doubles the advertising dollars. Here, the manager would like to know the nature of relationship that can be established between advertising and sales
- The testing of a hypothesis such as “More men than women are whistleblower” establishes the difference between two groups-men and women-in regard to whistle-blowing behaviour.

3.3. Research Strategies

In this section we turn our attention to the **research strategies** you may employ. Each strategy can be used for exploratory, descriptive and explanatory research (Yin 2003). However, often allocating strategies to one approach or the other is unduly simplistic. In addition, we must emphasise that no research strategy is inherently superior or inferior to any other. Consequently, what is most important is not the label that is attached to a particular strategy, but whether it will enable you to answer your particular research question(s) and meet your objectives. Your choice of research strategy will be guided by your research question(s) and objectives, the extent of existing knowledge, the amount of time and other resources you have available, as well as your own philosophical underpinnings. Finally, it must be remembered that these strategies should not be thought of as being mutually exclusive. For example, it is quite possible to use the survey strategy as part of a case study.

The strategies that we consider subsequently in this section are:

- | | |
|-------------------|----------------------|
| ○ experiment | ○ grounded theory |
| ○ survey | ○ ethnography |
| ○ case study | ○ Phenomenon logy |
| ○ action research | ○ Archival research. |

3.3.1 Experimental research Strategy

Definition

Experimentation is a research method that allows evaluation of causal relationship among variables. The purpose of experimental research is to allow the researcher to control the research situation so that causal relationships among variables may be evaluated.

Experiments differ from other research methods in terms of degree of control over the research situation in a typical experiment.

One variable (the independent variable) is manipulated and its effect on another variable (the dependant variable) is measured while all other variables that may confound such a r/ship are eliminated or controlled. The experiments either creates an artificial situation deliberately manipulate situation.

Examples

Dependant variable		Independent variables	
Training	Vs.	Performance	
Training program, Training style and Lecturers			
Productivity	Vs.	Change work situations	

Basic Issues in Experimental Design

1. Manipulation of the independent variable (s)
2. Experimental treatments and groups
3. Selection and measurement of the dependant variable
4. Selection and assignment of test units
5. Control over extraneous variables

1. Manipulation of the independent variable (s)

Independent variables: The variable that can be manipulated to be whatsoever the experiments wishes. Its value may be changed or altered independently of any other variable.

2. Experimental treatments and groups

Experimental treatments: An alternative manipulation of the independent variable being investigated

Experimental group: The group of subjects exposed to an experimental treatment

Control Group: A group of subjects who are exposed to the control condition in an experiment-that is they are subjects not exposed to the experimental treatment. By holding

conditions constant in the control group, the researcher controls for potential sources of error in the experiment.

3. Selection and measurement of the dependant variable

- The dependant variable is so named because its value is expected to be dependent on the experimenter's manipulation of the independent variable.
- The dependant variable is the criterion or standard by w/h the results are judged. It is assumed that changes in the dependant variable are a consequence of changes in the independent variable.

Examples

- Introducing a new pink-grape fruit tea mix
Dependent variable- sales volume
- Evaluating d/nt forms of training programs
Dependant variable- turnover, absenteeism, morale
- A company to introduce a new product
Dependent variables- brand awareness, trial purchase, repeat purchase

4. Selection and assignment of test units

- Test units are the subjects or entities whose responses to the experimental treatment are measured or observed. Individuals, organizational units, sales territories or other entities may be the test units.

Randomization: - a procedure in which the assignment of subjects and treatments of groups is based on chance.

Randomization assures the researcher

- The repetition of the experiment under the same conditions
- To assume that the groups are identical with respect to all variables except for the experimental treatment.

Random assignment of subjects to the various experimental groups is the most common technique used to prevent test units from differing each other on key variables; it assumes that all characteristics of these subjects have been similarly randomized.

Matching

If the experimenter believes that certain extraneous variables may affect the dependant variable, he/she can make sure that the subjects in each group are matched on these CXS.

Matching: - Is a procedure for the assignment of subjects to groups, it ensures each group of respondents is matched on the basis of pertinent CXS. Although matching assures that the

subjects in each group are similar on the matched characteristics, the researcher can never be sure that subjects have been matched on all of the characteristics that could be important to the experiment.

5. Control over extraneous variables

A number of extraneous variables may affect the dependant variables, these by distorting the experiment. Because an experimenter does not want extraneous variables to affect the results of an experiment, such variables must be controlled or eliminated.

Demand CXS

Refers to experimental design procedures that unintentionally give hints to subjects about the experimenter's hypothesis

Experimenter bias- An effect on an experimenter's results caused by the experimenter's presence, actions or comments

Guinea Pig effect: an effect on the results of the experiment caused by subjects changing their normal behavior or attitudes in order to cooperate with an experimenter.

Control Mechanisms

Constancy of conditions: Procedure in which subjects in experimental groups are exposed to situations identical except for different conditions of the independent variable.

Counter balancing: A technique to reduce error caused by order of presentation by varying the order of experimental treatments for different groups.

Blinding: A technique used to control subjects' knowledge of whether or not they have given a particular experimental treatment.

Double blind: A technique in which neither the subjects nor the experimenter knows which are the experimental and w/h are the controlled conditions.

Random assignment- an attempt to control extraneous variables caused by chance.

Issues of Experimental Validity

An experiment's quality is judged by two types of validity. These are known as internal and external validity.

Internal Validity

Internal validity exists to the extent that an experimental variable is truly responsible for any variance in the dependent variable. In other words, does the experimental manipulation truly cause changes in the specific outcome of interest? If the observed results were influenced or confounded by extraneous factors, the researcher will have problems making valid

conclusions about the relationship between the experimental treatment and the dependent variable. Thus, a lab experiment enhances internal validity because it maximizes control of outside forces.

Example

If we wish to know whether certain music causes increased productivity among workers, we may set up a task in a room with different music piped in (our experimental manipulation), but with the temperature, lighting, density, other sounds, and any other factors all controlled, which would be difficult or impossible to control outside of a lab environment. If the only thing that varies from subject to subject is the music, then we can safely say that any differences in performance must be attributable to human reactions to the music.

Extraneous variables can jeopardize internal validity. The six major ones are *history*, *maturation*, *testing*, *instrumentation*, *selection*, and *mortality*.

1. HISTORY

A **history effect** occurs when some change other than the experimental treatment occurs during the course of an experiment that affects the dependent variable. A common history effect occurs when competitors change their marketing strategies during a test marketing experiment. History effects are particularly prevalent in repeated measures experiments that take place over an extended time.

Example

If we wanted to assess how much a change in recipe improves individual subjects' consumption of a food product, we would first measure their consumption and then compare it with consumption after the change. Since several weeks may pass between the first and second measurement, there are many things that could occur that would also influence subjects' diets.

2. MATURATION

Maturation effects are effects that are a function of time and the naturally occurring events that coincide with growth and experience. Experiments taking place over longer time spans may see lower internal validity as subjects simply grow older or more experienced.

Example

Suppose an experiment were designed to test the impact of a new compensation program on sales productivity. If this program were tested over a year's time, some of the salespeople probably would mature as a result of more selling experience and gain increased knowledge

and skill. Their sales productivity might improve because of their knowledge and experience rather than the compensation program.

3. TESTING

Testing effects are also called *pretesting effects* because the initial measurement or test alerts or primes subjects in a way that affects their response to the experimental treatments. Testing effects only occur in a before-and-after study. A before-and-after study is one requiring an initial baseline measure be taken before an experimental treatment is administered. So, before-and-after experiments is a special case of a repeated measures design.

Example

Students taking standardized achievement and intelligence tests for the second time usually do better than those taking the tests for the first time. The effect of testing may increase awareness of socially appropriate answers, increase attention to experimental conditions (that is, the subject may watch more closely), or make the subject more conscious than usual of the dimensions of a problem.

4. INSTRUMENTATION

A change in the wording of questions, a change in interviewers, or a change in other procedures used to measure the dependent variable causes an **instrumentation effect**, which may jeopardize internal validity. Sometimes instrumentation effects are difficult to control.

Example

If the same interviewers are used to ask questions for both before and after measurement, some problems may arise. With practice, interviewers may acquire increased skill in interviewing, or they may become bored and decide to reword the questionnaire in their own terms. To avoid this problem, new interviewers could be hired. But this introduces another set of issues as different individuals are also a source of extraneous variation.

There are numerous other sources of instrument decay or variation. Again, instrumentation effects are problematic with any type of repeated measures design.

5. SELECTION

The **selection effect** is a sample bias that results from differential selection of respondents for the comparison groups, or sample selection error, discussed earlier. Researchers must make sure the characteristics of the research subjects accurately reflect the population of relevance. Furthermore, the key characteristics of the subjects must be distributed in such a way to create equal groups. That is, the subjects in the experimental and control groups, or in different experimental cells, must be equal across all variables of interest and those that could affect the dependent measure.

6. MORTALITY

If an experiment is conducted over a period of a few weeks or more, some sample bias may occur due to the **mortality effect (sample attrition)**. Sample attrition occurs when some subjects withdraw from the experiment before it is completed. Mortality effects may occur if subjects drop from one experimental treatment group disproportionately than from other groups.

Example

Consider a sales training experiment investigating the effects of close supervision of salespeople (high pressure) versus low supervision (low pressure). The high-pressure condition may misleadingly appear superior if those subjects who completed the experiment did very well. If, however, the high-pressure condition caused more subjects to drop out than the other conditions, this apparent superiority may be due to the fact that only very determined and/or talented salespeople stuck with the program.

External Validity

External validity is the accuracy with which experimental results can be generalized beyond the experimental subjects. External validity is increased when the subjects comprising the sample truly represent the population of interest and when the results extend to other market segments or groups of people. The higher the external validity, the more researchers and managers can count on the fact that any results observed in an experiment will also be seen in the “real world” (financial market, workplace, sales floor, and so on).

Trade-Offs Between Internal and External Validity

Naturalistic field experiments tend to have greater external validity than artificial laboratory experiments. Researchers often must trade internal validity for external validity. A researcher who wishes to test advertising effectiveness by manipulating treatments via a split-cable experiment has the assurance that the advertisement will be viewed in an externally valid situation, the subjects' homes. However, the researcher has no assurance that some interruption (for example, the telephone ringing, a child calling, or a pot boiling over on the stove) will not have some influence that will reduce the internal validity of the experiment. Laboratory experiments with many controlled factors usually are high in internal validity, while field experiments generally have less internal validity but greater external validity. Typically, it is best to establish internal validity first, and then focus on external validity. Thus, results from lab experiments would be followed up with some type of field test.

Classification of experimental Design

Experimental designs fall into two groups: basic design and complex experimental design. If only one variable is manipulated, the experiment has a basic experimental design. If the

experimenter wishes to investigate several levels of independent variable (e.g. four salary levels) or to investigate the interaction effects of two or more independent variables, then the experiment requires a complex experimental design.

Symbols to be used:

X= Exposure of a group to an experimental treatment

O= Observation or measurement of the dependant variable

R= Random assignment of test unit

A) Basic experimental designs

Pretest-post test control group design (Before-and-after with control group)

Experimental group: R O₁ x O₂

Control Group: R O₃ O₄

The subjects in the experimental group are tested before and after being exposed to the treatment. The control group is tested twice, at the same times as the experimental, group but these subjects are not exposed to the experimental treatment. The effect of the experimental treatment is calculated as follows:

$$E = (O_2 - O_1) - (O_4 - O_3)$$

Example Brand awareness

Before advertisement

after advertisement

Exper. O₁ = 0.20

O₂ = 0.35

Control O₃ = 0.20

O₄ = 0.22

$$\begin{aligned} \text{Treatment effect} &= (0.35 - 0.20) - (0.22 - 0.20) \\ &= 0.13 = \underline{13\%} \end{aligned}$$

Post test only group design (After- only with control group)

In some situations pretests are impossible or selection error is not expected.

Experimental group: R x O₁

Control group: R O₂

The experimental effect is measured by the difference b/n O₂ and O₁

$$E = (O_2 - O_1)$$

Example: - the manufacturer of an athlete's foot remedy better than a competing brand.

Post

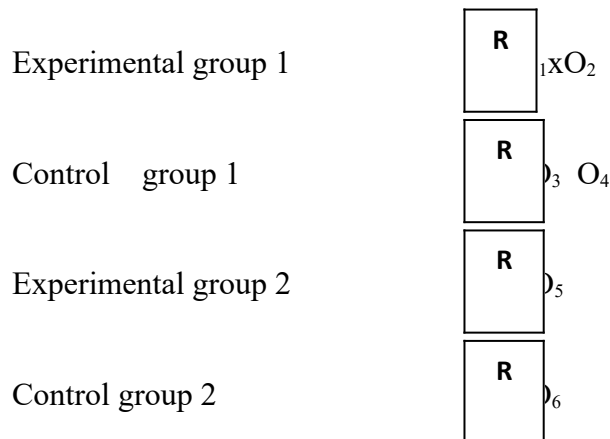
Experimental group: O₁ = 80%

Control group: O₂ = 70%

$$E = O_1 - O_2 = 10\%$$

Solomon four group design

Combining the before-after with control group design and the after only with control group design, the Solomon four group design provides a means for controlling the effects as well as other sources of extraneous variation.



$$E = (O_2 - O_1) - (O_3 - O_4) + (O_6 - O_5)$$

B) Complex Experimental Design

A complex experimental design is a statistical design that allows for the isolation of the effects of confounding extraneous variables or allows for manipulation of more than one independent variable in the experiment.

Completely randomized design (CRD)

It is an experimental design that uses a random process to assign experimental units to treatments. Randomization of experimental units is the researcher's attempt to control all extraneous variables while manipulating a single independent variable, treatment variable.

Example

Consider a financial institution that wants to increase their response to credit card offers. An experiment is constructed to examine the effects of various incentives on the percentage of potential customers that apply for a credit card with the institution. Thus, the experimental variable is the incentive. This can be manipulated over three treatment levels:

1. No incentive to the control group
2. No interest for the first 90 days with an approved application
3. A free MP3 player with an approved application

The financial institution rents a mailing list of 15,000 prospects. This sample frame is divided into three groups of 5,000 each ($n_1 + n_2 + n_3 = 15,000$). A random number process could be

used to assign subjects to one of the three groups. Suppose each of the 15,000 subjects is assigned a number ranging from 1 to 15,000. If a random number is selected between 1 and 15,000 (i.e., 1,201), that person can be assigned to the first group, with every third person afterward and before also assigned to the first group (1,204, 1,207, 1,210 . . . all the way back to 1,198). The process can be repeated with the remaining 10,000 subjects by selecting a random number between 1 and 10,000 and then selecting every other subject. At this point, only 5,000 subjects remain and will comprise the third group. All 15,000 subjects are now assigned to one of three groups. Each group corresponds to one of the three levels of incentive. A variable representing which group a subject belongs to becomes the independent variable. The dependent variable is measured for each of the three treatment groups and the number of respondents to the offer is determined. The analysis would compare differences across the number of respondents for each of the three treatment levels.

Randomized block design (RBD)

It is an extension of the CRD. A form of randomization is utilized to control for most extraneous variable. In RBD, the researcher has identified a single extraneous variable that might affect test units response to the treatment. The researcher will attempt to isolate the effects of this single variable by blocking its effects.

Example: Suppose that a manufacturer of Mexican food is considering two packaging alternatives. Marketers suspect that geographic region might confound the experiment. They have identified three regions where attitudes toward Mexican food may differ (the Southwest, the Midwest, and the Atlantic Coast). In a randomized-block design, each block must receive every treatment level. Assigning treatments to each block is a random process. In this example the two treatments will be randomly assigned to different cities within each region.

Percentage who purchase the product				
Treatment	Mountain	North East	South West	Mean for treatments
Package A	14% (Phoenix)	12%(St. Louis)	7%(Boston)	11%
Package B	16 %(Albuquerque)	15%(Peoria)	10% (New York)	13.6%
Mean for cites	19 %	13.5%	8.5 %	

Factorial design

This design is used in experiment where the effect of varying more than one factor is to be determined.

Main effect: - the influence on the dependent variable by each independent variable

Interaction effect: - the influence on a dependant variable by combinations of two or more independent variables.

Example

Suppose a researcher is comparing two magazine ads. The researcher is investigating the believability of ads on a scale from 0 to 100 and wishes to consider the sex of the reader as a blocking factor. The experiment has two independent variables: sex and ads.

This 2×2 factorial experiment permits the experimenter to test three hypotheses. Two hypotheses examine the main effects:

- Advertisement A is more believable than ad B.
- Men believe advertisements more than women.

However, the primary research question may deal with the interaction hypothesis:

- Advertisement A is more believable than ad B among women, but ad B is more believable than ad A among men.

	Ad A	Ad B	
Men	60	70	65
Women	80	50	65
	70	60	

Main effect of Ads

Analyses

A high score indicates a more believable ad. The table shows that the mean believability score for both sexes is 65. This suggests that there is no main sex effect. Men and women evaluate believability of the advertisements equally. The main effect for ads indicates that ad A is more believable than ad B (70 versus 60), supporting the first hypothesis. However, if we inspect the data and look within the levels of the factors, we find that men find ad B more believable and women find ad A more believable. This is an interaction effect because the believability score

of the advertising factor differs at different values of the other independent variable, sex. Thus, the interaction hypothesis is supported

Latin square Design

The Latin square design attempts to control or block out the effect of two or more confounding extraneous factors. This design is so named because of the layout of the table that represents the design. A Latin square is a balanced, two classification scheme.

Example: A taste test might be confounded by the order of tasting and individual preference.

		Order of usage		
		1	2	3
Subject 1		A	B	C
	2	B	C	A
	3	C	A	B

The capital letters A, B & C identify the three treatments

3.3.2 Survey Research Strategy

Survey studies ask large numbers of people questions about their behaviours, attitudes, and opinions. Some surveys merely describe what people say they think and do. Other survey studies attempt to find relationships between the characteristics of the respondents and their reported behaviours and opinions. For example, a survey could examine whether there is a relationship between gender and people's attitudes about some social issue. When surveys are conducted to determine relationships, as for this second purpose, they are referred to as *correlation studies*.

- Survey is defined as a method of gathering primary data based on communication, either verbal or written means, with a representative sample of individual.
- Survey investigations attempt to describe what is happening or to learn the reasons for a particular business activity. Most survey research is therefore descriptive research.

The principal advantage of survey studies is that they provide information on large groups of people, with very little effort, and in a cost effective manner. Surveys allow researchers to assess a wider variety of behaviours and other phenomena than can be studied in a typical naturalistic observation study.

Errors in survey research

The two major sources of survey error are random sampling error and systematic error.

Random sampling error

Most surveys try to portray a representative cross section of a particular target population. Even with technically proper random probability samples, statistical errors will occur because of chance variation in the elements selected for the sample. Unless the sample size is increased, these statistical problems are unavoidable.

Systematic error /non sampling errors

Results from some imperfect aspect of the research design or from a mistake in the execution of the research .Includes all sources of errors other than introduced by the random sampling procedure

The many sources of error that in some way systematically influence answers can be classified under two general categories: respondent error and administrative error.

A. Respondent Error

Surveys are based on asking people for answers. If respondents cooperate and give truthful answers, a survey will likely accomplish its goal. Two problems that arise if these two conditions are not met are non response error and response bias.

- a) **Non-response error:** The statistical difference between a survey that includes only those who responded and a perfect survey that would also include those who failed to respond.

To utilize results a researcher must be sure that those who did respond to the questionnaire were representative of those who did not.

This problem is especially acute in mail and internet surveys but is also exists in telephone and face-to-face interviews.

Non respondents: People who are not contacted or who refuse to cooperate

No contact: A potential respondent who is not at home or who is otherwise inaccessible on the first and second attempts to contact

Refusals: Occurs when people are unwilling to participate in the research

- b) Response Bias:** A response bias occurs when respondents tend to answer in a certain direction, i.e., when they consciously or unconsciously misrepresent the truth.

There are five specific categories of response bias

Acquiescence bias: A category of response bias in w/c individuals have a tendency to agree with all questions or to indicate of positive connotation to a new idea. They are “yea sayers”. It is particularly prominent in research on new products, new programs or ideas previously unfamiliar to the respondents. Disagree with all questions or to indicate a negative connotation to a new idea.

Extremity Bias a tendency to use extremes when responding to question others; others avoid extreme positions and tend to respond neutrally.

Interviewer Bias: Bias in the responses of subjects due to the influence of the interviewer Dressing, age, gender, tone of voice, facial expressions

Auspices Bias: Bias in responses subjects caused by their being influenced by the organization conducting the study

Social desirability Bias: Bias in the responses of subjects caused by their desire, either conscious or unconscious to gain prestige or to appear in a different role. **Example** Recreation

- Visiting tourist sites and attractions
- Education is overstated
- Charity contributions
- “respectable” answers given in order to gain prestige
- Attempt to create a favorable impression or “save face”

B. Administrative Errors

Result of improper administration or execution of the research task

Such errors can be caused by carelessness, confusion, neglect, omission or some other blunder. Four types of administrative errors are data processing error, sample selection error, interviewer error and interviewer cheating.

Data Processing error: A category of administrative error that occurs because of in correct data entry, incorrect computer programming and other errors during data analysis. data processing errors can be minimized by establishing careful procedure for verifying each step in the data processing stage.

Sample selection error: It is a systematic error that because of an error in either the sample design or execution of the sampling procedure results in an unrepresentative sample.

Interviewer error Interviewer error is introduced when interviewers record answers and check the wrong response. Interviewers are not able to write fast enough to record answers verbalism. Interviewers misreport answers

Interviewer cheating: Occurs when an interviewer falsifies entire questionnaire or fills in the answers to certain questions.

Classifying survey research Methods

Surveys may be classified according to the method of communication, the degree of structure and amount of disguise in a questionnaire and the time frame in w/c the data are collected.

Classification by the method of communication

- **Questionnaire**
 - Self administered questionnaire
 - Mail questionnaire
 - Electronic questionnaire
- **Interview**
 - Personal interview
 - Telephone interview
 - Interview schedule
 - In-depth interview
 - Focused group discussions

Structured and Disguised questions

In designing a questionnaire (or an interview schedule), the researcher must decide how much structure or standardization is needed.

A structured question- limits the number of responses available

Unstructured questions- do not restrict the answers the respond is allowed. Example:- what do you think are the two most important problems facing your organization?

Disguised questions: are indirect type of question that assumes that the purpose of the study must be hidden from the respondents.

Undisguised question -Direct question

Questionnaires can be categorized by their degree of structure and degree of disguise.

	Disguise	Un disguise
--	-----------------	--------------------

Structure	Structured disguise	Structure un disguised
Unstructured	Unstructured disguise	Structured un disguised

Limitations of this classification

- Degree of structure and degree of disguise vary; they are not clear cut strategies
- Most surveys are by brids , asking both structured and unstructured questions

Recognition of the degree of structure and disguise to meet survey objectives will help in the selection of the appropriate communication medium for conducting the survey.

3.3.3. Case Research Strategy

Definition and Importance

Case studies simply refer to the documented history of a particular person, group, organization, or event. Typically, a case study may describe the events of a specific company as it faces an important decision or situation, such as introducing a new product or dealing with some management crisis. Textbook cases typify this kind of case study. Clinical interviews of managers, employees, or customers can represent a case study.

Robson (2002:178) defines **case study** as ‘a strategy for doing research which involves an empirical investigation of a particular contemporary phenomenon within its real life context using multiple sources of evidence’. Yin (2003) also highlights the importance of context, adding that, within a case study, the boundaries between the phenomenon being studied and the context within which it is being studied are not clearly evident. This is the complete opposite of the experimental strategy we outlined earlier, where the research is undertaken within a highly controlled context. It also differs from the survey strategy where, although the research is undertaken in context, the ability to explore and understand this context is limited by the number of variables for which data can be collected.

The case study strategy will be of particular interest to you if you wish to gain a rich understanding of the context of the research and the processes being enacted (Morris and Wood 1991). The case study strategy also has considerable ability to generate answers to the question ‘why?’ as well as the ‘what?’ and ‘how?’ questions, although ‘what?’ and ‘how?’ questions tend to be more the concern of the survey strategy. For this reason the case study strategy is most often used in explanatory and exploratory research. The data collection techniques employed may be various and are likely to be used in combination.

They may include, for example, interviews, observation, and documentary analysis and (as if to emphasise the dangers of constructing neat boxes in which to categorise approaches,

Strategies and techniques) questionnaires. Consequently, if you are using a case study strategy you are likely to need to use and triangulate multiple sources of data.

Triangulation refers to the use of different data collection techniques within one study in order to ensure that the data are telling you what you think they are telling you. For example, qualitative data collected using semi-structured group interviews may be a valuable way of triangulating quantitative data collected by other means such as a questionnaire.

According to Kazdin (1982), the major characteristics of case studies are the following:

- They involve the intensive study of an individual, family, group, institution, or other level that can be conceived of as a single unit.
- The information is highly detailed, comprehensive, and typically reported in narrative form as opposed to the quantified scores on a dependent measure.
- They attempt to convey the nuances of the case, including specific contexts, extraneous influences, and special idiosyncratic details.
- The information they examine may be retrospective or archival.

Types of case studies

Yin (2003) distinguishes between four case study strategies based upon two discrete dimensions:

- single case v. multiple case;
 - Holistic case v. embedded case.
1. **A single case** is often used where it represents a critical case or, alternatively, an extreme or unique case. Conversely, a single case may be selected because it is typical or because it provides you with an opportunity to observe and analyse a phenomenon that few have considered before. Inevitably, an important aspect of using a single case is defining the actual case. Where you choose to use a single case study, you will need to have a strong justification for this choice.
 2. A case study strategy can also incorporate **multiple cases**, that is, more than one case. The rationale for using multiple cases focuses upon the need to establish whether the findings of the first case occur in other cases and, as a consequence, the need to generalise from these findings. For this reason Yin (2003) argues that multiple case studies may be preferable to a single case study and that.
 3. **Holistic case study**: you may well have chosen to use an organisation by which you have been employed or are currently employed as your case. If your research is concerned only with the organisation as a whole then you are treating the organisation as a holistic case study.

4. ***Embedded case:*** when you are researching and are concerned with a single organisation as a whole, if you wish to examine also a number of logical sub-units within the organisation, perhaps departments or work groups, then your case will inevitably involve more than one unit of analysis. Whatever way you select these units, this would be called an embedded case study

You may be suspicious of using a case study strategy because of the ‘unscientific’ feel it has. We would argue that a case study strategy can be a very worthwhile way of exploring existing theory. In addition, a well-constructed case study strategy can enable you to challenge an existing theory and also provide a source of new research questions.

A primary advantage of the case study is that an entire organization or entity can be investigated in depth with meticulous attention to detail. This highly focused attention enables the researcher to carefully study the order of events as they occur or to concentrate on identifying the relationships among functions, individuals, or entities. Conducting a case study often requires the cooperation of the party whose history is being studied. This freedom to search for whatever data an investigator deems important makes the success of any case study highly dependent on the alertness, creativity, intelligence, and motivation of the individual performing the case analysis.

3.3.4 Action research Strategy

Lewin first used the term **action research** in 1946. It has been interpreted subsequently by management researchers in a variety of ways, but there are four common themes within the literature.

1. The first focuses upon and emphasises ***the purpose of the research:*** research in action rather than research about action (Coghlan and Brannick 2005) so that, for example, the research is concerned with the resolution of organisational issues such as the implications of change together with those who experience the issues directly.
2. The second relates to ***the involvement of practitioners*** in the research and, in particular, a collaborative democratic partnership between practitioners and researchers, be they academics, other practitioners or internal or external consultants. Eden and Huxham (1996:75) argue that the findings of action research result from ‘involvement with members of an organization over a matter which is of genuine concern to them’. Therefore, the researcher is part of the organisation within which the research and the change process are

taking place (Coghlan and Brannick 2005) rather than more typical research or consultancy where, for example, employees are subjects or objects of study.

3. The third theme emphasises *the iterative nature of the process of* diagnosing, planning, taking action and evaluating. The action research spiral commences within a specific context and with a clear purpose. This is likely to be expressed as an objective (Robson 2002). Diagnosis sometimes referred to as fact finding and analysis, is undertaken to enable action planning and a decision about the actions to be taken. These are then taken and the actions evaluated (cycle 1). Subsequent cycles involve further diagnosis, taking into account previous evaluations, planning further actions, taking these actions and evaluating.
4. The final theme suggests that action research should have *implications beyond the immediate project*; in other words, it must be clear that the results could inform other contexts.

3.3.5. Grounded Theory

Grounded theory represents an inductive investigation in which the researcher poses questions about information provided by respondents or taken from historical records. The researcher asks the questions to him or herself and repeatedly questions the responses to derive deeper explanations. Grounded theory is particularly applicable in highly dynamic situations involving rapid and significant change. Two key questions asked by the grounded theory researcher are “What is happening here?” and “How is it different?” The distinguishing characteristic of grounded theory is that it does not begin with a theory but instead extracts one from whatever emerges from an area of inquiry.

A grounded theory strategy is particularly helpful for research to predict and explain behaviour, the emphasis being upon developing and building theory. As much of business and management is about people’s behaviours, for example consumers’ or employees’, a grounded theory strategy can be used to explore a wide range of business and management issues.

Example: Consider a company that approaches a researcher to study whether or not its sales force is as effective as it has been over the past five years. The researcher uses grounded theory to discover a potential explanation. A theory is inductively developed based on text analysis of dozens of sales meetings that had been recorded over the previous five years. By questioning the events discussed in the sales interviews and analyzing differences in the situations that may have led to the discussion, the researcher is able to develop a theory. The theory suggests that with an increasing reliance on e-mail and other technological devices for

communication, the salespeople do not communicate with each other informally as much as they did five years previously. As a result, the salespeople had failed to bond into a close-knit “community.”

Misconceptions of grounded theory (suddy, 2006)

1. Grounded theory is not an excuse to ignore the literature, or defer reading existing theory until the data are collected and analysed.
2. Grounded theory is not presentation of raw data. It is essential that the data collected are considered at a conceptual level in order to draw conclusions which contain theoretical insights.
3. Grounded theory is not theory testing, content analysis, or word counts.
4. Grounded theory is not simply routine application of formulaic procedures to data. It is not a mechanical involving techniques and procedures such as prescribed amounts of interviews, computer software packages to analyse data or elaborate data sorting into categories.
5. Grounded theory is not perfect. By its nature it is ‘messy’. It requires researchers to develop a tacit knowledge of, or feel for, their data.
6. Finally, he cautions against assuming that grounded theory is easy. He argues that ‘the seamless craft of a well-executed grounded theory study, however, is the product of considerable experience, hard work, creativity and, occasionally, a healthy dose of good luck

3.3.6. Ethnography Research Strategy

Ethnography represents ways of studying cultures through methods that involve becoming highly active within that culture. The purpose is to describe and explain the social world the research subjects inhabit in the way in which they would describe and explain it. This is obviously a research strategy that is very time consuming and takes place over an extended time period as the researcher needs to immerse her or himself in the social world being researched as completely as possible.

In adopting an ethnographic strategy, you will be researching the phenomenon within the context in which it occurs and, in addition, not using data collection techniques that oversimplify the complexities of everyday life. Given this, it is not surprising that most ethnographic strategies involve extended participant observation

Ethnographic study can be particularly useful when a certain culture is comprised of individuals who cannot or will not verbalize their thoughts and feelings. For instance, ethnography has advantages for discovering insights among children since it does not rely largely on their answers to questions. Instead, the researcher can simply become part of the environment, allow the children to do what they do naturally, and record their behaviour.

3.3.7 Phenomenology Research Strategy

Phenomenology represents a philosophical approach to studying human experiences based on the idea that human experience itself is inherently subjective and determined by the context in which people live. The phenomenological researcher focuses on how a person's behaviour is shaped by the relationship he or she has with the physical environment, objects, people, and situations. Phenomenological inquiry seeks to describe, reflect upon, and interpret experiences.

Researchers with a phenomenological orientation rely largely on conversational interview tools. When conversational interviews are face to face, they are recorded either with video or audiotape and then interpreted by the researcher. The phenomenological interviewer is careful to avoid asking direct questions when at all possible. Instead, the research respondent is asked to tell a story about some experience. In addition, the researcher must do everything possible to make sure a respondent is comfortable telling his or her story. One way to accomplish this is to become a member of the group. Another way may be to avoid having the person use his or her real name. This might be particularly necessary in studying potentially sensitive topics such as smoking, drug usage, shoplifting, or employee theft.

3.3.8 Archival research

The final strategy we wish to consider, **archival research**, makes use of administrative records and documents as the principal source of data. Although the term archival has historical connotations, it can refer to recent as well as historical documents.

An archival research strategy allows research questions which focus upon the past and changes over time to be answered, be they exploratory, descriptive or explanatory.

However, your ability to answer such questions will inevitably be constrained by the nature of the administrative records and documents.

3.4. Type of Investigation

A manager (a researcher) should determine whether a causal or a correlation study is needed to find an answer to the issue at hand. While determining this, you are also determining the

extent of your interference in the study and the study setting. As usual, whether a study is a causal or a correlation one depends on the type of research questions asked and how the problem is defined.

1. Casual Research

- necessary to establish a definite cause-and-effect relationship
- Researcher is keen on delineating one or more factors that are undoubtedly causing the problem. In other words, the intention of the researcher conducting a casual study is to be able to state that variable X causes variable Y. So, when variable X is removed or altered in some way, problem Y is solved.
- The researcher tries to **manipulate** certain variables so as to study the effects such manipulations on the dependant variable of interest. In other words, the researcher deliberately changes certain variables in the setting and interferes with the events as they normally occur.
- They can be carried out in two studies settings- in the natural environment where work proceeds normally (non contrived) and or in artificial environment (contrived). Most rigorous causal studies, however, are done in artificial, contrived settings.
- Studies conducted to establish cause-and-effect relationship using the same natural environment in which employees normally function are called **field experiments**. Here, the researcher interferes with the natural occurrence events in as much as the independent variable is manipulated.
- Experiments done to establish cause- and-effect relationships beyond the possibility the least doubt require the creation of an artificial, contrived environment in which all the extraneous factors are strictly controlled. Similar subjects are chosen carefully to respond to certain manipulated stimuli. These studies are referred to as **lab experiments**.

Example 1

A bank manager wants to determine the cause-and-effect relationship between interest rate and inducements it offers to clients to save and deposit money in the bank. She selects four branches within a 60-mile radius for the experiment. For 1 week only, she advertises the annual rate for a new certificate of deposits received during the week in the following manner: the interest rate would be 9% in one branch, 8% in another, and 10% in the third. In the fourth branch, the interest rate remains unchanged at 5%. Within the week, she would be able determine the effects, if any, of interest rates on deposit mobilization.

Points about this research design

- This is a field experiment with non contrived setting- all activities occur in the normal and natural work environment.
- The researcher has interfered in only manipulating (altering) the interest rates.
- More likely, all four branches chosen would be more or less compatible in size, number of depositors, and the like, so that the interest-savings relationships are not influenced by some third factor. But it is possible that some other factors might affect the findings.

Example 2

The bank manager now wants to establish the causal connection between interest rates and savings, beyond doubt. For this purpose, she recruits 40 students who are all business majors in their final years of study and are more or less of the same age. She splits them into four groups and gives each one of them chips that counts for \$ 1000, which they are told they might utilize to buy their needs or save for the future, or both. She offers them by way of incentive; interest on what they save but manipulates the rates by offering a 6% interest rate on savings for group 1, 8% for group 2, 9% for group 3 and keeps the interest at the low rate of 1% for group 4.

Points about this research design

- An artificial laboratory environment is created. Thus, a contrived setting is chosen.
- The researcher interference has been maximal, in as much as the setting is different, the independent variable is manipulated, and most external factors such as age and experience have been controlled.
- If the banker finds that the savings by the four groups increase progressively, keeping in step with the increasing rates of interest, she would be establish a cause-and-effect relationship b/n interest and the disposition to save.

2. Correlation Research

- A researcher wants to delineate the important variables associated with the problem.
- A correlation study is conducted in the natural environment of organization (non contrived setting) with minimal researcher interference.
- Correlation studies conducted in organizations are called *field studies*.

Example

The bank manager wants to analyze the relationship b/n interest rates and bank deposit patterns of clients. She tries to correlate the two by looking at deposits into different kinds of accounts as interest rates changed.

Points on this research design

- This a correlation field study
- Research is done in a non contrived setting with no interference with the normal work routine

3.5. Time Horizon

An important question to be asked in planning your research is ‘Do I want my research to be a “snapshot” taken at a particular time or do I want it to be more akin to a diary or a series of snapshots and be a representation of events over a given period?’ (As always, of course, the answer should be ‘It depends on the research question.’) The ‘snapshot’ time horizon is what we call here **cross-sectional** while the ‘diary’ perspective we call **longitudinal**.

We should emphasise here that these time horizons to research design are independent of which research strategy you are pursuing or your choice of method. So, for example, you may be studying the change in manufacturing processes in one company over a period of a year. This would be a longitudinal case study.

1. Cross-sectional studies

- the study of a particular phenomenon (or phenomena) at a particular time
- most research projects undertaken for academic courses
- Cross-sectional studies often employ the survey strategy
- Most field studies are cross sectional studies in nature

Example 1: data were collected from stock brokers between April and June of last year to study their concerns in a turbulent stock market. Data with respect to this particular research had not been collected before, nor will they be collected again from them for this research.

Example 2: a drug company desirous of investing in research for a new obesity (reduction) pill conducted a survey among obese people to see how many of them would be interested in trying the new bill.

The purpose of both of the studies in the two examples was to collect data that would be pertinent to find the answer to a research question. Data collection at one point in time was sufficient. Both were cross-sectional designs.

2. Longitudinal studies

- In longitudinal studies the basic question is ‘Has there been any change over a period of time?’
- the ‘diary’ perspective
- Take more and time effort and cost more than cross-sectional studies.

- Help to identify cause-and-effect relationships
- Experimental designs are invariably longitudinal studies since data are collected both before and after the manipulation.
- Field studies can be also longitudinal. For example, a study of the comparison data pertaining to the reactions of managers in a company toward working women now and 10 years later will be a longitudinal study.
- Necessary if a manager wants to keep track of certain factors over a period of time, to detect possible casual connections.

Example 1: UPS experienced a shut down for 15 days during the teamsters' walkout and their clients shifted their business to other carriers such as FedEx and the U.S. postal service. After a number of the strike, UPS tried to woo their customer back through several strategies and collected data month after month to see what progress they were making in this regard.

Here, data were collected every month to assess whether UPS had regained the business volume. Since data were collected at various points in time to answer the same research question (have we gained lost background?), the study is a longitudinal one.

Example 2: A marketing manager is interested in tracing the pattern of sales of a particular product in four different regions of the country on a quarterly basis for the next 2 years. Since data are collected several times to answer the same issues (tracing pattern of sales), the study falls under the longitudinal category.

3.6. Research Methods Choice

	QUALITATIVE	VERSUS	QUANTITATIVE RESEARCH
Purpose	To understand & interpret social interactions.		To test hypotheses, look at cause & effect, & make predictions.
Group Studied	Smaller & not randomly selected.		Larger & randomly selected.
Variables	Study of the whole, not variables.		Specific variables studied
Type of Data Collected	Words, images, or objects.		Numbers and statistics.
Form of Data Collected	Qualitative data such as open- ended responses, interviews, participant observations, field notes, & reflections.		Quantitative data based on precise measurements using structured & validated data-collection instruments.
Type of Data Analysis	Identify patterns, features, themes.		Identify statistical relationships.
Objectivity and Subjectivity	Subjectivity is expected.		Objectivity is critical.
Role of Researcher	Researcher & their biases may be known to participants in the study, & participant characteristics may be known to the		Researcher & their biases are not known to participants in the study, & participant

	researcher.	characteristics are deliberately hidden from the researcher (double blind studies).
Results	Particular or specialized findings that is less generalizable.	Generalizable findings that can be applied to other populations.
Scientific Method	Exploratory or bottom-up: the researcher generates a new hypothesis and theory from the data collected.	Confirmatory or top-down: the researcher tests the hypothesis and theory with the data.
View of Human Behavior	Dynamic, situational, social, & personal.	Regular & predictable.
Most Common Research Objectives	Explore, discover, & construct.	Describe, explain, & predict.
Focus	Wide-angle lens; examines the breadth & depth of phenomena.	Narrow-angle lens; tests specific hypotheses.
Nature of Observation	Study behaviour in a natural environment.	Study behaviour under controlled conditions; isolate causal effects.
Nature of Reality	Multiple realities; subjective.	Single reality; objective.
Final Report	Narrative report with contextual description & direct quotations from research participants.	Statistical report with correlations, comparisons of means, & statistical significance of findings.

- a) **Quantitative** is predominantly used as a synonym for any data collection technique (such as a questionnaire) or data analysis procedure (such as graphs or statistics) that generates or uses numerical data.
- b) **Qualitative** is used predominantly as a synonym for any data collection technique (such as an interview) or data analysis procedure (such as categorising data) that generates or use non-numerical data. Qualitative therefore can refer to data other than words, such as pictures and video clips.

In choosing your research methods you will use a single data collection technique and corresponding analysis procedures (**mono method**) or use more than one data collection technique and analysis procedures to answer your research question (**multiple methods**).

1. Mono Method

If you choose to use a mono method you will combine either a single quantitative data collection technique, such as questionnaires, with quantitative data analysis procedures; or a single qualitative data collection technique, such as in-depth interviews, with qualitative data analysis procedures.

2. Multiple Methods

If you choose to combine data collection techniques and procedures using some form of multiple methods design, there are four different possibilities.

- a) The term **multi-method** refers to those combinations where more than one data collection technique is used with associated analysis techniques, but this is restricted within either a

quantitative or qualitative world view. Thus you might choose to collect quantitative data using, for example, both questionnaires and structured observation analysing these data using statistical (quantitative) procedures, a **multi-method quantitative study**. Alternatively, you might choose to collect qualitative data using, for example, in-depth interviews and diary accounts and analyse these data using non-numerical (qualitative) procedures, a **multi-method qualitative**. Therefore, if you adopted multi-methods you would not mix quantitative and qualitative techniques and procedures.

- b) **Mixed methods approach** is the general term for when both quantitative and qualitative data collection techniques and analysis procedures are used in a research design. It is subdivided into two types. **Mixed method research** uses quantitative and qualitative data collection techniques and analysis procedures either at the same time (parallel) or one after the other (sequential) but does not combine them. This means that, although mixed method research uses both quantitative and qualitative world views at the research methods stage, quantitative data are analysed quantitatively and qualitative data are analysed qualitatively. In addition, often either quantitative or qualitative techniques or procedures predominate. In contrast, **mixed-model research** combines quantitative and qualitative data collection techniques and analysis procedures as well as combining quantitative and qualitative approaches at other phases of the research such as research question generation. This means that you may take quantitative data and **qualitise** it, which is, convert it into narrative that can be analysed qualitatively. Alternatively, you may **quantitise** your qualitative data, converting it into numerical codes so that it can be analysed statistically

Reasons for using mixed-methods designs

Reasons	Explanation
Triangulation	Use of two or more independent sources of data or data collection methods to corroborate research findings within a study.
Facilitation	Use of one data collection method or research strategy to aid research using another data collection method or research strategy within a study (e.g. qualitative/quantitative providing hypotheses, aiding measurement, quantitative/qualitative participant or case selection)
Complementarity	Use of two or more research strategies in order that different aspects of an investigation can be dovetailed (e.g. qualitative plus quantitative questionnaire to fill in gaps quantitative plus

Generality	qualitative questionnaire for issues, interview for meaning) Use of independent source of data to contextualise main study or use quantitative analysis to provide sense of relative importance (e.g. qualitative plus quantitative to set case in broader context; qualitative × quantitative analysis is to provide sense of relative importance)
Aid interpretation	Use of qualitative data to help explain relationships between quantitative variables (e.g quantitative/qualitative)
Study different aspects	Quantitative to look at macro aspects and qualitative to look at micro aspects
Solving a puzzle	Use of an alternative data collection method when the initial method reveals unexplainable results or insufficient data

Chapter 4

Research Proposal

Chapter outline

4.1. Functions of research proposal

4.2. Importance of research proposal

4.3. Elements of Research Proposal

Chapter Objectives

Up on the completion of the chapter, you will be able to

- Describe the functions and importance of research proposal
 - Identify the components of research proposal
 - Explain the components of research proposal
-

Introduction

The term "research proposal" is coined from two words, research and proposal. Research has been defined in first chapter. Proposal lexically means a course of, activity etc. proposed. Therefore, a research proposal can be defined as a written document requesting both authorization and funds to undertake a specific research project. It always includes a statement explaining the purpose of the study or a definition of the problem. It systematically outlines the particular research methodology and details the process that will be utilized at each stage of the research process

4.1 Functions of Research Proposal

Research proposal may function at least in three ways, namely, as a means of communication, as a plan and as a contract

1. ***Proposal as a means of communication*** serves to communicate the investigator's research plan to those who-give consultations and / or disburse fund. Proposal is the primary source on which the graduate students' **thesis or dissertation** committee must base the function of review, consultation and more importantly, approval for implementation of the research project. In general, the quality of assistance and the probability of financial support will all depend directly on the clarity and justification of the proposal.
2. ***Research proposal as plan*** helps the researcher organize his/her ideas in a systematic manner and to look for strengths and flaws. A successful proposal sets out the plan in step-by-step detail. That is, it provides an inventory of what must be done and which materials have to be collected as a preliminary step.

Generally, the acceptability of results is judged exclusively in terms of the adequacy of the methods employed in recording, analyzing and interpreting the data in the planned study.

3. ***Proposal also functions as a contract.*** A completed proposal, approved for execution and signed by all members of the sponsoring committee, constitute a bond of agreement between the researcher and that committee. An approved grant proposal results in a contract between the investigator (and often the university) and funding sources.

Proposal for thesis and dissertation should be in final form prior to the collection of data. Once document is approved in final form, neither the student nor the sponsoring faculty

members should be free to alter the fundamental terms of the contract by unilateral decision.

4.2 Importance of Research Proposal

The research proposal has the following importance.

- It serves as a basis for determining the feasibility of the research project.
- It provides a systematic plan of procedure for the researcher to follow.
- It gives the research supervisor a basis for guiding the researcher while conducting the study.
- It reduces the probability of costly mistake.

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1. *Title*

- The title should be as explicit as possible and transparent
- It should be clear and short

- It should capture and reflect the content of the proposal. It should enable the readers to understand the concepts of the study
- Try to get the title down to one line or two

2. Abstract:

- Is summary, which reflects the whole content of the proposal (most of the time less than 300 words?)
- Abstract should be concise, informative and should provide brief information about the whole problem to be investigated
- The abstract of a proposal should contain the following points
 - 📖 Title or topic of the research
 - 📖 Statement of the problem and objectives
 - 📖 Methodology of investigation
 - 📖 Expected result (tentative only if a researcher starts with a formulated hypothesis)

3. Background of the study

This section in research proposal should give the general background of the study and the need for undertaking the study. It reviews the previous work to reveal what is known of the subject.

Some questions to be addressed in this part may include:

- What are current and previous studies that have been made on the issues to be studied?
- What is the present gap in knowledge and current trend?
- What makes worth studying?
- Why has the problem not been solved yet?
- What do we intend to fill the gap or solve the problem?

If the researcher finds this section to be very vast he can add (split this section) a section (heading) called “rationale” and/or the problem statement.

4. The Rationale and/or Statement of the problem

The Rationale provides some of the reasons for doing the study and its importance. Citations from past research may be relevant here; likewise recent political and economic events, previous efforts to solve the problem that did not succeed can be mentioned. This section of the research proposal can be merged with the above section or in the objective part of the proposal. In some

cases where wider explanation and justification as to the rationale of the proposed study is needed, it can be shown in a separate section.

5. Objectives of the study

In this section, the proposal can indicate what is expected to occur and why, suggest the variables that determine a relationship, or provide an explanation for a phenomenon that has been observed, which strengthens the proposal and research agenda.

Generally this section should focus on the points

- What is to be achieved by the proposed investigation has to be stated plainly and concisely
- Should be to the point and logically arranged
- Should be stated in a form which shows the relations between variables
- Must clearly state the target of the research activity, i.e., what kind of results are expected or sought from conducting research.

6. Significance of the proposed study

By including this section, the writer creates a rationale for conducting the study and a statement why the results will be important. A significance section elaborates on the importance and implications of a study for researchers, practitioners, and policy makers. In designing this section, one might include

- Three or four ways in which the study adds to the scholarly research and literature in the field
- Three or four ways in which the study helps improve practice.
- Three or four reasons why the study will improve policy.

7. Delimitations and Limitations

Two more parameters for a research study establish the boundaries, exceptions, reservations, and qualifications inherent in every study: delimitations and limitations (Castetter & Heisler, 1977).

They are found in proposals for qualitative, quantitative, and mixed methods studies.

- Use delimitations to narrow the scope of a study. For example, the scope may focus on specific variables or a central phenomenon, delimited to specific participants or sites, or narrowed to one type of research design (e.g., ethnography or experimental research).
- Provide limitations to identify potential weaknesses of the study. At the proposal stage, it is often difficult to identify weaknesses in the study before it has begun. However,

advisers like students to anticipate the potential weaknesses in their studies, an student can identify limitations related to the research methods of data collection and analysis. For example, all statistical procedures have limitations; so also do research strategies, such as surveys or grounded theory studies. In introductory discussions about these strategies, authors typically mention both their strengths and their weaknesses (e.g., see Creswell, 2002)

In journal articles, researchers incorporate delimitations into the method or procedure sections, and they write limitations into the final section of their studies. In proposals, authors may include them in a separate section: they also may separate them into two subsections, one on delimitations and the other on limitations.

8. The definition of terms

Researchers define terms so that readers can understand their precise meaning. The definitions can be elaborated in a separate section of the research proposal

Terms to Define

Define terms that individuals outside the field of study may not understand and that go beyond common language (Locke et al., 2000). Clearly, whether a term should be defined is a matter of judgment. A rule of thumb is to define a term if there is likelihood that readers will not know its meaning. Also, define terms when they first appear, so that a reader does not read ahead in the proposal operating with one set of definitions only to find out later that the author is using a different set.

As Wilkinson (1991) commented, “scientists have sharply defined terms with which to think clearly about their research and to communicate their findings and ideas accurately” (p.22).

Defining terms also adds precision to a scientific study, as Firestone (1987) stated:

The words of everyday language are rich in multiple meanings. Like other symbols their power comes from the combination of meaning in a specific setting ... Scientific language ostensibly strips this multiplicity of meaning from words in the interest of precision. This is the reason common terms are given “technical meanings” for scientific purposes. (p.17)

In consideration of this need for precision, one finds terms stated early in the introduction to articles. In dissertations and thesis proposals, terms typically are defined in a special section of the study. The rationale is that in formal research, students must be precise in how they use

language and terms. The need to ground thoughts in authoritative definitions constitutes good science. Define terms as they are introduced in all sections of the research plan. Define terms not understood by readers as found in these sections:

- The title of the study
- The statement of the problem
- The research questions, hypotheses, or objectives
- The literature review
- The methods section

9. *Literature review*

Literature review means locating literature in a variety of sources, reading it carefully and thoroughly, and organizing it into themes along with the line of investigation.

Objectives of the literature review

- To establish a familiarity with a body of knowledge and establishes credibility: A review tells a reader that he knows the research in an area and knows the major issue.
- To show the path of prior research and how a current project is linked to it: A review outlines the direction of research on a question and shows the development of knowledge. A good review places a research project in a context and demonstrates its relevance by making connections to a body of knowledge.
- To integrate and summarize what is known in the area. A review pulls together and synthesizes different results. A good review points out areas where prior studies agree, where they disagree, and major questions remain. It collects what is known up to a point in time and indicates the direction for future research.
- To learn from others and stimulate new ideas. A review tells what others have found so that a researcher can benefit from the efforts of others. A good review identifies blind alleys and suggests hypotheses for replication. It divulges procedures, techniques, and research design worth copying so that a researcher can better focus hypotheses and gain new insights.

Types of reviewed literature can be

- Theoretical works and review
- Methodological review
- Original findings and reviews of them

It takes a researcher over year to complete an extensive professional summary review of all the literature on a broad question. Reading or review for research could take place in three stages of the research project. Theses include the following

1. At the beginning of the research: the objective here is
 - To check what other research has been done
 - To focus your idea, and
 - To explore the context for your project
2. During your research: with the objectives of
 - keeping your interest and up to date with development
 - helping you better understand the method you are using and the field you are researching and,
 - serving as a source of data
3. After your research, with the objectives of
 - seeing what impact you own work has had, and
 - helping you develop ideas for further research project

It is not necessary to have a separate heading for the review of literature in a proposal. In many respects it is more useful to show that such a review has been made by citing useful literature in problem statement (introductory part), objective, methodology and where ever this information has been helpful in developing the research proposal.

The researcher may decide to have a separate heading for review literature when he wants to develop theoretical frameworks reviewed from different sources, detailed explanation and extensive references of the research outcomes related to the proposed study.

10. Material and Methods (Methodology) or Data and Methodology

The methodologies are the core and the largest part of the scientific research proposal. The methodologies are largely descriptive. This section is an essential prerequisite for validating the results and hence acceptability of the proposal.

An accurate, clear and valid methodology is required so that peers can:

- ◆ replicate the experiments described in an experimental research
- ◆ modify them with the assurance that the modification is different from the original in particular way
- ◆ apply them under different conditions

Methodology should give full details to show how the research activity is going to be carried out. Since this section is the largest and the most important part of the research proposal, researchers are advised to split it in to sub-sections. Such effort makes the proposal transparent and clear to the readers as well as to the reviewing committee and therefore, minimizes the risk of being rejected by a research committee (examiners board).

The suggested sub-sections can be:

- Data (materials)
- Methodology (methods) used

Data (some authors prefer Materials)

In this subsection of the methodology, information required to attack the problem is explained. Each variable or indicator identified by a researcher is explained and justified to show how it is relevant to the current study. The source of the data, the site (location) of the study and time that will be included (relevant time period) should also be specified.

Methods (Methodology)

In this subsection the details of all technical matters will be explained, which includes,

Method of data collection: How the data are obtained from earlier specified sources? There are different methods to collecting primary data. All available methods are expected to be considered by a researcher and the best one is selected. The researcher explains and justifies how his/her selection of a particular method is relevant and appropriate to his/her study.

Method of sampling: From which elements will the data be collected? The researcher has to specify the elements from which the desired information is collected and how these elements are selected. There are different sampling techniques. The researcher is expected to consider all of them and select the one which is appropriate for the current study. Justification of his/her choice of a particular sampling technique is relevant here. The sample size should also be specified with its method of sample size determination.

Method of analysis: how data will be processed and analyzed should be explained in this part. Particularly if the researcher uses mathematical models like simple or multiple regression, time series or liner programming models, he/she has to show their mathematical derivations and relations briefly to demonstrate their relevancy for the current study.

Method of presenting the outcome: Sometimes it is worth to mention how the outcome of the research will be presented to consumers (general public). This is important if the research is not a

part of a particular study program (like, master or doctoral study).

11. Work plan

A timetable explaining how the researcher expects to carry out his/her project and when each of the important phases will be completed is helpful to both the researcher and the reviewer. It is a plan in terms of numbers of weeks or months and expected completion dates. Commonly researcher presents it in a table form.

The length of time required for postgraduate study (Doctoral study) is determined by the budget of the time presented by students. Thus, researcher should be realist in budgeting his time.

	Research Activity	Duration	Feb. W1 W2 W3 W4	Mar. W1 W2 W3 W4
1	Problem identification ✓ Statement of the problem ✓ Setting objectives ✓ Identifying variables ✓ Setting research questions ✓ Title confirmation			
2	Research design :specification of methods and procedures			
3	Developing review of literature			
4	writing research proposal			
5	Data collection instrument dev't			
6	Pre testing			
7	Distribution and collection of data			
8	Data Processing: Data editing and coding			
9	Data analysis & interpretation			
10	Writing the research report			
11	Submission & presentation			

12. Cost budget

Most proposals put together with the expectation that funding will be necessary and an itemized list of items needed to carry out the methodology is listed in some detail. A carefully developed budget reflects the seriousness of the proposal and the degree to which it is a realistic assessment of what is needed.

S/N	Items	Unit of measurement	Amount unit	Unit cost ETB	Total cost ETB	Remark
1	Stationary expense					
2	Advisor ship honorable payment					
3	Examiners					
4	Backup disk CD-RW (Sony)					
5	Flash disk					
6	Printing expense for download articles & journals					
7	Printing and typing the draft					
8	Travel expense					
9	Enumerators Allowance					
10	Researchers' allowance					
11	Miscellaneous expense					
12	Contingency 2%					
Total						

13. Reference (Bibliography)

All works cited in the proposal are listed either alphabetically or numerically at the end of the document usually under the heading of “**References or Bibliography** “

Some conservative authors would like to make distinction between references and bibliography as mentioned below.

- **References:** are lists of literature, which have been cited in the text of a document
- **Bibliography:** includes also items, which were not cited but are relevant to the document.

There are several ways of making citation and recording references. The two main systems of referring are:

A) Harvard System: This system uses author's name, date, and page reference. This system is used mainly by Anthropology, Social science and in life science research. Harvard system is commonly used in business research.

- The reference list must include all published works referred to in the text.
- Each reference to a periodical publication must include, in order, the name(s) of the author(s), the year of publication, the full title of the article, the publication in which it appears, and the volume and inclusive page number

e.g.

Porter, Michael E. (1994), '**competitive advantage of nations**', Harvard Business Review, 68(2), pp. 73-83

- Reference to a book must give the author or authors' name(s), the year, the title, the edition, if other than the first; the publisher, the city of publication, and the number of the volume (if two or more).

e.g.

Davidson, Alan. B. (1994), **the Pursuit of Business**, London, Chapman & Halls

- It is possible to cite quote or paraphrase unpublished work if they are relevant to the study.

Unpublished works are:

- 📌 Personal communication with experts (professionals).
- 📌 Unpublished data collected by other researchers
- 📌 Unpublished research works
- 📌 Public speeches
- 📌 Conferences discussion and the like

- The researcher should not put such an unpublished work in reference list unless it has been fully approved and a date set for its publication.

- Unpublished references are not usually found or catalogued in libraries. And such unpublished reference should be placed in the body of the text, not in the reference list. Use parenthesis to enclose the complete name of the author, the year (if available) and appropriate title, location, or other information needed to establish the authenticity of the reference or help the reader locate it.

Citation under the Harvard system:

In an investigation conducted in large factories, Gibbs (1993) reported that absenteeism was higher on the night shift than the day

Citing more than one author

A number of research studies have been conducted into the effect of the recession on small business (Smith, 1984; Anderson, 1990; Jones, 1995)

b) Vancouver System: uses in-text number instead of an author, date, and page reference. *The same superscript or bracketed text number is given in the text each time the source is cited.* It is used in the applied science such as chemistry, physics and mathematics. *Example:*

“In an investigation conducted in large factories, Gibbs¹ reported that absenteeism was higher on the night shift than the day”

Or

“In an investigation conducted in large factories, it has been reported that absenteeism on the night shift was measured and found to be significantly higher...Gibbs¹

14. Appendix (Annex)

It is not mandatory to have this section. If the researcher thinks that having this section will increase the quality of the research proposal he is free to do so.

Information types provided in this section are those additional details, which are difficult to accommodate within the standard headings

Stuffs to be included in appendix:

- Detail Questionnaires and interview schedules
- Detailed experimental design (in experimental research)

¹ Gibbs, Jennifer M. (1993) ‘Absenteeism in SME’s’, International Small Business Journal, 13 (1), pp. 13-26.

- Detailed statistical procedures
- Samples of research materials (chemical and some biological materials)
- Survey and other geographic maps

Chapter 5

Sample Design and Sampling Procedures

Chapter Outline

- 5.1. Terms and Definitions
- 5.2. The Need for Sampling ○
- 5.3. Overview of sampling Techniques ○
- 5.4 Probability Sampling ○
- 5.5. Non-probability sampling ○

5.1 **Terms and Definitions**

1. Population/Universe: Any complete group: People, Sales territories, Stores, units of production

Remark: In sampling, the term ‘population’ is not used in its normal sense, as the full set of cases need not necessarily be people. A finite group is called population whereas a non-finite (infinite) group is called universe

2. Elements/cases : Members of the population
3. Census: Investigation of all individual elements that make up a population
4. Sample: Subset/portion of a larger population
5. *Sampling*: the process of gathering data from some portion of the population.

Sampling enables researchers to make estimates/test hypothesis of some unknown characteristics of the population in question- Statistical inference

5.2. **Needs for sampling**

1. Budget and time Constraints (in case of large populations)
2. Physical impossibility of checking all elements
3. Destructive nature of some census
4. Adequacy of sample results: High degree of accuracy and reliability (if sample is representative of population)

5.3. Overview of sampling techniques

The sampling techniques available to you can be divided into two types:

- probability or representative sampling;
- non-probability or subjective sampling

This and the next sections discuss- the *advantages and disadvantages of, how and when* to use- the most frequently used sampling techniques

- For many research projects, you will need to use a variety of sampling techniques at different stages
- Some projects involving both probability and non-probability sampling techniques.

5.4. Probability Sampling

- the chance, or probability, of each case/element being selected from the population is known and is usually equal for all cases
- Probability sampling is often associated with survey and experimental research strategies.
- The process of probability sampling can be divided into four stages:
 1. Identify a suitable sampling frame based on your research question(s) or objectives.
 2. Decide on a suitable sample size.
 3. Select the most appropriate sampling technique and select the sample.
 4. Check that the sample is representative of the population

Identification of Sampling Frame

- The **sampling frame** for any probability sample is a complete list of all the cases/elements in the population from which your sample will be drawn.

Examples

- student telephone directory (for the student population),
- the list of companies on the stock exchange,

- the directory of medical doctors and specialists,
- the yellow pages (for businesses)

Selecting your sampling frame: Checklist

- Are cases listed in the sampling frame relevant to your research topic, in other words will they enable you to answer your research question and meet your objectives?
- How recently was the sampling frame compiled, in particular is it up to date?
- Does the sampling frame contain the correct information, in other words is it accurate?
- Does the sampling frame exclude irrelevant cases, in other words is it precise?
- Does the sampling frame include all cases, in other words is it complete?

Deciding on a suitable sample size

Choice of sample size is governed by:

1. the confidence you need to have in your data and thereby the findings
2. the margin of error that you can tolerate – that is, the accuracy you require for any estimates made from your sample;
3. Variability observed in the target population, and to lesser extent
4. the types of analyses you are going to undertake
5. the size of the total population from which your sample is being drawn

Remarks

- sample size is almost always a matter of judgement as well as of calculation
- a minimum number of 30 for statistical analyses (central limit theorem)

Sample Size for Estimating Population Mean

$$n = \frac{(z_{\alpha/2})^2 \sigma^2}{E^2}$$

Sample Size to Estimate Population Proportion

$$n = \frac{(z_{\alpha/2})^2 p(1-p)}{E^2}$$

Selecting the most appropriate sampling Technique

Five main techniques can be used to select a probability sample:

- ☐ simple random;
- ☐ Systematic random ;
- ☐ stratified random;
- ☐ Cluster random ;
- ☐ and multi-stage random

- 1. Simple random sampling (sometimes called just random sampling) involves you selecting the sample at random from the sampling frame using either random number tables or lottery system**

To do so,

- Number each of the cases in your sampling frame with a unique number.
- **Select** cases using random numbers until your actual sample size is reached.

When to Use? It is recommended when

- Statistical inferences are to be made
- Research does/ does not require face to face contact
- The population is geographically concentrated
- The sampling frame does not have relevant strata
- The sampling frame contains periodic patterns

Advantages

- only minimal advance knowledge of population needed
- Easy to analyze data and computer error
- simple to use and understand
- Requires only one stage of sample selection

Disadvantages

- Requires sampling frame to work from
- Does not use knowledge of population that researcher may have
- Larger errors for sample size than with stratified sampling
- Respondents may be widely dispersed , hence costs may be higher

Example 1: The human resources director of a company with 82 people on its payroll has been asked by the vice president to consider formulating an implementable flex time policy. The director feels that such a policy is not necessary since everyone seems happy with the 9 to 5 hours, and no one has complained. Formulating such a policy now, in the opinion of the director, runs the risk of creating domestic problems for the staff and scheduling problems for the company. She wants, however, to resort to a simple random sampling procedure to do an initial survey, and with the results, convince the V.P. that there is no need for flex time, and urge him to drop the matter. Since simple random sampling offers the greatest generalizability of the results to the entire population, and the V.P. needs to be convinced, it is important to resort to this sampling design.

Example 2 : The regional director of sales operations of a medium-sized company having 20 retail stores in each of its four geographical regions of operation, wants to know what types of sales gimmicks worked best for the company overall during the past year. This is to help formulate some general policies for the company as a whole and prioritize sales promotion strategies for the coming year.

Instead of studying each of the 80 stores, some dependable (i.e., representative and generalizable) information can be had, based on the study of a few stores drawn through a simple random sampling procedure. That is, each one of the 80 stores would have an equal chance of being included in the sample, and the results of the study would be the most generalizable.

2. Systematic sampling

Systematic sampling involves you selecting the sample at regular intervals from the sampling frame. To do this you:

1. Number each of the cases in your sampling frame with a unique number.
2. Select the first case using a random number.
3. Calculate the sampling fraction.
4. Select subsequent cases systematically using the sampling fraction to determine the frequency of selection.

When to use?

- Statistical inferences are to be made

- Research require does/does not require face to face contact
- The population is geographically concentrated
- The sampling frame does not contain relevant strata
- The sampling frame does not contain periodic patterns

Disadvantages

- | | |
|---|---|
| <h4>Advantages</h4> <ul style="list-style-type: none"> ○ Simple to draw samples ○ Easy to check | <ul style="list-style-type: none"> ○ periodicity: a problem that occurs when the original list has a systematic pattern ○ May introduced increased biased |
|---|---|

Example1: An administrator wants to assess the reactions of employees to a new and improved health benefits scheme that requires a modest increase in the premiums to be paid by the employees for their families. The administrator can assess the enthusiasm for the new scheme by using a systematic sampling design. The company's records will provide the population frame, and every nth employee can be sampled. A stratified plan is not called for here since the policy is for the entire company.

Example 2: If customers' interest in a highly sophisticated telephone is to be gauged by an entrepreneur, a systematic sampling procedure with the telephone directory as the population frame will be the easiest and quickest way to obtain the information, while still ensuring representativeness of the population studied.

Note: Systematic sampling will be inadvisable where systematic biases can be anticipated to be present. For example, systematic sampling from the personnel directory of a company (especially when it has an equal number of employees in each department), which lists the names of the individuals department-wise, with the head of the department listed first, and the secretary listed next, has inherent biases. The possibility of systematic biases creeping into the data cannot be ruled out in this case, since the selection process may end up picking each of the heads of the department or the departmentally secretaries as the sample subjects. The results from such a sample will clearly be biased and not generalizable, despite the use of a probability sampling procedure. Systematic sampling will have to be scrupulously avoided in cases where known systematic biases are possible.

3. *Stratified random sampling*

It involves dividing the population into two or more relevant and significant strata based on one or a number of attributes. To do so, you

1. Choose the stratification variable or variables.
2. Divide the sampling frame into the discrete strata
3. Number each of the cases within each stratum with a unique number, as discussed earlier
4. Select your sample using either simple random or systematic sampling, as discussed earlier.

When to use?

- Statistical inferences are to be made
- Research require does/does not require face to face contact
- The population is geographically concentrated
- The sampling contains relevant strata
- The sampling frame does/does not contain periodic patterns

Advantages

- Assures representation of all groups in sample
- Characterization of each stratum can be estimated and comparisons made
- Reduces variability for same sample size

Disadvantages

- requires accurate information on proportion in each stratum
- If stratified lists are not already available, they can be costly to prepare

Example 1: The director of human resources of a manufacturing firm wants to offer stress management seminars to the personnel who experience high levels of stress he conjectures that three groups are most prone to stress: the workmen who constantly handle dangerous chemicals, the foremen who are held responsible for production quotas, and the counsellors who, day in and day out, listen to the problems of the employees, internalize them, and offer them counsel, with no idea of how much they have really helped the clients.

To get a feel for the experienced level of stress within each of the three groups and the rest of the firm, the director would stratify the sample into four distinct categories: (1) the workmen

handling the dangerous chemicals, (2) the foremen, (3) the counsellors, and (4) all the rest. He would then choose a disproportionate random sampling procedure [since group (3) can be expected to be very small, and groups (2) and (1) are much smaller than group (4)].

This is the only sampling design that would allow the designing of stress management seminars in a meaningful way, targeted at the right groups.

Example 2: If in Example 2 (Simple random Sampling) the regional director had wanted to know which sales promotion gimmick offered the best results for each of the geographical areas, so that different sales promotion strategies (according to regional preferences) could be developed, then first the 80 stores would be stratified on the basis of the geographical region, and then a representative sample of stores would be drawn from each of the geographical regions (strata) through a simple random sampling procedure. In this case, since each of the regions has 20 stores, a proportionate stratified random sampling process (say, five stores from each region) would be appropriate. If, however, the northern region had only 3 stores, the southern had 15, and the eastern and western regions had 24 and 38 stores, respectively, then a disproportionate stratified random sampling procedure would be the right choice, with all three stores in the northern region being studied, because of the small number of elements in that population.

If the sample size is retained at 20, then the north, south, east, and west regions will probably have samples respectively of three, four, five and eight.

It is interesting to note that sometimes when stratified random sampling might seem logical, it might not really be necessary. For example, when test-marketing results show that Cubans, Puerto Ricans, and Mexicans perceive and consume a particular product the same way, there is no need to segment the market and study each of the three groups using a stratified sampling procedure.

4. Cluster sampling

Cluster sampling is similar to stratified sampling as you need to divide the population into discrete groups prior to sampling

- The groups are termed **clusters** in this form of sampling and can be based on any naturally occurring grouping
- For cluster sampling your **sampling frame** is the complete list of clusters rather than a complete list of individual cases within the population

When to use?

- Statistical inferences are to be made
- Research require does/does not require face to face contact
- The population is in discrete geographic clusters
- The sampling frame contains relevant clusters

Advantages

- Yields lowest field cost if clusters are geographically defined
- Requires listing of all clusters but of individuals only within clusters
- Can estimate characteristics of clusters as well as of population

Disadvantages

- larger error for comparable size than other probability samples
- Researcher must able to assign population members to unique clusters
- Duplication or omission of individual elements may occur

Example 1: A human resources director is interested in knowing why staff resign. Cluster sampling design will be useful in this case for conducting exit interviews of all members completing their final papers in the human resources department on the same day (cluster), before resigning. The clusters chosen for interviews will be based on a simple random sampling of the various clusters of personnel resigning on different days.

The interviews would help to understand the reasons for turnover of a heterogeneous group of individuals (i.e., from various departments), and the study could be conducted at a low cost.

Example 2: A financial analyst desires to study the lending practices of pawnbrokers in same Clara Country, California. All the pawnbrokers in each city would form a cluster. By randomly sampling the clusters, the analyst would be able to draw conclusions on the lending practices.

5. *Multi-stage sampling*

- Development of cluster sampling
- It is normally used
 - to overcome problems associated with a geographically dispersed population
 - when face-to-face contact is needed

- where it is expensive and time consuming to construct a sampling frame for a large geographical area
- like cluster sampling, you can use it for any discrete groups, including those that are not geographically based
- The technique involves taking a series of cluster samples, each involving some form of random sampling

5.5. Non-probability sampling

- the probability of each case being selected from the total population is not known
- It is impossible to answer research questions or to address objectives that require you to make statistical inferences about the characteristics of the population
- limited resources or the inability to specify a sampling frame
- No appropriate statistical techniques for measuring random sampling error
- Often used in qualitative researches

Stages

1. Identify a suitable sampling frame based on your research question(s) or objectives.
2. Decide on a suitable sample size.
3. Select the most appropriate sampling technique and select the sample.
4. Check that the sample is representative of the population

Identification of Sampling Frame

- generally used when there is no suitable sampling frame
- Inability or any other plausible reason to prepare sampling frame
- the issue of sample size, except for quota sampling, is ambiguous

Deciding on a suitable sample size

- unlike probability sampling, there are no rules

- the logical relationship between your sample selection technique and the purpose and focus of your research is important
- your sample size is dependent on your research question(s) and objectives – in particular
 - what you need to find out,
 - what will be useful,
 - what will have credibility and
 - what can be done within your available resources

Selecting the most appropriate sampling technique and the sample

Five main techniques can be used to select a probability sample:

- | | |
|--|---|
| ☐ Quota sampling; | ☐ Self selection sampling; |
| ☐ Purposive sampling; | ☐ convenience sampling |
| ☐ Snowball sampling ; | |

1. Quota sampling

- ☐ Based on the premise that your sample will represent the population as the variability in your sample for various quota variables is the same as that in the population.
- ☐ Selection of cases within strata is entirely non-random.
- ☐ Procedures:
 1. Divide the population into specific groups.
 2. Calculate a quota for each group based on relevant and available data.
 3. Combine the data collected by interviewers to provide the full sample

When to use

- | | |
|---|--|
| ○ Statistical inferences are not required. | ○ Relevant quota variables are available |
| ○ The sample is required to be representative | Advantages |
| | ○ Less costly and can be done quickly |
| | ○ Does not require sampling frame |

- Introduces some stratification of population
- Introduces bias as interviewers may diffuse
- Sampling error cannot be estimated
- Projecting data beyond sample is inappropriate

Example: A company is considering operating an on-site kindergarten facility. But before taking further steps, it wants to get the reactions of four groups to the idea: (1) Employees who are parents of kindergarten-age children, and where both are working outside of the home, (2) employees who are parents of kindergarten-age children, but where one of them is not working outside of the home, (3) single parents with kindergarten-age children, and (4) all those without children of kindergarten age. If the four groups are expected to represent 60%, 70%, 23%, and 10% respectively, in the population of 420 employees in the company, then a quota sampling will be appropriate to represent the four groups.

Note: The last group should also be included in the sample since there is a possibility that they may perceive this as a facility that favours only the parents of kindergarten children, and therefore resent the idea. It is easy to see that resorting to quota sampling would be important in case such as this.

2. Purposive sampling

- Purposive or judgemental sampling enables you to use your judgement to select cases that will best enable you to answer your research question(s) and to meet your objectives.
 - Useful when
 - statistical inferences are not to be made
 - the sample is not required to be representative
 - no difficulty of identifying individual cases
 - the sample to be selected are very small
- Advantages
- Sample guaranteed to meet a specific objective
 - Easy to understand and undertake
 - No sampling frame is required

Disadvantages

- Bias due to experts' beliefs may make sample unrepresentative
- Projecting data beyond sample is inappropriate

Example: A pharmaceutical company wants to trace the effects of a new drug on patients with specific health problems (muscular dystrophy, sickle cell anemia, rheumatoid arthritis, etc.). It then contacts such individuals and, with the group of voluntarily consenting patients, tests the drug. This is a judgment sample because data are collected from appropriate special groups.

3. *Snowball sampling*

- Snowball sampling is commonly used when it is difficult to identify members of the desired population, for example people who are working while claiming unemployment benefit

Procedures:

1. Make contact with one or two cases in the population.
2. Ask these cases to identify further cases.
3. Ask these new cases to identify further new cases (and so on).
4. Stop when either no new cases are given or the sample is as large as is manageable

Advantages

- Useful in locating members of rare populations
- No sampling frame is required
- problems of bias are huge, as respondents are most likely to identify other potential respondents who are similar to themselves, resulting in a homogeneous sample

Disadvantages

- The main problem is making initial contact. Once you have done this, these cases identify further members of the population, who then identify further members
- Projecting data beyond sample is inappropriate

4. *Self-selection sampling*

- Self-selection sampling occurs when you allow each case, usually individuals, to identify their desire to take part in the research. You therefore:

1. Publicise your need for cases, either by advertising through appropriate media or by asking them to take part.
 2. Collect data from those who respond
- Publicity for convenience samples can take many forms: articles and advertisements in magazines, postings on appropriate Internet newsgroups and discussion groups
 - Cases that self-select often do so because of their feelings or opinions about the research question(s) or stated objectives
 - Often appropriate when the purpose of the investigation is just exploratory.

5. *Convenience sampling*

- ☐ Convenience sampling (or haphazard sampling) involves selecting haphazardly those cases that are easiest to obtain for your sample
- ☐ The sample selection process is continued until your required sample size has been reached.
- ☐ useful when
 - ✓ there are little variation in the population
 - ✓ no difficulty of identifying individual cases
 - ✓ large sample is required

Advantages

- Easy to understand and undertake
- Very low cost
- No sampling frame is required

Disadvantages

- prone to bias and influences that are beyond your control
- projecting beyond sample is inappropriate

Example: The accounts executive has established a new accounting system that maximally utilizes computer technology. Before making further changes, he would like to get a feel for how the accounting clerks react to the new system without making it seem that he has doubts about their acceptability. He may then “casually” talk to the first five accounting personnel that walk into his office, trying to gauge their reactions.

Note: Convenience sampling should be resorted to in the interest of expediency, with the full knowledge that the results are not generalizable at all.

Chapter 6

Methods of Data Collection

6 1	
6 2	○
6 3	○
6 4	○
6 5	○
6 6	○

6.1 Sources of Data

Any research is based on various data

- The more information (data) the researcher has about the phenomenon, the better will be the investigation.
- The more valid is the source of data, the more reliable will be the data collected which in turn will lead to correct and reliable conclusion
- Data are building blocks of research

Different authors have classified the source of data differently. Some of the most important views are of the following.

1. According to PV young
 - a. Documentary sources
 - Books - Diaries - Magazines – news paper
 - Manuscripts –letters -Journals -archives
 - b. Field sources
2. According to WA Bagley
 - a. Primary source

- Include the actual information received from individuals directly concerning the problem of the study.
- b. Secondary sources
- Includes all types of published and unpublished public or private documents and other such types of data.

3. According to G.A Lun berg

- a. Historic sources- includes sources representing the past incident – documents, papers, stone inscriptions
- b. Field sources

Generally, the sources of data are of two types: primary sources and secondary sources

1. Primary sources of data

- Primary data
- Gathered by the researcher him/her self for the first time
- The sources are individuals and the incidence around them, thus happening to be original
- Means: survey ,observation and experiment

2. Secondary sources of data

- Secondary data
- They have already gone through statistical process and are collected by someone else.
- Means, books, manuscripts, records, census reports, magazine, news paper, diaries, letters, magazines.

Note

- Before deciding the method of data collection, the researcher should be aware of the existence of two types of data- primary and secondary data.
- The researcher has to decide w/c source of data to use for the study
- The methods of collecting primary and secondary data are different

6.2 Personal interview

Definition and Features

A personal interview is a form of direct communication in w/c an interviewer asks respondents in a face to face situation. This versatile and flexible method is a two way conversation between an interviewer and a respondent.

- Personal interviews can take place in a factory, café homeowners doorway, in an executive office or in other settings.

Advantages

- The opportunity for feed back
The interviewer can provide feedback in clarifying any questions any respondent has about the instruction or questions.
- Probing complex answers
If a respondent answer is brief or unclear, the researcher may ask for a clearer or more comprehensive explanation. Asking “can you tell me more about what you had in mind?” is an example of probing.
- There is greater flexibility under this method as the opportunity to restructure questions is always involved.
- Item non response: - failure to provide an answer to a question is less likely to occur with an experienced interviewer and a face to face interaction.
- High participation:- this is because
 - Respondents are generally not required to do any reading or writing- all they have to do is to talk.
 - Most people enjoy sharing information and insights with friendly and sympathy interviewers.
- More information and that too in greater detail can be obtained
- Observation method as well can be applied
- The interviewer can be adapted to the ability of or the educational level of the respondent and as such misinterpretation can be avoided.
- Supplementary information about the respondent can be obtained
- Illiterate and functionally participants can be reached
- Interviewer can prescreen participant to ensure he/she fits the population profile.

Disadvantages

- Respondents are not guaranteed **anonymity** and therefore may be reluctant to provide confidential information to another person.

Example: - “Do you see any major internal instabilities..?”

Many managers may be reluctant to answer this sensitive question honestly in a personal interview where their identity is known

- **Interviewer Bias**

- The rephrasing of a question, the interviewer tone of voice and the interviewer's appearance may influence the respondents answer.
- Considerable interview variability

- **Cost**

Personal interviews are generally more expensive than mail internet and telephone surveys. The geographic proximity of respondents, the length and complexity of the questionnaire and the sample size all influence the cost of the personal interview.

- **Social desirability bias may occur**
- **Need for highly trained interviewers**
- **Not all participants are available or accessible**

6.3. Telephone interview

Telephone interview- Contacting respondents via telephone to gather responses to survey questions

Advantages

- **Speed**

Where as data collection with mail or personal interviews can take several weeks, hundreds of telephone interviews can be conducted literally over night.

- **Cost**

- lower contacts than personal interview
- It is estimated that the cost of telephone interviews is less than 25% of the cost of door to door personal interviews.
- Travel time and cost are eliminated
- Typical internet survey is less expensive than a telephone survey

- **Absence of face to face contact**

- Respondent may answer embarrassing or confidential questions more willingly in a telephone interview than in personal interview
- Reduction of refusal rate

- Interviewer bias is reduced

- **Call backs**

An unanswered call, a busy signal or a respondent who is not at home requires a call back.

Telephone call backs are substantially easier and less expensive than personal interview call backs.

- Better **access to hard -to reach participants** through repeated call backs
- **Representative and wider distribution** of sample is possible
- **No field staff** is required

Disadvantages

- Little time is given to respondents for considered answers; interview period is not likely to exceed five minutes in most cases.
- Higher costs if interviewing geographically dispersed sample is involved
- Surveys are restricted to respondents who have telephone facilities
- It is not suitable for intensive surveys where comprehensive answers are required to various questions.
- Problems are difficult to handle

Principles of good interviewing

The basics

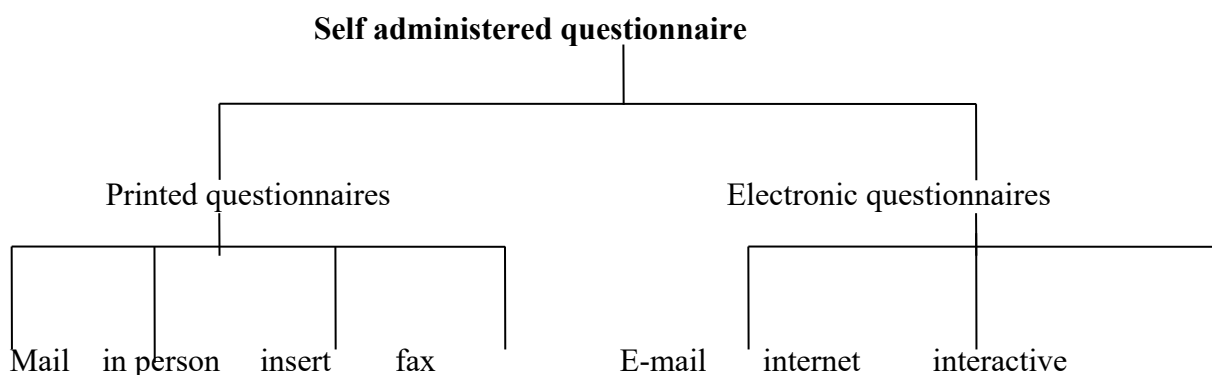
2. Have integrity and be honest
3. Have patience and tact
4. Pay attention to accuracy and detail
 - Not to “mislead or misunderstand” a client
 - Probe for clarification, detail and full answers
- Record responses verbatim
- Never assume you know what a respondent is thinking
5. Exhibit a real interest in the inquiry at hand but keep your own opinions to yourself
 - You are asker and a recorder of other people’s opinions (**not a contributor !**) to the study data
6. Keep the inquiry and respondents responses confidential
 - Do not discuss the studies you are doing with relatives, friends or associates

- Never quote one respondents opinion to another-i.e., greatest violation of privacy of call
- 7. Complete the number of interviews according to the sample plan
 - The study will benefit from having available the amount and type of information originally specified
- 8. Follow the directions provided
- 9. Make every effort to keep schedules
- 10. Keep control of interview you do
 - Establish average length of interview
 - Always get the whole story from the respondent and writ it all down in the respondents own words.
 - Remember to keep the interviewer focused on the subject at hand and not let it wander off into unnecessary small talk.
 - Avoid offending the respondent by being too talkative to yourself
- 11. Check over each question you have completed
- 12. Compose your sample execution with the sample plan
- 13. Active listener

6.4. Self administered questionnaire

Self administered questionnaire

- A questionnaire that is filled in by the respondent rather than by an interviewer. Self administered questionnaires can be either printed or electronic



Drop off

kiosk

Self administered questionnaire present a challenge to the business researcher because they rely on the clarity of the written word rather than on the skills of the interviewer.

Questionnaires are:-

- Mailed faxed or couriered to be self administered with return mechanism generally included
- Computer delivered via intranet , internet and online service
- People are intercepted in a central location and studied via paper or computerized instrument.

Advantages

1. Cost

-often they are the lowest cost option

- Mail questionnaires are relatively in expensive compared to personal interviews and telephone surveys
- However mail surveys often include a follow up mailing requires additional postage and the print of additional questionnaires.

Note: It usually isn't cost effective to try to minimize costs on printing- questionnaire that are duplicated on poor quality paper have a greater likelihood of being thrown in the waste basket than those prepared with more expensive high quality printing.

2. Geographic coverage

- Mail questionnaires can reach a geographically dispersed sample simultaneously and at a low cost because interviewers are not required.

3. Allows contact with other wise inaccessible participants

Some groups such as major corporate executives and physicians are difficult to reach in person or by phone, as gate keepers (secretaries, office managers and assistants) limit access but researcher can often access these special participants by mail or computer. Self administered questionnaires can be widely distributed to a large number of employees so organizational problems can be assessed quickly and inexpensively.

4. Time

Respondents have adequate time to give well thought answers. In mail surveys, the participant can take more time to collect facts, talk with others or consider replies at length than is possible in a survey employing the telephone or in a personal interview.

5. Interviewer absence

- It is free from bias of the interviewer. Answers are in respondents own words.
 - Mail surveys are typically perceived as more impersonal, providing more anonymity than the other communication modes-thus respondents tend to reveal sensitive or socially undesirable information.
 - Computer delivered surveys still enjoy the same perceived anonymity although increased concerns about privacy may erode this perception in the future.
6. Requires minimal staff
7. Incentives may be used to increase response rate

Disadvantages

1. Low response rates in some modes

- Non response error
- Respondents who answer the questionnaire may not be typically of all people in the sample

2. No interviewer intervention

- No feed back
- No probing explanation / classification
- Misunderstanding /mislead /misperceive
- Incomplete answers
- Misunderstanding by the interviewer

3. It can be used only for **literate respondents**

4. Difficulty of stating **standardized questions**

5. **Speed of data collection**

Mail surveys specially are the slowest method of all.

Principles and Guidelines for Questionnaire Construction

1. What should be asked?

2. How should each question be phrased?
3. In what sequence should the questions be arranged?
4. What questionnaire layout will best serve the research objectives?
5. How should the questionnaire be pretested? The questionnaire needs to be revised?

1. Questionnaire relevancy

- A questionnaire is relevant if no unnecessary information is collected and if the information that is needed to solve the business problem is obtained.

Asking the wrong or an irrelevant question is a pitfall to be avoided.

To ensure information relevancy,

-the researcher must be specific about data needs and there should be rationale for each item.

- When planning the questionnaire design it is essential to think about possible omissions
- Is information collected on relevant demographic and psychographic variable?
- Are there any questions that might classify the answers to other questions?
- Will the results of the study provide the solution to the manager's problem?

2. Response strategy

Open ended response vs fixed alternative questions

open ended questions

- Pose some problem or topic and ask the respondent to answer in his or her own words

Closed end questions

- A question in which the respondent is given specific limited alternative responses and asked to choose the one (or more) closest to his or her own view point.

Advantages

Open ended

- Respondents are free to answer what ever is uppermost in their thinking

Closed end

- require less interview skill
- Takes less time and easier for the

Disadvantages

- Difficulty of coding editing and analyzing

respondent to answer

Note:-

Multiple choice questions can be efficient but they also present unique design and analysis problems

- The list of alternatives is not exhaustive. Fixed alternative question may tempt respondents to check an answer that is untrue. As a matter of convenience, they may select a given alternative rather than think of the most correct alternative.
- We may add the category “other (please specify)” as a safe guard to provide the participant with an acceptable alternative for all other options.
- Another challenge occurs when the choices are not mutually exclusive (the participant thinks two or more responses overlap)
- Most questionnaires include a mixture of open ended and closed questions. Each form has unique benefits; in addition a change of pace can eliminate respondent boredom and fatigue.

Forms of fixed alternative questions

- i. Simple dichotomy or dichotomous alternative question a fixed alternative question that requires the respondents to choose one of two alternatives the answer can be a simple “yes” or “no” or a choice between “this” and “that”
- ii. Multiple choice question(X)
They are appropriate where there are more than two alternatives or where we seek gradation of preference interest or agreement.
- iii. Determinant choice question
 - Requires the respondent to choose one and only one response from among several possible alternatives.
- iv. Attitude rating scales
Ask the respondent to position each factor on a companion scale
- v. Checklist question allows the respondent to provide multiple answers to a single question the respondent indicates past experience preference and the like merely by checking off an item in many cases the choices are adjectives that describe a particular object.

3. Wording of Questions

- A. Avoid complexity: use simple conversational language
 - Words in questionnaires should be readily understandable to all respondents.

- Remember not all people have the same vocabulary set
- Avoid jargon /technical terms

B. Avoid leading and loaded question

Leading questions:- suggest or imply certain answers

Loaded question:- suggest a socially desirable answer or are emotionally charged.

C. Avoid Ambiguity :-

- be as specific as possible
- Indefinite words often, occasionally, usually , regularly, frequently, many, good , fair and poor, fair

D. Avoid Double barreled Items

A question that may induce bias because it covers two issues at once

E. Avoid making assumptions

F. Avoid burdensome questions that may tax the respondents memory

Examples

1. *When effluents from a paper mill can be drunk and exhaust from factory smokestacks can be breathed, then humankind will have done a good job in saving the environment. . . . Don't you agree that what we want is zero toxicity: any effluents? (*
2. *Many people are using dry cleaning less because of improved wash-and-wear clothes. How do you feel wash-and-wear clothes have affected your use of dry cleaning facilities in the past 4 years?*
3. *Where did you rank academically in your high school graduating class?*

Use less	No change	Use more	
Top quarter	2nd quarter	3rd quarter	4th quarter
4. *I feel capable of handling myself in most social situations.*

Agree	Disagree
-------	----------
5. *What media do you rely on most?*

☐ Television
☐ Radio
☐ Internet
☐ Newspapers
6. *"Do you feel our hospital emergency room waiting area is clean and comfortable?"*
7. *Should General Electric continue to pay its outstanding quarterly dividends?*

☐ Yes ☐ No
--
8. *"Do you recall any commercials on that program?"*

4. Question Sequence

Order bias

- Bias caused by the influence of earlier questions in a questionnaire or by answers position in a set of answers.
- Question sequence can encourage or discourage commitment and promote or hinder the development or researcher participant rapport.

The basic principle used to guide sequence decisions is this: the nature and needs of the participant must determine the sequence of decisions. Four guide lines are suggested to implement this principle.

A. Awaken interest and motivation

The question sequence must quickly awaken and motivate to participate in the interview.

- Put the more interesting topical target questions early
- Leave classification questions (e.g age, family size, income) not used filters or screens to the end of the survey.

B. Simple to complex

Deferring complex question or simple questions that require much thought can help reduce the number of “don’t know” responses that are prevalent early in interviews.

C. Sensitive and ego involving information

Put questions that might influence the participant to discontinue or terminate questioning process near the end.

D. General to specific funnel approach

- The procedure of moving from general to more specific questions is sometimes called funnel approach.
- The funnel techniques allows the researcher to understand the respondents frame of response before asking more specific questions about the level of the respondents information and the intensity of her or her opinions.

E. Question groups and transitions

- Most questionnaires that cover a range of topics are divided into sections with clearly defined transitions between sections to alert the participant to the change in frame of reference.

5. Questionnaire Pretesting

Testing /trying out a questionnaire on a group that is selected on a convenience basis and that is similar in make up to the one that ultimately will be sampled.

Researchers should select a group that is not too divergent from the actual respondents but it is not necessary to get a statistical sample for pretesting.

Importance of pretesting

- Allows the researchers to determine if the respondents have any difficulty understanding the questionnaire and whether there are any ambiguous or biased questions.
- Pretests are typically conducted to answer questions about the questionnaire such as the following.
- Can the questionnaire format be followed by the interviewers?
- Does the questionnaire flow naturally and conversationally?
- Can respondents answer the questions easily?
- Which alternative question work best?
- Provides the means to test the sampling procedure.
- Pretest may determine if interviewers follow the sampling instructions properly and if the procedure is efficient. They also provide estimates of the response rate for mail surveys and the completion rate for telephone surveys.

6. Instructions

Instructions to the interviewer or respondent attempt to ensure that all respondents are treated equally. Two principles form the foundation for good instructions: clarify and courtesy.

7. Good introduction

A proper introduction that clearly discloses the identity of the researcher and conveys the purpose of the survey is absolutely necessary. It is also essential to establish some rapport with the respondents and motivate them to respond to the questions in the questionnaire wholeheartedly and enthusiastically. Assurance of confidentiality of the information provided by them will allow for less biased answers. The introduction section should end on a courteous note, thanking the respondent for taking the time to respond to the survey. The following is an example of an appropriate introduction

6.5. Interview Schedule

Schedule, which is very much like self administered questionnaire, are filled in by enumerators who are specially appointed for the purpose. These enumerators along with schedules go to respondent.

1. Put to them questions to the respondent from the pro forma in the order questions are listed and record the replies in the space meant for the same in the pro forma.
2. Handed over to the respondents and enumerator help them in recording their answers to various questions. Enumerators explain the aims and objectives of the investigation and also removes difficulties any respondent may feel in understanding the implication of a particular question.

Difference between questionnaires and schedules

Questionnaire	Schedule
○ Filled in by the respondent	- filled in by the respondent enumerator
○ Cost relatively cheap and economical	- relatively more expensive
○ No field staff required	- field staff (enumerator) are required
○ Non response is high bias remains	- Non response is low
○ Indeterminate interviewer bias	- danger of interviewer bias
○ It is not clear as to why replied	- Identify of the respondent is known
○ Slow	- information is collected well time
○ Probe/ feedback impossible	- probe feedback possible
○ Literate	- illiterate
○ Wider and more representative	- difficulty in sending enumerators

Distribution of sample over a wider area

- | | |
|-------------------------------------|-------------------------------------|
| ○ Risk of incomplete and wrong inf. | – complete and accurate information |
| ○ Success depends more on the | - honesty and competence of |

Quality of the questionnaire the interviewer

- | | |
|----------------------------------|------------------------|
| ○ Observation is also impossible | - observation possible |
|----------------------------------|------------------------|

6.5. Observation Methods

Observation becomes a tool for scientific inquiry when it

- Serves a formulated research purpose/question
- Systematically planned and executed
- Is recorded systematically

- Related to general propositions rather than being presented as reflecting a set of interesting curiosities
- Is subjected to checks and controls on validity and reliability

Scientific observation is the systematic process of recording the behavioral patterns of people, objects and occurrences as they are witnessed. No questioning of or communicating with people occurs.

Classification of observation methods

1. On the basis of the subject “to be observed”

- a. Behavioral observation and
- b. Non-behavioral observation

a. Behavioral observation

The observation study of persons can be classified into four major categories.

Non-verbal behavior – the most prevalent and includes body movement, motor expressions, head nods, smiles raised eye brows and other facial expressions

Examples : Body movement-

- how sales person travel a territory
- share of time workers spend in various works
- Indicator of interest or boredom anger or pleasure.

Facial expression – sign of emotional states

Eye blink- are studied as indicators of interest

Linguistic behavior (verbal behavior)

The second frequently used form of behavior observation

Examples

- Statements made by airline travelers waiting in line
- “ahs” or other sounds(words) a professor makes or uses during a class

Extra linguistic (expressive behavior) one author has suggested four dimensions of extra linguistic activity

Local – pitch, loudness and timbre

Temporal- rate of speaking, duration of utterance and rhythm

Interaction- tendencies to interrupt, dominate or inhibit

Verbal stylistic- vocabulary and pronunciation peculiarities, dialect and cxs expression

Spatial relationships-especially how a person relates physically to others

b. Non behavioral observation

Record analysis: This may involve historical or current records and public or private records. They may be written, printed, sound recorded, photographed or videotaped.

Example

- Historical statistical data
- Analysis of current financial records and economic data
- Content analysis of competitive advantage.
- Analysis of personnel records

Physical condition analysis

- Store audits of merchandise availability
- Studies of plant safety compliance
- Analysis of inventory conditions
- Analysis of financial statements

Process or activity analysis includes time/motion studies of manufacturing processes and analysis of

- Traffic flows in a distribution system
- Paperwork flows in an office
- Financial flows in the banking system

Note: Although the observation method may be used to describe a wide variety of behaviors, cognitive phenomena such as attitudes, motivations, expectations, intentions and preferences cannot be observed.

2. On the basis of observer- participant relationship

a. Direct vs. indirect observation

Direct observation occurs when the observer is physically present and personally monitors what takes place. This approach is very flexible because it allows the observer to react to and report subtle aspects of

events and behaviors as they occur. He/She is free to shift places, change the focus of the observation or concentrate on unexpected events if they occur.

Limitations

- Observer bias
- The observer may not record every detail that describe the persons, objects and events in a given situations w/h may suffer accuracies.
- Interpretation of observation data. Facial expressions and other non-verbal communication may have several meanings.

Indirect observation (mechanical observation) Occurs when the recording is done by mechanical, photographic or electronic means. Indirect observation is less flexible than direct observation but is also less biasing and may be less erratic in accuracy. Another advantage of indirect observation is that the permanent record can be reanalyzed to include many different aspects of an event.

b. Visible vs. hidden observation

A second factor affecting observer-participant r/ship concerns whether the participant should know of the observer's presence.

Visible observation

- An observation situation in which the observer's presence is known to the subject (participants).
- The initial entry of an observer in to a situation often upsets the activity patterns of the participants. The potential bias from participant awareness of observers is always a matter of concern.

Hidden Observation – an observation situation in which the subject (participants) unaware that the observation is taking place.

- Often technical means such as one way mirrors, hidden cameras or microphones are used.
- These methods reduce observer bias. Furthermore, asking subjects to participate in the research is not required.
- Hidden observation raises an ethical issue concerning respondents privacy.

Partial concealment

- The presence of the observer is not concealed but the objective and participant of interest are.

C. Participant vs. Non participant observation

The third observer-participant issue is whether the observer should participate in the situation while observing.

Participant observation exists when the observer enters the social setting and acts as both an observer and a participant. While reducing the potential for bias this again raises ethical issues.

Participant observation makes a dual demand on the observer. Recording can interface with participation and participation can interface with observation. The observer's role may influence the way others act. Because of these problems, participant observation is used less in business research.

Non-Participant observation

Observation in a detached emissary without any attempt on his/ her part to experience through participation what others feel.

3. On the basis of degree of structure in the environmental setting

a. Structured observation

Structured observation employs standardized procedures, trained observers, schedule for recording and other devices for the observer that mirror the scientific procedures of other primary data collection methods.

Structured observation is considered appropriate in descriptive studies.

- b. Unstructured observation that takes place without thinking in advance about the above matters. In exploratory study, the observational procedure is most likely to be relatively unstructured

Advantages

1. We can collect the **original data** at the time they occur
 - We need not depend on reports by others
 - Forgetting may occur in answering questions
 - Respondents may not want to report fully and fairly
 - Every respondent filters information

Thus observation overcomes many of these deficiencies of questioning.

2. Observations is the **only method of available** to gather certain types of information

Examples

- The study of records, mechanical process
 - Young children, inarticulate participants
3. It alone can capture the whole event as they occur in its natural environment
 4. Participants seem to accept an observational intrusion better than they respond to questioning. b/c
 - Observation is less demanding of them
 - Observation has less biasing effect on their behaviors
 - It is possible to conduct disguised and unobtrusive observation studies much more easily than disguised questioning.

Disadvantages

1. The observer normally must be at the scene of the event when it takes place, yet it is often impossible to predict where and when the event will occur.
2. Observation is a slow and expensive process that requires either human observer or costly surveillance equipment.
3. The research environment is more likely suited to subjective assessment and recording of data than to controls and quantification of events
4. The observation is limited as way to learn about the past
 - It is similarly limited a method by w/c to learn what is going on in the present at some distant place
5. It is difficult to gather information on such matters as intentions, attitudes, opinions or preferences.

However, any consideration of the merits of observation confirms its value when used with care and understand.

6.6. Secondary Data Collection

Secondary information is information, which has already been collected by some one and which, have already been passed through statistical process. The researcher does not obtain them directly rather he gathered them from published and unpublished material.

Why a researchers use data from secondary sources? In most cases researchers use secondary data for different reasons like;

Secondary data may solve problem: If adequate data are available from secondary sources, primary data collection will not be necessary.

Search costs are substantially lower than primary sources. It is cheap and less time consuming than primary sources. A comprehensive search of secondary sources can almost always be made in a fraction of the time and cost required for collection of primary data.

It has also important supplementary use, these includes:

- It helps to define the problem and formulate hypothesis about its solution. The assembly of available secondary data (information) always provides a better understanding of the problem and it may suggest reformulation and solution not considered previously.
- It will help to plan the collection of primary data. Examination of the methods and techniques employed by other investigators in similar studies may be useful in planning the present one.
- It will help to define the population and select the sample in primary data collection.

However, the researcher must be careful when using secondary data, particularly as the only sources of data. Secondary data to be useful, it must be:

- Available
- Relevant to the information needed
- Accurate and sufficient to meet data requirement.

Before using secondary data, we must see that they posses the following characters.

1. Reliability of the data

The reliability can be tested by finding answers to the following questions about the said data:

- Who collect the data? (How is the reputation of the collector?)
- What were the sources of the data?
- Were they collected using proper method?
- At what time were they collected?
- Was there any bias of the collector?
- What level of accuracy was desired?
- Was it achieved?

2. Suitability

The data that are suitable for one inquiry may not necessarily be found suitable in another inquiry. The researcher must very carefully scrutinize the definition of various terms and units of collection used at the time of collecting the data from the primary sources originally. The object, scope and nature of the original inquiry must be studied. If the researcher finds difference in these, the data will remain unsuitable for the present inquiry and should not be used

3. Adequacy

If the area and scope of the study is narrower than the area of present study the data are considered as inadequate. Because of the fact that information will be missed and the researcher needs additional sources to make it complete and usable. .

Secondary data may be obtained either from:

- Internal (with in the company or organization)
- External sources (outside sources)

Internal secondary data: All companies or organizations or any other organs keep record of the every course of conducting business or any other activities.

- Orders received are filled
- Cost are recorded
- Sales' people reports are submitted. Such and the like data are some of the many sources of information collected by the company for other purpose and are useful to the researcher

External sources of secondary data: The major sources of external information are:

- Government (federal, state, local)
- Trade association and trade press
- Periodicals
- Institution (research, academic)
- Commercial services and the like

The secondary sources of information can be classified into

- i. Personal Document
- ii. Public document

1. Personal document

These include the entire published and unpublished information document by the individuals for different purpose:

- Personal documents are not written in scientific manner (style) nor do they have an objective basis.
- They generally represent some ideas, values and feeling etc.

In spite of being subjective unscientific they have been very much useful in social research. Personal documents have been classified in to four types

A. Biography / Autobiography

Some great political, business leaders, social reformers and eminent persons write their own autobiographies or some other people (biographer) write their biography. Such persons provide useful information concerning, social, economical, political, religious and cultural conditions and incidents of their time.

Even, in some cases it is possible to get some information, which could not be known from any other sources.

Autobiographies are life histories, life experience and incidents of prominent individuals written by them. Biographies are life history, experience and events of some prominent person written by some biographer. Autobiography/ biography can be written for different purpose. It can be written in order to preserve specific feelings, incidents and events of writer's life. Such autobiography has objective basis and are useful source of information, if it is available indeed. Autobiography can also be written in order to publish and get financial benefit and popularity. Such autobiographies are less reliable as they are full of exaggeration and are full of deliberate distortion of facts.

B. Diaries:

Many educated people have a habit of writing diaries concerning their social relationship and life incidents. Most of the dairy writer has no intention to publish them. Some people daily write diaries to include program of the day.

Diaries are written with different intentions. Some write diaries to remember important incidents of their life. Some on the other hand write diaries to note things, which they cannot tell to anyone. They may write it to release their tension. Diary writing is a confidential and private performance. Therefore, if they are available they can be considered as a reliable.

C. Letters

Letters are often used by researchers as evidence especially in studies like, history and sociology and psychology. Most people write a number of letters during the course of their lives. Letters are the medium

of expressions of feelings likes and dislikes attitudes, desires, emotion, ambitions and important incidents of life. Letters are in general, personal and very reliable. Thus, they constitute one of the most reliable sources of information about the inter relationship, such as, friendship, love, marital affairs, family tension and divorce and the like.

Limitations

- Difficult to collect private letters
- The letters do not provide detailed and complete description of incidents
- If letters of only one person is secured, they hardly provide scientific data since no comparable information is available.

D. Memories

Some people are interested in writing down memories for their travels, participation in different activities. Such memories provide useful information in social research.

Limitations of personal documents

- Difficult in collection.
- They are rarely recorded and more seldom preserved
- Problem associated with the reliability of data: There is almost no criteria to judge their validity since they are written from subjective view point.
- It is not possible to arrive at some definite conclusion say about a social problem on the basis of personal document.

2. Public and Official document

In many research studies these sources of information are commonly and widely used. Public documents are information gathered from some governmental or non-governmental institutions.

Public document can be either unpublished or published documents. Unpublished documents are like reports of some big companies, confidential records of government departments and non-governmental organization.

Published document are those published documents in the general interest of the public. Such documents may be prepared by government or any other organs.

Some common public documents are of the following types

A. Records: These are the most important sources of secondary data. Most government and non-government institution preserves many types of records consisting of important information. For instant

each police department keeps record of criminals. Company records like no employees, wage paid, growth rate sale etc.

B. Census report and statistics: A very important source of data collection is census report and the annual digest of statistics and statistical report of various state departments and national bodies. These reports are reported by experts and, therefore, their data are highly reliable. The data collected in these reports are arranged according to the administrative objective,

- Health (concerning birth and death rate, number of hospitals, number of beds, number of doctors and the like)
- Economic and related data: Annual production, average income, foreign trade, wage, hours of work, price level (inflation rate), interest rate, exchange rate etc.)
- Education (number of schools, students, teachers, number of graduates and the like)

The Importance of census data can be summarized as:

- **Political importance:** It provides population statistics about the whole country and this information help government in formulating various plan and policies
- **Economic importance:** Economic plan of the country is based on population statistics. It provides data concerning income, expenditure and the like.
- **Sociological importance:** It also shows social condition, social evils, customs and tradition etc.
- **Commercial importance:** Commercial plans (business plans) require different types of information and statistical data. This is very much supplied by census report. Based on this information they will estimate the consumption (demand) and supply of different commodities.

C. Journals and magazines: These are common and widely used sources of information. Information obtained from these sources is very much reliable.

D. Newspapers: Published news (discussion on contemporary issues, reports of meetings and conferences) is very important sources. Economic news paper usually report data like,

- Daily stock price
- Exchange rate
- Interest rate
- Daily commodity price
- Price indices (report on quarterly basis)

E. Other documents: Under this category we can have information sources like, television, film, radio and public speech. The investigator, however, should be capable of sorting out the reliable material and distinguish it from unreliable one.

Generally, information through secondary data is relatively easy to obtain but difficult to use them in social science. Thus, the information should be carefully judged before using it.

Judgment of such information involves

- Verification of reliability of facts before using them in any study.
- Verification of honesty, ability, methods and tools of the investigator himself.
- Verification of the method of data collection
- Verification of representativeness of sample used in the primary data collection
- The organization for collection of data should be scrutinized as well

Chapter 7

Measurement and Scaling Concepts

7.1 The Nature of Measurement and scaling

- ☞ *In every day usage, measurement occurs when an established index verifies the height, weight or other feature of a physical object. How well you like a song, a painting or the personality of a friend is also measurement.*
- ☞ *To measure is to discover the extent, dimensions, quantity or capacity of something especially by comparison with a standard.*

What is measured?

Variables being studied in research may be classified as objects or as properties.

- a) *Objects include the concepts of ordinary experience, such as tangible items like furniture, laundry, detergent, people or automobiles. Objects also include things that are not as concrete such as genes, attitudes and peer-group pressures.*

- b) *Properties are the characteristics of the object. A person's physical properties may be stated in terms of weight, height and posture, among others. Psychological properties include attitudes and intelligence. Social properties include leadership ability class affiliation and status.*

These and many other properties of an individual can be measured in a research study. Scale

- ***A scale** may be defined as any series of items that are arranged progressively according to value or magnitude in to which an item can be placed according to its quantification.*
- *The purpose of scaling is to present, usually quantitatively, an item's, a person's or an event's place in the spectrum.*

Levels of Measurement Scales

- *In business research there are a great many scales. It is traditional to classify scales of measurement on the basis of the mathematical comparisons that are allowable with these scales. The four types of scales are nominal, ordinal, interval and ratio.*

1. Nominal scale

- *A nominal scale is the simplest type of scale. Numbers or letters assigned to objects serve as labels for identification or classification. These are scales "in name only."*
- *The data are sorted in to categories with no particular order to the categories.*

Example

- *Colour*
- *Gender*
- *Nationality*
- *Region*
- *Religion*

- *For the nominal level the only measurement involved consists of counts.*

2. Ordinal scale

- *An ordinal scale arranges objects or alternatives according to their magnitudes in an ordered relationship. The ordinal level of measurement presumes that one classification is ranked higher than another.*
- *A typical ordinal scale in business research asks respondents to rate career opportunities, brands, companies, or the like as “excellent”, “fair”, or “poor” but they do not know by how much.*

3. Interval scale

- *Interval scales not only indicate order, they also measure order (or distance) in units of equal intervals.*
- *The classic example of an interval scale is the Fahrenheit temperature scale. If a temperature is 80°, it cannot be said that it is twice as hot as 40°. The reason for this is that 0° does not represent the lack of temperature, but a relative point on the Fahrenheit scale.*
- *Similarly, when an interval scale is used to measure psychological attributes, the researcher can comment on the magnitude of difference or compare average differences on attributes that are measured but cannot determine the actual strength of attitudes toward an object.*

4. Ratio scale

- *Ratio scales have absolute rather than relative quantities. For example, money and weight are measured with ratio scale that possesses an absolute zero and interval properties. The absolute zero represents a point on the scale where there is an absence of the given attribute.*

Mathematical and Statistical Analysis of Scale

The type of scale that is utilized in business research will determine the form of statistical analysis.

a. Nominal scale

- The most sophisticated form of statistical analysis for nominal scale data is counting. Because the numbers in such a scale are merely labels for classification purposes, they have no quantitative meaning.*
- The researcher tallies the frequency in each category and identifies which category contains the highest number of observation.*

b. Ordinal Level

- Observation may be associated with a percentile rank such as median.*

c. Interval level

- An interval scale's property of equal intervals allows researchers to compare differences between scale values and to perform arithmetic operations such as addition and subtraction. Numbers may be changed, but the numerical operation must preserve order and relative magnitude of differences.*
- The mean and standard deviations (variance) may be calculated when true interval- scale data are obtained.*

d. Ratio level

- Researchers may make comparison of absolute magnitude because the scale has an absolute zero point.*
- Arithmetic operations on actual quantities are permissible. The ratio of scale values are meaning full.*

Type of scale	Numerical Operation	Descriptive statistics
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Nominal	Counting	Frequency in each category, Percentage in each category, Mode
Ordinal	Rank ordering	Median, range, percentile Ranking
Interval	Arithmetic operations On intervals b/n number	Mean, standard deviation, variance
Ratio	Arithmetic operation on Actual quantities	Geometric mean, coefficient of variation.

Remark: All statistics that are appropriate for lower order scales (nominal is the lowest) are appropriate for higher order scales (ratio is the highest)

7.2. Techniques for Measuring Attitudes

Measurement scales in business research are generally constructed to measure behaviour, knowledge and attitudes. Attitude scales are among the most difficult to construct so we will use attitudes to develop understanding of scaling.

The Nature of attitudes

An attitude is usually viewed as an enduring disposition to respond consistently in a given manner to various aspects of the world including persons, events and objects. There are three components of attitude: the affective, the cognitive and the behavioural.

- i. Affective component reflects an individual's general feelings or emotions toward an object. Statements such as "I love my job", "I liked that book" and "I hate Bete juice" reflect the emotional characteristics of attitudes.
- ii. The cognitive component represents one's awareness of and knowledge about an object.

Example: A woman might feel happy about her job because she "believes that the pay is great or because she knows" that my job is the biggest challenge."

- iii. Behavioural component intentions /behavioural expectation/ predisposition to action.

7.2.1. Rating scales

Using rating scales to measure attitude is perhaps the most common practice in business research. This section discusses many rating scales designed to enable respondents to report the intensity of responses.

1. Category scales

- Some rating scales have only two response categories, agree and disagree. Expanding the response categories provides the respondent more flexibility in the rating task.

Example

If you could choose, how much longer would you stay at your present job? Less

☐ Less than six months

☐ Six months to one year

☐ Longer than one year

2. Summated Ratings method: The Likert scale

With the Likert scale, respondents indicate their attitudes by checking how strongly they agree or disagree with carefully constructed statements that range from very positive to very negative toward the attitudinal object.

Individuals generally choose from five alternatives: Strongly agree, agree, uncertain, disagree and strongly disagree. But the number of alternatives may range from three to nine.

Example

Mergers and acquisitions provide a faster means of growth than internal expansion.

Strongly disagree	Disagree	Uncertain	Agree	strongly disagree
(1)	(2)	(3)	(4)	(5)

- To measure the attitude researcher assigns scores or weights to the alternative responses. In this example, weights of 5,4,3,2 and 1 are assigned to the answers.

- If a negative statement toward the object were given, the weights would be reversed and “strongly disagree” would be assigned the weight of 5.
- A single scale item on a summated rating scale is an ordinal scale.

3. Semantic differential

It is an attitude measure consisting of a series of seven point bipolar rating scale showing response to a concept. Bipolar adjectives such as “good” and “bad” , “modern” and “old fashioned”, or “clean” and “dirty”, anchor the beginning and end (or poles) of the scale.

Example

Modern_____, _____, _____, _____, _____, _____, _____ old fashioned

- Respondents are instructed to check the place that indicates the nearest appropriate adjective. From left to right, the scale intervals are interpreted as extremely modern, very modern, slightly modern, both modern and old fashioned, slightly old fashioned, very old fashioned and extremely old fashioned.
- A weight is assigned to each position on the rating scale. Traditionally scores are 7, 6, 5, 4, 3, 2, 1 or +3, +2, +1, 0, -1, -2, -3
- Many researchers find it desirable to assume that the semantic differential provides interval data. This assumption, although widely accepted, has its critics, who argue that the data have only ordinal properties because the weights are arbitrary.

4. Numerical Scales

Numerical scales have numbers, rather than “semantic space” or verbal descriptions, as response options, to identify categories (response options). If the scales have five response options, the scale is called as 5-point numerical scale; with seven response positions, it is called a 7-point numerical scale and so on.

Example

Now that you’ve had your automobile for about one year. Please tell us how satisfied you are with ford Taurus.

Extremely satisfied 7 6 5 4 3 2 1 Extremely dissatisfied

5. Constant sum scale

Respondents are asked to divide a constant sum to indicate the relative importance of the attributes

Example

Suppose you had \$3000 in benefits per month. How much would you like to allocate to salary, medical insurance and retirement plan? Divide the \$3000 according to your preference.

Salary _____

Medical insurance _____

Retirement plan _____

6. Staple scale

It places a single adjective in the center of an even number of numerical values (for example, ranging from +3 to -3). It measures how close or how distant from the objective a given stimulus is perceived to be.

The following example illustrates a staple scale used in a measurement of attitudes towards a supervisor.

+3 +2 +1 supportive -1 -2 -3

7. Graphic rating scales

- A graphic rating scale presents respondents with a graphic continuum to indicate their attitudes*

Example

Please evaluate each attribute in terms of how important it is to you by placing an "X" at the position on the horizontal line that most reflects your feelings.

Seating comfort Not important _____ very important

In flight meals Not important _____ very important

Air fare Not important _____ very important

Typically, respondent's score is determined by measuring the length (in millimetres) from one end of the graphic continuum to the point marked by the respondent.

Many researchers believe scoring in this manner strengthens the assumption that graphic rating scales of this type are interval scales. Alternatively, the researcher may divide the line in to predetermined scoring categories (lengths) and record respondents marks accordingly.

Summary of the advantages and disadvantages of rating scales

<i>Rating measure</i>	<i>Subject must</i>	<i>Advantages</i>	<i>Disadvantages</i>
<i>Category scale</i>	<i>Indicate response</i>	○ <i>Flexible</i> ○ <i>Easy to respond to</i>	○ <i>Items may be ambiguous</i> ○ <i>Only gross distinctions can be made</i>
<i>Likert scale</i>	<i>Evaluate statements on a scale that typically contains five points</i>	○ <i>Easiest scale to construct</i>	○ <i>Hard to judge what a single score means</i>
<i>Semantic differential or numerical scale</i>	<i>Choose points between bipolar adjectives on relative dimensions</i>	-<i>easy to construct</i> -<i>norms exist for comparison</i>	-<i>Bipolar adjectives must be found</i> -<i>data may be ordinal, not interval</i>
<i>Constant-sum scale</i>	<i>Divide a constant sum among response alternatives</i>	-<i>approximates interval measure</i>	<i>Difficult for respondents with low educational level</i>
<i>Staple scale</i>	<i>Choose a point on a scale with a single adjective in centre</i>	-<i>Easier to construct than semantic differential</i> -<i>easy to administer</i>	<i>- end points are not verbals, not labels</i>

Graphic scale	Choose a point on a continuum	-visual impact -Unlimited scale points	-No standard answers
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7.2.2 Ranking

The participant directly compares two or more objects and makes choices among them from most preferred to least preferred.

Example

Assume participants are asked to select the most preferred to among three models of a product. In response

40% choose model A

30% choose model B

30 % choose model C

Q- Which is the referred model?

Analysis: *the analyst would be taking a risk to suggest that A is most preferred. Perhaps all B and C voters would place A last preferring either B or C to A.*

Solution: *This ambiguity can be avoided by using the following techniques*

1. Paired comparison scale

It is a measurement technique that involves presenting the respondent with two objects and asking the respondent to pick the preferred object. More than two objects may be presented, but comparisons are made in pairs. No. of paired comparison for n stimuli/ objectives to be judged by

$$cn2 = \frac{n!}{2(n-2)!} = \frac{n(n-1)}{2}$$

Example: five designs

No. of comparisons = C_5^2

$$2 = 10$$

	A	B	C	D	E
A	--	164*	138	50	10
B	36	--	54	14	30
C	62	146	--	32	50
D	150	186	168	--	118
E	130	170	150	82	--
Total	378	666	510	178	268
Rank	3	1	2	5	4

164 of 200 customers preferred suggested design B

** If the data collection consists only of paired comparisons, as many as 10 stimuli are reasonable.*

2. The forced ranking scale

This scale Lists attributes that are ranked relative to each other. This method is faster paired comparisons and is usually easier and more motivating to the participant. Also, this ranking has no transitivity problem.

A drawback to forced ranking is the no. of stimuli that can be handled by this method.

3. Comparative Scale

This method calls for a standard by which other programs, processes, brands, products or people can be compared. The comparative scale is ideal for such comparisons if the participants are familiar with the standard.

Example

“Compared to your previous hair dryer’s performance, the new one is”

<i>Superior</i>		<i>about the same</i>		<i>inferior</i>
<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>

7.2.3 Sorting

Sorting tasks require that respondents indicate their attitudes or beliefs by arranging items.

Q- sorts

- ☞ Require sorting a deck of cards into piles that represent points along a continuum. The participant or judge groups the cards based on his/her response to the concept written on the card.*

Researchers using Q-sort resolve three special problems.

- 1. Item selection*
- 2. Structured or unstructured choices in sorting, and*
- 3. Data analysis*

Procedure

- 1. Selection of set of verbal statements, phrases, words or photos related to the concept related 60-120*
- 2. Shuffle the cards*
- 3. Sort the cards into a set of piers (usually 7 to 11) each pile representing a point on the judgment continuum. The left most pile represents the concepts statements which are “most valuable” “favorable”, “agreeable” and so forth. The right most pile contains the least favorable cards.*

7.2.4 Choice technique

It is a measurement task that identifies preference by requiring respondents to choose between two or more alternatives. If a respondent chooses one object over another, the researcher can assume that the respondent reference the chosen object over the other.

Chapter 8

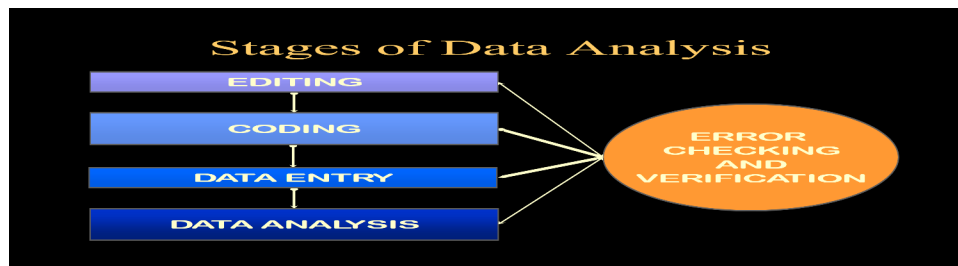
Methods of Data Analysis

8.1. Stages of Data Analysis

A very common phrase that is used by researchers is “garbage in, garbage out.” This refers to the idea that if data is collected improperly, or coded incorrectly, your results are “garbage,” because that is what was entered into the data set to begin with. This chapter focuses on the critical key step of data entry and coding. Like any part of the business research process, care and attention to detail are important requirements for data editing and coding.

The first two stages result in an electronic file suitable for data analysis. This file can then be used in the application of various statistical routines including those associated with descriptive, univariate, bivariate, or multivariate analysis.

Each of these data analysis approaches will be briefly outlined in the subsequent sections. An important part of the editing, coding, and filing stages is checking for errors. As long as error remains in the data, the process of transformation from raw data to intelligence will be made more risky and more difficult. Editing and coding are the first two stages in the data analysis process.



8.2. Editing and Coding

Editing

Editing is the process of checking and adjusting data for omissions, consistency, and legibility. In this way, the data become ready for analysis by a computer.

So, the editor's task is to check for errors and omissions on questionnaires or other data collection forms.

When the editor discovers a problem, he or she adjusts the data to make them more complete, consistent, or readable.

Types of Editing

A. Field Editing

Field supervisors often are responsible for conducting preliminary **field editing** on the same day as the interview. Field editing is used to

1. Identify technical omissions such as a blank page on an interview form
2. Check legibility of handwriting for open-ended responses
3. Clarify responses that are logically or conceptually inconsistent.

Field editing is particularly useful when personal interviews have been used to gather data. In these cases, a daily field edit allows supervisors to deal with some questions by asking interviewers, who may still be able to remember the interviews, about facts that may allow errors to be identified and perhaps corrected.

In addition, the number of unanswered questions or incomplete responses can be reduced with rapid follow up. A daily field edit allows fieldworkers to identify respondents who should be recontacted to fill in omissions in a timely fashion.

Although simultaneous field editing is highly desirable, in many situations (particularly with mail questionnaires) early reviewing of the data is not always possible. **In-house editing** rigorously investigates the results of data collection. The research supplier or research department normally has a centralized office staff perform the editing and coding function.

Reasons for Editing

1. Illustrating inconsistency fact or fiction?

Consider a situation in which a telephone interviewer has been instructed to interview only registered voters in a state that requires voters to be at least 18 years old. If the editor's review of a questionnaire indicates that the respondent was only 17 years old, the editor's task is to correct this mistake by deleting this response because this respondent should never have been considered as a sampling unit. The sampling units (respondents) should all be consistent with the defined population.

Example

A survey on out-shopping behaviour (shopping in towns other than the one in which the person resides) might have a question such as the following:

In which of the following cities have you shopped for clothing during the last year?

- *San Francisco*
- *Sacramento*
- *San José*
- *Los Angeles*
- *Other* _____

Please list the clothing stores where you have shopped during the last two months.

Suppose a respondent checks Sacramento and San Francisco to the first question. If the same respondent lists a store that has a location only in Los Angeles in the second question, an error is indicated. Either the respondent failed to list Los Angeles in the first question or listed an erroneous store in the second question. These answers are obviously inconsistent.

Taking action when response is obviously an error

What should the editor do? If solid evidence exists that points to the fact that the respondent simply failed to check Los Angeles, then the response to the first question can be changed to indicate that the person shopped in that city as well. Since Los Angeles is not listed next to Sacramento or San Francisco, it is unlikely that the respondent checked the wrong city inadvertently. Perhaps the question about the stores triggered a memory that did not come to the respondent when checking off the cities. This seems quite possible, and if another question can also point strongly to the fact that the respondent actually shopped at the store in Los Angeles, then the change should be made. However, perhaps the respondent placed a mail order with the store in Los Angeles and simply did not physically shop in the store. If other evidence suggests this possibility, then the researcher should not make an adjustment to the first question. For example, a later question may have the respondent list any clothing orders placed via mail order (or by telephone or Internet order).

Responses should be logically consistent, but the researcher should not jump to the conclusion that a change should be made at the first site of an inconsistency. In all but the most obvious situations, a change should only be made when multiple pieces of evidence exist that some response is in error and when the likely true response is obvious.

In some cases the respondent may have answered only the second portion of a two-part question. The following question creates a situation in which an in-house editor may have to adjust answers for completeness:

Does your organization have more than one computer network server?

_ Yes _ No

If yes, how many? _____

If the respondent checked neither yes nor no but indicated three computer installations, the editor should change the first response to a “Yes” as long as other information doesn’t indicate otherwise.

Here again, a computerized questionnaire may either not allow a response to the “how many” question if someone checked yes or require the respondent to go back to the previous question once he or she tries to enter a number for the “how many” question.

2. Editing Questions Answered Out of Order

Another task an editor may face is rearranging the answers given to open-ended questions such as may occur in a focus group interview. The respondent may have provided the answer to a subsequent question in his or her comments to an earlier open-ended question. Because the respondent already had clearly

identified the answer, the interviewer may not have asked the subsequent question, wishing to avoid hearing “I already answered that earlier” and to maintain interview rapport.

If the editor is asked to list answers to all questions in a specific order, the editor may move certain answers to the section related to the skipped question.

Coding

Editing may be differentiated from **coding**, which is the assignment of numerical scores or classifying symbols to previously edited data. Careful editing makes the coding job easier. Codes are meant to represent the meaning in the data.

Assigning numerical symbols permits the transfer of data from questionnaires or interview forms to a computer. **Codes** often, but not always, are numerical symbols. However, they are more broadly defined as rules for interpreting, classifying, and recording data. In qualitative research, numbers are seldom used for codes.

8.3. BASIC DATA ANALYSIS: Descriptive Statistics

Perhaps the most basic statistical analysis is descriptive analysis. Descriptive statistics can summarize responses from large numbers of respondents in a few simple statistics. When a sample is obtained, the sample descriptive statistics are used to make inferences about characteristics of the entire population of interest. This section introduces basic descriptive statistics, which are simple but powerful. This section also provides the foundation for the subsequent sections, which will extend basic statistics into the area of univariate statistical analysis. **Descriptive analysis** is the elementary transformation of data in a way that describes the basic characteristics such as central tendency, distribution, and variability.

Qualitative Data Summary Techniques: Graphical and Tabular

- Frequency Distribution
 - Relative Frequency
 - Percent Frequency Distribution
 - Bar Graph
 - Pie Chart
1. A frequency distribution is a tabular summary of data showing the frequency (or number) of items in each of several no overlapping classes.

The objective is to provide insights about the data that cannot be quickly obtained by looking only at the original data.

2. Relative Frequency Distribution:

The relative frequency of a class is the fraction or proportion of the total number of data items belonging to the class.

A relative frequency distribution is a tabular summary of a set of data showing the relative frequency for each class.

3. Percent Frequency Distribution

The percent frequency of a class is the relative frequency multiplied by 100.

A percent frequency distribution is a tabular summary of a set of data showing the percent frequency for each class.

4. Bar Graph

- A bar graph is a graphical device for depicting qualitative data that have been summarized in a frequency, relative frequency, or percent frequency distribution.
- On the horizontal axis we specify the labels that are used for each of the classes.
- A frequency, relative frequency, or percent frequency scale can be used for the vertical axis.
- Using a bar of fixed width drawn above each class label, we extend the height appropriately.
- The bars are separated to emphasize the fact that each class is a separate category.

5. Pie Chart

- The pie chart is a commonly used graphical device for presenting relative frequency distributions for qualitative data.
- First draw a circle; then use the relative frequencies to subdivide the circle into sectors that correspond to the relative frequency for each class.
- Since there are 360 degrees in a circle, a class with a relative frequency of .25 would consume $.25(360) = 90$ degrees of the circle.

Quantitative Data Summary Techniques: Graphical and tabular

- Frequency Distribution
- Relative Frequency and Percent Frequency Distributions
- Dot Plot
- Histogram
- Cumulative Distributions
- Ogive

1. Frequency Distribution

Guidelines for Selecting Number of Classes

- a. Use between 5 and 20 classes.
- b. Data sets with a larger number of elements usually require a larger number of classes.
- c. Smaller data sets usually require fewer classes.
- d. Use classes of equal width. Approximate Class Width =

2. Dot Plot

- One of the simplest graphical summaries of data is a dot plot.
- A horizontal axis shows the range of data values.
- Then each data value is represented by a dot placed above the axis

3. Histogram

- Another common graphical presentation of quantitative data is a histogram.
- The variable of interest is placed on the horizontal axis and the frequency, relative frequency, or percent frequency is placed on the vertical axis.
- A rectangle is drawn above each class interval with its height corresponding to the interval's frequency, relative frequency, or percent frequency.
- Unlike a bar graph, a histogram has no natural separation between rectangles of adjacent classes.

4. Cumulative Distribution

- The cumulative frequency distribution shows the number of items with values less than or equal to the upper limit of each class.
- The cumulative relative frequency distribution shows the proportion of items with values less than or equal to the upper limit of each class.
- The cumulative percent frequency distribution shows the percentage of items with values less than or equal to the upper limit of each class.

5. Ogive

- An ogive is a graph of a cumulative distribution.
- The data values are shown on the horizontal axis.
- Shown on the vertical axis are the:
 - cumulative frequencies, or
 - cumulative relative frequencies, or
 - cumulative percent frequencies
- The frequency (one of the above) of each class is plotted as a point.
- The plotted points are connected by straight lines.

Measures of Central Tendency (Location): Numerical Methods

- Mean
- Median
- Mode
- Percentiles
- Quartiles

1. Mean: The mean of a data set is the average of all the data values.

2. Median

- The median is the measure of location most often reported for annual income and property value data.
- A few extremely large incomes or property values can inflate the mean.
- The median of a data set is the value in the middle when the data items are arranged in ascending order.
- For an odd number of observations, the median is the middle value.
- For an even number of observations, the median is the average of the two middle values

3. Mode

- The mode of a data set is the value that occurs with greatest frequency.
- The greatest frequency can occur at two or more different values.
- If the data have exactly two modes, the data are bimodal.

- If the data have more than two modes, the data are multimodal.
- 4. Percentiles
 - A percentile provides information about how the data are spread over the interval from the smallest value to the largest value.
 - Admission test scores for colleges and universities are frequently reported in terms of percentiles.
- 5. Quartiles
 - Quartiles are specific percentiles
 - First Quartile = 25th Percentile
 - Second Quartile = 50th Percentile = Median
 - Third Quartile = 75th Percentile

Measures of Variability: Numerical Methods

- It is often desirable to consider measures of variability (dispersion), as well as measures of location.
- For example, in choosing supplier A or supplier B we might consider not only the average delivery time for each, but also the variability in delivery time for each.
 - Range
 - Interquartile Range
 - Variance
 - Standard Deviation
 - Coefficient of Variation

1. Range

- The range of a data set is the difference between the largest and smallest data values.
- It is the simplest measure of variability.
- It is very sensitive to the smallest and largest data values.

2. Interquartile Range

- The interquartile range of a data set is the difference between the third quartile and the first quartile.
- It is the range for the middle 50% of the data.
- It overcomes the sensitivity to extreme data values.

3. Variance

- The variance is a measure of variability that utilizes all the data.
- It is based on the difference between the value of each observation (x_i) and the mean ($\bar{x}$ for a sample, μ for a population).
- The variance is the average of the squared differences between each data value and the mean.

4. Standard Deviation

- The standard deviation of a data set is the positive square root of the variance.
- It is measured in the same units as the data, making it more easily comparable, than the variance, to the mean.
- If the data set is a sample, the standard deviation is denoted s .
- If the data set is a population, the standard deviation is denoted by σ (sigma).

5. Coefficient of Variation

The coefficient of variation indicates how large the standard deviation is in relation to the mean.

8.4. Univariate Analysis

Descriptive research and causal research designs often climax with hypothesis tests. Hypotheses are defined as formal statements of explanations stated in a testable form. Generally, hypotheses should be

stated in concrete fashion so that the method of empirical testing seems almost obvious.

Empirical testing typically involves inferential statistics. This means that an inference will be drawn about some population based on observations of a sample representing that population. Statistical analysis can be divided into several groups:

- **Univariate statistical analysis** tests hypotheses involving only one variable.
- **Bivariate statistical analysis** tests hypotheses involving two variables.
- **Multivariate statistical analysis** tests hypotheses and models involving multiple (three or more) variables or sets of variables.

Choosing the Appropriate Statistical Technique

Numerous statistical techniques are available to assist the researcher in interpreting data. Choosing the right tool for the job is just as important to the researcher as to the mechanic. Making the correct choice can be determined by considering

1. The type of question to be answered
2. The number of variables involved
3. The level of scale measurement

Today, the researcher rarely has to perform a paper and pencil calculation. Hypotheses are tested by using a correct click-through sequence in a statistical software package. The mathematics of these packages is highly reliable. Therefore, if the researcher can choose the right statistic, know the right click-through sequence, and read the output that results, the right statistical conclusion should be easy to reach.

1. Type of Question to Be Answered

The type of question the researcher is attempting to answer is a consideration in the choice of statistical technique. For example, a researcher may be concerned simply with the central tendency of a variable or with the distribution of a variable. Comparison of different business divisions' sales results with some target level will require a one-sample *t*-test. Comparison of two salespeople's average monthly sales will require a *t*-test of two means, but a comparison of quarterly sales distributions will require a chi-square test. The researcher should consider the method of statistical analysis before choosing the research design and before determining the type of data to collect. Once the data have been collected, the initial orientation toward analysis of the problem will be reflected in the research design.

2. Number of Variables

The number of variables that will be simultaneously investigated is a primary consideration in the choice of statistical technique. A researcher who is interested only in the average number of times

3. Level of Scale of Measurement

The scale measurement level helps choose the most appropriate statistical techniques and appropriate empirical operations. Testing a hypothesis about a mean, as we have just illustrated, is appropriate for interval scaled or ratio scaled data. Suppose a researcher is working with a nominal scale that identifies users versus nonusers of bank credit cards. Because of the type of scale, the researcher may use only the mode as a measure of central tendency. In other situations, where data are measured on an ordinal scale, the median may be used as the average or a percentile may be used as a measure of dispersion. For example, ranking brand preferences generally employs an ordinal scale. Nominal and ordinal data are often analyzed using frequencies or cross-tabulation.

Parametric versus Nonparametric Hypothesis Tests

The terms **parametric statistics** and **nonparametric statistics** refer to the two major groupings of statistical procedures. The major distinction between them lies in the underlying assumptions about the data to be analyzed. Parametric statistics involve numbers with known, continuous distributions.

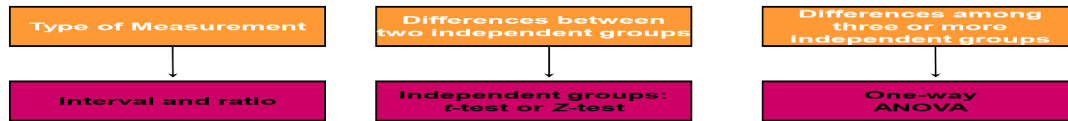
When the data are interval or ratio scaled and the sample size is large, parametric statistical procedures are appropriate. Nonparametric statistics are appropriate when the numbers do not conform to a known distribution.

Parametric statistics are based on the assumption that the data in the study are drawn from a population with a normal (bell-shaped) distribution and/or normal sampling distribution. For example, if an investigator has two interval-scaled measures, such as gross national product (GNP) and industry sales volume, parametric tests are appropriate. Possible statistical tests might include product-moment correlation analysis, analysis of variance, regression, or a *t*-test for a hypothesis about a mean.

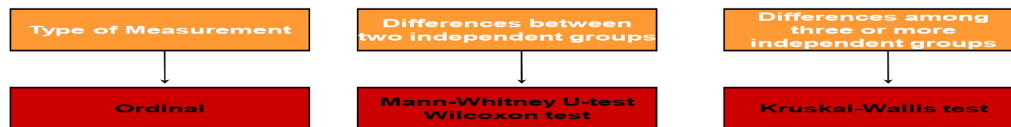
Nonparametric methods are used when the researcher does not know how the data are distributed. Making the assumption that the population distribution or sampling distribution is normal generally is inappropriate when data are either ordinal or nominal. Thus, nonparametric statistics are referred to as distribution free. Data analysis of both nominal and ordinal scales typically uses nonparametric statistical tests.

8.5. Bivariate statistical analysis: differences between two variables

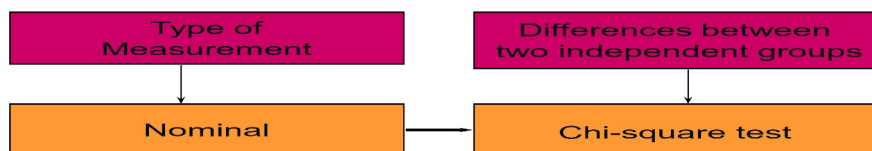
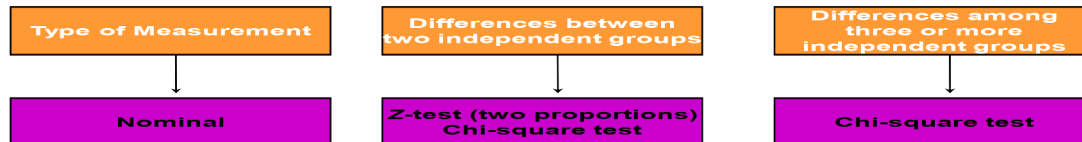
Common Bivariate Tests



Common Bivariate Tests



Common Bivariate Tests

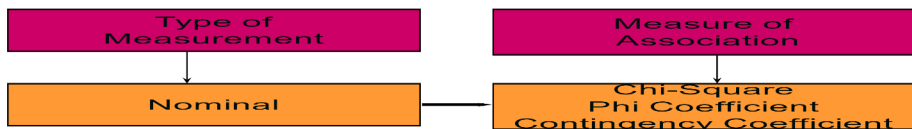
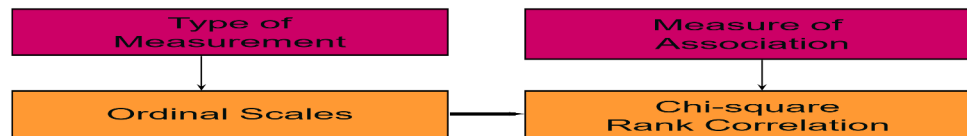
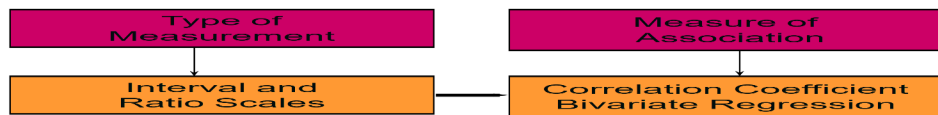


Bivariate Analysis: Measures of Associations

Measures of Association: A general term that refers to a number of bivariate statistical techniques used to measure the strength of a relationship between two variables.

Relationships among Variables

- Correlation analysis
- Bivariate regression analysis

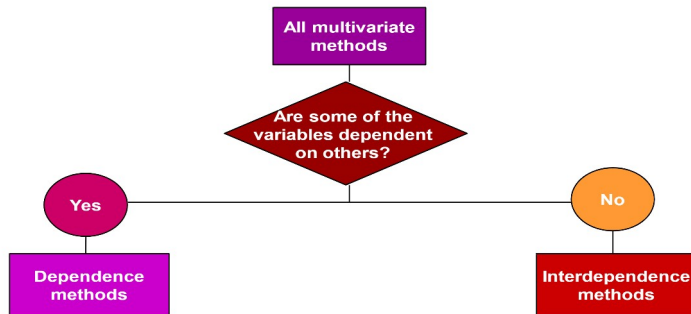


1. Correlation Coefficient
 - A statistical measure of the covariation or association between two variables.
 - Are dollar sales associated with advertising dollar expenditures?
2. Bivariate Regression
 - A measure of linear association that investigates a straight line relationship
 - Useful in forecasting

8.6. Multivariate Analysis

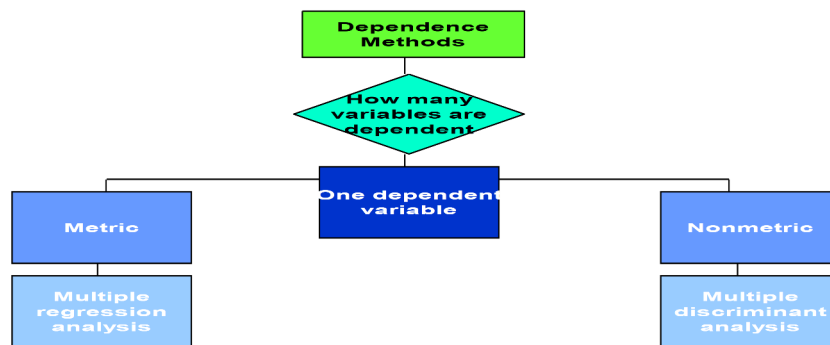
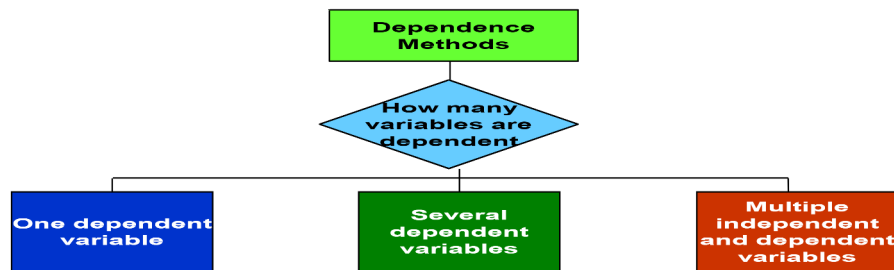
- Statistical methods that allow the simultaneous investigation of more than two variables

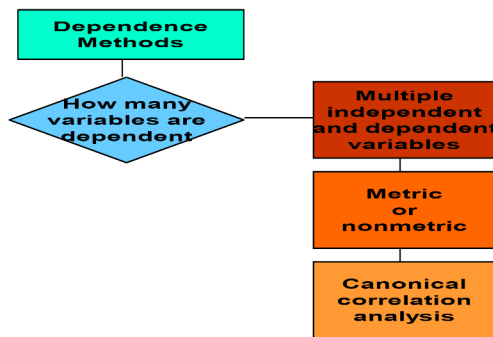
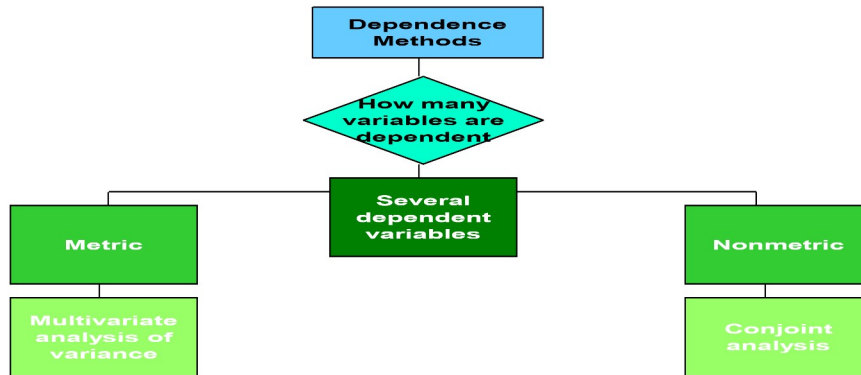
A Classification of Selected Multivariate Methods



Dependence Methods

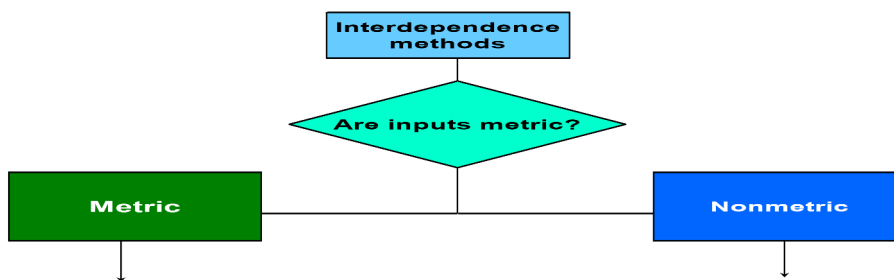
A category of multivariate statistical techniques; dependence methods explain or predict a dependent variable(s) on the basis of two or more independent variables

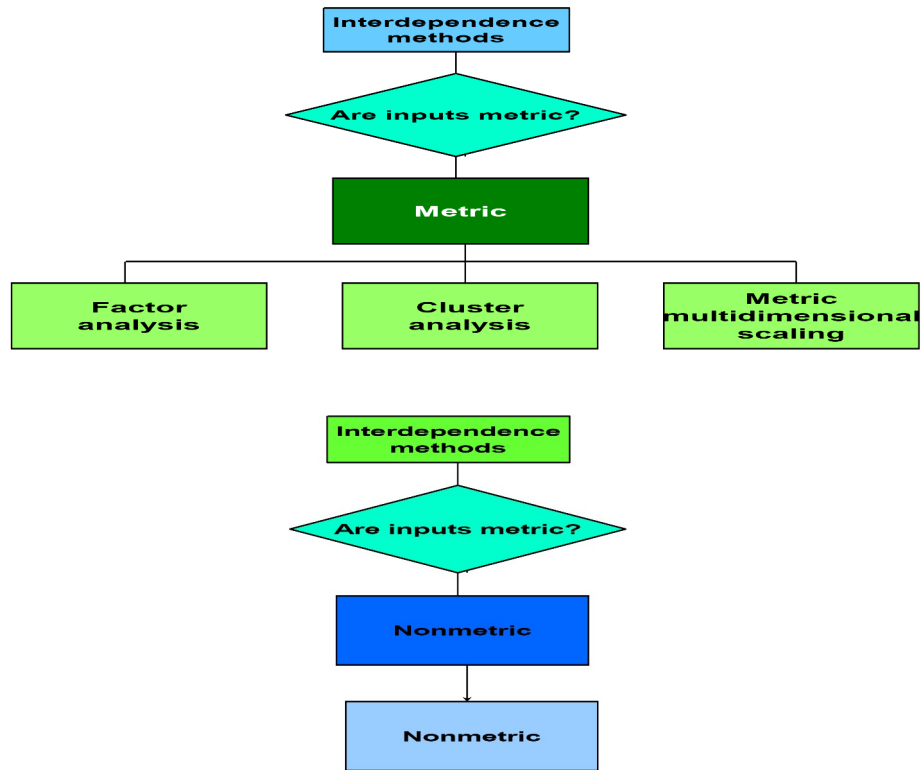




Interdependence Methods

- A category of multivariate statistical techniques; interdependence methods give meaning to a set of variables or seek to group things together





Chapter 9

Writing Research Report

9.1. THE RESEARCH REPORT

A research report deals with results of completed research work. After completing a research work, it is generally produced in the written 'form, and is called research report or thesis. A detailed description of research activities are provided in it. It has a variety of formats and varies as to its length.

It is written in past tense and in third person. It is the final form of the research work. A research report includes usually the following chapters-Introductory or theoretical background, Review of related literature, Methodology, Data collection, Analysis of data, Discussion of results and findings of the study, Bibliography and Appendices. It is also submitted for evaluating its contributions. It serves the purpose of communicating the results of a research work done.

Since the written report is an account of research project, the organization of the report follows quite closely the organization of the research project. The writing of the report is usually associated with the close of the research for project, few portions of writing may be done while the research study is in progress. The preliminary drafts of the review of the literature can be written and revised and brought upto-date later. Research procedures can be recorded as the project is conducted and while it is fresh in the researcher's mind. It is difficult to write from memory and avoid serious omissions.

The writer should assemble the available information before taking to the writing work. Mechanical procedures for presenting foot notes, references, figures, tables and the like should be well in mind. In the interest of efficiency and continuity, relatively lot of time should be reserved for the writing work.

The research report should contain all the necessary data and at the same time it should be brief and to the point. There is usually considerable changing and rewriting before the final draft. The report is also known as thesis or dissertation.

9.2. NEED OF RESEARCH REPORT

Writing a research report is very challenging task for the researcher. A good research report requires sufficient-experience and insight about his research activities. A research report is needed due to the following reasons:

- The research must be reported in full and its results should be subjected to a criticism and verification.
- The research work is done for the benefit of human being, therefore, it must be communicable to the general public for the practical use.
- The research should be considered to be the culminating act for reflective thinking. It encourages other persons to take up some problem for further investigation.
- The research report requires the creative thinking of a researcher to review the related studies and discussing the result of the study and also suggest some new problems for further studies.
- The research report is also necessary for giving shape and form to the investigation and solidifying it.
- It is needed for providing a clear picture of research method, sample and techniques used in conducting the research work.
- The research report is meant for popularizing the new contributions in the discipline.

9.3. GENERAL FORMAT OF RESEARCH REPORT

A written format of a research work is known as thesis or research report. All such works may differ considerably in scope of treatment and details of presentation. Even then all types of research reports are expected to follow a general uniform, common pattern of format, style and structure. The general format of research report is evolved and it has become a tradition in academic area. A research report or thesis is an organized format of research work done. It is viewed in three major categories:

- A. Preliminaries,
- B. Textual Body, and
- C. References.

Each category has been outlined further as follows:

- A. Preliminary Section

1. Title page
2. Preface or acknowledgements
3. Table of content
4. List of tables (if any)
5. List of figures (if any).
6. Acronyms

B. Main Body of Report or Textual Body

1. Introduction

- 1.1. Statement of the problem
- 1.2. Objectives of the study
- 1.3. Research questions/Hypotheses to be tested
- 1.4. Significance of the study
- 1.5. Assumptions and delimitations.
- 1.6 Definitions of Important terms used.

2. Review of related literature.

3. Design of the study

- 3.1. *variables of the study*
- 3.2. *Research Strategies*
- 3.3. *Sources of data*
- 3.4. *Methods of sampling*
- 3.5. *Methods of Data Collection*
- 3.6. *Methods of Data processing and analysis.*

4. Analysis and presentation of data

- (a) Analysis of data
- (b) Tables and interpretation
- (c) Figures and interpretation.

5. Conclusions

- (a) Discussion of results
- (b) Main Findings and inferences
- (c) Implication of the findings and limitations
- (d) Suggestions for further studies.

C. Reference Section

1. Bibliography
2. Appendices (if any)
3. Index or glossary (if any).

The detailed explanation of each aspect is given here.

9.3.1. PRELIMINARY SECTION

As the preliminaries form a significant part of the whole thesis report, due care should be taken in preparing them. If the specifications are already laid down by some colleges or universities they should be observed. However, a general standard pattern suggested here in each case will be helpful for a researcher.

1. Title Page

This is the first page of a thesis or a dissertation. It includes:

- (a) Title of thesis.
- (b) Name of the candidate.
- (c) Purpose or relationship of the thesis to the course or degree requirement.
- (d) College and/or department in which the candidate has been admitted for the degree.
- (e) Name of the university to which it is submitted.
- (f) Month and year of submission or acceptance.

The title should be accurate, concise and clearly printed in capital letters. It should convey the main theme of the problem investigated and if possible one should give a clue about the method or type of research involved. A specimen of title page has been presented here.

2. Preface or Acknowledgement

A preface is different from introduction. It is a brief account of the purport or the origin and the utility of the study for which the thesis is presented. It also includes the acknowledgement to the persons and sources that have been helpful to the investigator. If the researcher does not want to mention anything about the study on this page except acknowledging debt to others, it will be desirable to use the title simple and restrained without flattery and effusive recognition for help by the family members and others. The preface should not be too long with too many details about the research work or its organization, which can appear in introduction. The word PREFACE or ACKNOWLEDGEMENT should be typed in capital letters. It should be written in an impressive way.

3. Table of Contents

This section lists all the main chapter headings and the essential sub-heading in each with the appropriate page numbers against each. The listing of main chapters is generally preceded by some preliminaries like

preface or acknowledgement, list of tables, list of figures, abstract or synopsis and their respective pages in small Roman numbers and followed at the end by appendices, and Indexes.

Contents should neither be too detailed nor should too sketchy the table of contents serve an important purpose in providing an outline of the contents of the report. The capitalized title ‘Contents’ should be the central heading of the page and the capitalized word ‘CHAPTER’ and ‘PAGE’ should lead to the numbers of chapters and those of pages respectively on the left and right margins. An example’ has’ been given in the tabular of form.

4. List of Tables

The table of contents is followed by the list of tables on a separate page. This list of tables consists of the titles or captions of the tables included “in the thesis along with the page number where these can be located. It has been illustrated here. The capitalized title ‘LIST OF TABLES’ should be the central heading of the page and the capital words ‘TABLE’ and ‘PAGE’ should lead to the numbers and those of pages respectively at left and right margins.

5. List of Figures and Illustrations

If any charts graphs or any other illustrations are used in the thesis, a list of figures on a separate page is prepared in the same form as the list of tables except that they are numbered with Arabic numbers. An example has been given here for this -purpose.

Table of Contents

CONTENTS

Preface

List of Tables

List of Figures

I. INTRODUCTION

(a) Statement of Problem

(b) Objectives

(c) Hypotheses

(d) Assumptions and Limitations

II. REVIEW OF LITERATURE

III. DESIGN OF RESEARCH

(a) Method of Sample

(b) Procedure and Technique	
(c) Statistical Technique	
IV. ANALYSIS OF DATA	
V. CONCLUSIONS	
Bibliography	
Appendix	

LIST OF TABLES

Table	Page
1. Sample Structure	20
2. Distribution of Academic Qualification	22
3. Distribution of Aptitude Scores	23
4. Regression Weights	28

Similarly list-of figures is prepared. The page number of figures is given facing the page number of the report.

9.3.2. MAIN BODY OF REPORT OR TEXTUAL BODY

The text of the thesis is the most important section in the organization of research report. The quality of worth of thesis is mainly examined. It is the original production of the researcher. The report of the main body serves the function of demonstrating the competence of the researcher. If any sentence, paragraph, concept fails to serve the single function within a given section or chapter, it is irrelevant

The subject matter of any chapter should be relevant to that point. Generally the main body of the research reports consists of five or six chapters.

Chapter

- I. Introduction or Theoretical Frame Work
- II. Review of Related Literature
- III. Design or Methodology
- IV. Data Collection or Administration of Tools and Scoring.
- V. Analysis and Interpretation of Data.
- VI. Conclusions and Suggestions for the Further Researches.

Chapter 1: Introduction or Theoretical Frame Work

The main purpose of this chapter is to indicate the need and scope of the study. It consists essentially of the statement of research inquiry. It is reported in past tense form of work completed. The problem objectives,

hypotheses, assumptions and delimitations of the study are reported precisely. If an introduction is required, the researcher should make certain that it is an introduction that generates an interest and appropriate mental set which introductions are regarded as capable of producing. It must be long enough to do its jobs and nothing more.

Chapter 2: Review of Related Literature

This chapter is essential in most of the research studies. It presents the comprehensive development of the problem background. It indicates what has already been studied by others, which has a bearing upon the present study.

The review of literature stresses two aspects: the first is the consideration of the subject-matter and it is likely more important than the other. The second is related to methodology and design. The review chapter is devoted to the development of the problem statement or the object of the inquiry. The review is utilized to retain a direct relevancy to the study in hand. It is the balancing chapter of the research report.

Chapter 3: Design or Methodology of Research

This chapter indicates the line of approach of the study. The first aspect deals with the method, population and sample of the study and second part provides the tools and techniques employed in the research. It also presents the procedure of the study. The whole plan of the study is discussed in detail under this chapter. Administration of tools and scoring procedure are reported systematically. The data organization and presentation should be given in this section. It may be reported in a separate chapter of the report.

Chapter 4: Analysis and Interpretation of the Data

In this chapter analysis and results are reported so as to draw the inferences of the study. The analyses of data are presented in tabular form and in figures or pictorial presentation. The results are interpreted at length. This chapter provides the original work or contribution by the researcher. The communicative inaccuracy is required in this chapter. The text must be developed to ensure an effective ordering of the evidences.

Chapter 5: Conclusions and Suggestions

This is most important chapter of the report. It requires the creative and reflective aspect of the researcher. The results are discussed to make them more meaningful comparison of the results with the evidence in the review section should be woven into the text whenever such a discussion can serve to clarify the points being reported. This is the final chapter of a report, thus findings of the study are summarized and suggestions for the further studies are also given. The implications and delimitations of the findings are

also mentioned in this section. The main thrust in the section is the answer of the question or solution of the problem. The validity of the findings should be mentioned.

9.3.3 REFERENCE SECTION

This is the third section of a research report. It consists of generally the bibliography and appendices. It is also essential to include glossary and index for the convenience of the readers. The bibliography, appendix, glossary and index all these are written on a separate page - in the centre with capital letters.

1. Bibliography

The bibliography is a list of the printed sources utilized in the research work. The publications used for information-yield but not quoted in the report may also be included in the bibliography. The formats of the bibliography depend on the footnote style. If the foot-notes reference in the text are numbered to refer to the source in the bibliography, the entries must be numerically listed in the order of appearance in the text. The various format manuals include information on form for the bibliography. If the list of sources is too large the bibliography should be categorized in the following sections:

Books, monographs, documents and reports, periodicals and journals, essay and articles, unpublished thesis and material and newspapers.

If selected sources are reported the words 'Selected Bibliography' should be written. In writing bibliography the surname is written first than initials, year of publication, title of the book, publishers name, place and total number of pages.

2. Appendix

An appendix is the important reference materials category. It includes the material which cannot be logically included in the main body or textual body of the research report or the relevant materials too unwieldy to include in the main body. The appendix usually includes: tools of research, statistical tables and sometime raw-data (when data were processed through computer). Even the material of minor importance e.g. forms, letters, reminders, interview sheets, blank questionnaires, charts, tables, lengthy questions, report of cases (if follow-up or case studies have been conducted). The tools and other material should be placed first and tables at the end and page numbers should be assigned in Roman Numbers (i, ii, xxi). The appendix serves the function of providing greater clarity and authenticity for the readers or consumers of the thesis. The items of the appendix are very essential for a good research report.

3. Index and Glossary

When a research report is published in index, must be given. The index includes authors and subjects and topics or words in alphabetical order. In the report glossary should be provided. It includes the meanings or

definitions of some words and terms ‘used in the research report. Some notations symbols or abbreviations should be explained what actually they mean or indicate in the study.